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A Reproducing-Kernel-Based Nonparametric Test for Conditional Independence of Functional Data

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Abstract

Conditional independence is a fundamental concept in many areas of statistical research, including, for example, sufficient dimension reduction, causal inference, and statistical graphical models. In many modern applications, data arise in the form of random functions, making it important to determine whether two random functions are conditionally independent given a third. However, to the best of our knowledge, existing conditional independence tests in the literature apply only to multivariate data, and extensions to the functional setting are not available. To fill this gap, we develop a reproducing-kernel-based test for conditional independence of random functions based on the conjoined conditional covariance operator (CCCO). We rigorously derive the asymptotic distribution of the CCCO estimator using a recently established sharpened convergence rate for the regression operator (Choi et al., 2026). Based on this result, we construct a test statistic using the spectral decomposition of the operator appearing in the asymptotic distribution. The proposed method is illustrated through applications to an activity and biometrics dataset and a macroeconomic dataset.

Keywords: conditional independence, functional data, conjoined conditional covariance operator, reproducing kernel Hilbert space

1 Introduction

Conditional independence is a fundamental concept underlying a wide range of multivariate data analysis methods. For instance, in Sufficient Dimension Reduction, we need to find β such that $Y \perp\!\!\!\perp X|\beta^{\mathsf{T}}X$ (Li, 2018b). In graphical models, nodes i and j are disconnected if and only if

$$X_i \perp\!\!\!\perp X_j|\{X_1, \ldots, X_p\} \setminus \{X_i, X_j\},$$

where $X_1, \ldots, X_p$ represent the random variables on nodes $1, \ldots, p$. See Lauritzen (1996) and Koller and Friedman (2009). In causal inference, the common cause principle indicates that two dependent random variables X and Y may have a common cause Z such that $X \perp\!\!\!\perp Y|Z$ (Peters et al., 2017). Conditional independence plays a central role in all three applications.

It is thus of great importance to test whether two random variables are conditionally independent given a third. Li and Fan (2020) provides a comprehensive review of nonparametric conditional independence (CI) tests for continuous variables. For example, under the Gaussian assumption, partial correlation and conditional correlation can be used to characterize the conditional dependency between two random variables. In fact, the Gaussian assumption can be relaxed as indicated in Baba et al. (2004). Representative works in this area include the discretization-based test (Huang, 2010), metric-based tests (Su and White, 2008; Huang et al., 2016), the permutation-based two-sample test (Doran et al., 2014), and the mutual independence test under transformations (Cai et al., 2022). Also, some regression-based methods can be employed for this purpose under some additivity or linearity assumptions (see, for example, Zhang et al. (2017)).

To further relax modeling assumptions, reproducing-kernel-based methods have recently gained prominence by embedding probability distributions into feature spaces induced by kernels. Within the reproducing-kernel Hilbert space (RKHS) framework, conditional independence can be characterized via the Conjoined Conditional Covariance Operator (CCCO); see, for example, Fukumizu et al. (2004, 2007b); Lee et al. (2016). Building on this characterization, Zhang et al. (2011) proposed a kernel-based conditional independence test whose statistic is closely related to the Hilbert–Schmidt norm of the CCCO introduced in Fukumizu et al. (2007b). Related developments include Huang et al. (2022), which introduces a kernel partial correlation coefficient as a measure of conditional independence, and Sheng and Sriperumbudur (2023), which investigates the connection between distance-based and reproducing-kernel-based measures of conditional independence. As a sidenote, we emphasize that the reproducing kernels considered in this paper are fundamentally different from the smoothing kernels used in functional nonparametric regression and estimation (Ling and Vieu, 2018).

Our conditional independence test for functional data has wide applications in nonparametric Functional Data Analysis. For example, it could be used for variable selection in the function-on-function nonparametric regression setting (see, for example, Sang and Li (2026)), where we can drop a predictor X_2 if $Y \perp\!\!\!\perp X_2 \mid X_1$. Here, X_1, X_2 and Y can all be random functions or vectors of random functions. Another example is the functional graphical model (Qiao et al., 2019; Li and Solea, 2018), where we can determine the number of edges in a statistical graphical model, with the vertex variables being random functions.

Although a rich literature exists on conditional independence testing for multivariate data, analogous methods for random functions have not, to the best of our knowledge, been developed. This paper fills this gap by developing a reproducing-kernel-based conditional independence test for functional data, where the three random elements involved in the test are random functions or vectors of random functions. Such functional observations are increasingly prevalent in modern data analysis (Goia and Vieu, 2016; Aneiros et al., 2019, 2022), particularly with the rapid growth of applications involving smart wearable devices. See, for example, Ramsay and Silverman (2007) and Acar-Denizli and Delicado (2025). A flexible conditional independence testing procedure would constitute a valuable addition to the toolbox of functional data analysis, enabling the study of complex interconnections among functional variables. Such interconnections arise in a wide range of problems, including function-on-function regression (Sun et al., 2018; Sang and Li, 2026), functional graphical models (Qiao et al., 2019; Li and Solea, 2018), and functional sufficient dimension

reduction (Ferré and Yao, 2003; Li and Song, 2017, 2022).

In addition to the importance of conditional independence testing for functional data, we would like to highlight a particularly important theoretical contribution of our extension. As the pioneering work on reproducing-kernel-based conditional independence testing, Zhang et al. (2011) provided only a sketch of the proof for the asymptotic null distribution, without a detailed list of conditions under which the asymptotic distribution holds.

A key difficulty arises from the fact that the test statistic involves the inverse of a covariance operator. In practice, the inverse must be estimated using Tikhonov regularization, which slows down the convergence rate of a related regression operator. Until recently, the best known convergence rate for this operator was

$$\epsilon_n^{\beta\wedge 1} + n^{-1/2}\epsilon_n^{-1}, \tag{1}$$

where n is the sample size, ϵ_n is the Tikhonov regularization parameter, and $\beta > 0$ characterizes the smoothness of the regression relationship between the two random variables involved. See, for example, Li and Song (2017). The optimal rate implied by (1) is $n^{-1/4}$, which can be achieved when $\beta \geq 1$ with the optimal tuning $\epsilon_n \asymp n^{-1/4}$. However, our calculations (see Section 5) indicate that this rate is not sufficiently fast to guarantee the asymptotic distribution claimed in Zhang et al. (2011). Recently, under a slightly different setting, Choi et al. (2026) sharpened the convergence rate of a similar regression operator to

$$n^{-1/2}\epsilon_n^{(\beta\wedge 1)-1} + \epsilon_n^{\beta\wedge 1} + n^{-1}\epsilon_n^{-(3\alpha+1)/(2\alpha)} + n^{-1/2}\epsilon_n^{-(\alpha+1)/(2\alpha)}, \tag{2}$$

where $\alpha > 1$ characterizes the decay rate of the eigenvalues of the covariance operator. It turns out that we can use that approach to sharpen the convergence rate of the regression operator in the present context, and with the sharpened rate, we can rigorously establish the asymptotic distribution of our test statistics parallel to that claimed in Zhang et al. (2011) in the multivariate setting.

Although our analysis is carried out in the functional setting, the above result also applies to the multivariate setting. Consequently, Theorem 2 provides the first rigorous justification of the asymptotic null distribution for the reproducing-kernel-based conditional independence test. Given the widespread use of this test, establishing its asymptotic validity is both important and significant.

The rest of the paper is organized as follows. In Section 2, we review the background on RKHS and several linear operators needed for constructing the test statistic at the population level. In Section 3, we introduce the test statistic for conditional independence. In Section 4, we rigorously develop the asymptotic distribution of the proposed reproducing-kernel-based test statistic for conditional independence for functional data. We also carefully discuss the key conditions needed in the asymptotic development. In Section 5, we introduce practical methods for approximating random functions involved in constructing the test statistic. Section 6 presents the sample-level implementation, including estimation of the eigenvalues appearing in the asymptotic distribution using an acceleration method, selection of tuning parameters, and the resulting algorithms. Section 7 reports results from simulation studies, and Section 8 illustrates the proposed method through an application to the WISDM dataset. Due to the page limit, another application to the WDI dataset (World Bank, 2025)

is placed in the Supplementary Material. Some illustrative examples and proofs of most results are also deferred to Appendix A and Appendix B in the Supplementary Material to avoid interrupting the main exposition.

2 Background of RKHS

2.1 Construction of nested Hilbert spaces

Let $(\Omega, \mathcal{F}, P)$ be a probability space, let T be an interval in $\mathbb{R}$ which, without loss of generality, is assumed to be $[0,1]$, and let $\mathcal{H}_X, \mathcal{H}_Y, \mathcal{H}_Z$ be a separable Hilbert spaces of functions defined on T. Let $\mathcal{F}_X, \mathcal{F}_Y, \mathcal{F}_Z$ be the Borel σ-fields generated by the open sets in $\mathcal{H}_X, \mathcal{H}_Y, \mathcal{H}_Z$, respectively. Let $\mathcal{F}_X \times \mathcal{F}_Y \times \mathcal{F}_Z$ be the product σ-field and let $(\mathcal{H}_X \times \mathcal{H}_Y \times \mathcal{H}_Z, \mathcal{F}_X \times \mathcal{F}_Y \times \mathcal{F}_Z)$ be the product measurable space. Furthermore, let $(X, Y, Z) : \Omega \to \Omega_X \times \Omega_Y \times \Omega_Z$ be a random element measurable with respect to $\mathcal{F}/(\mathcal{F}_X \times \mathcal{F}_Y \times \mathcal{F}_Z)$, and let $P_X = P \circ X^{-1}$, $P_Y = P \circ Y^{-1}$, $P_Z = P \circ Z^{-1}$, and $P_{XYZ} = P \circ (X, Y, Z)^{-1}$ denote the distributions of X, Y, Z, and (X, Y, Z), respectively. Our goal is to construct a test for whether $X \perp\!\!\!\perp Y | Z$ holds.

Having defined the random elements X, Y, Z, we then introduce positive definite kernels

$$\kappa_X : \mathcal{H}_X \times \mathcal{H}_X \to \mathbb{R}, \quad \kappa_Y : \mathcal{H}_Y \times \mathcal{H}_Y \to \mathbb{R}, \quad \kappa_Z : \mathcal{H}_Z \times \mathcal{H}_Z \to \mathbb{R}.$$

Let $\mathcal{G}_X, \mathcal{G}_Y, \mathcal{G}_Z$ be the RKHS's generated by $\kappa_X, \kappa_Y, \kappa_Z$, respectively. For further information about RKHS, see, for example, Aronszajn (1950); Berlinet and Thomas-Agnan (2004). We assume that κ_X is uniquely determined by the inner product in $\mathcal{H}_X$; that is, for any $f, g \in \mathcal{H}_X$, $\kappa_X(f, g)$ is a function of $\langle f, f \rangle_{\mathcal{H}_X}$, $\langle g, g \rangle_{\mathcal{H}_X}$ and $\langle f, g \rangle_{\mathcal{H}_X}$. Li and Song (2017); Li and Solea (2018) refer to this special structure as nested Hilbert spaces: $\mathcal{H}_X$ is nested in $\mathcal{G}_X$ in the sense that the inner product in $\mathcal{H}_X$ determines the kernel in $\mathcal{G}_X$. We make the same assumptions on κ_Y, κ_Z, $\mathcal{G}_Y$, $\mathcal{G}_Z$ as well.

2.2 Mean element and covariance operator

Before introducing the mean element and covariance operator, we first make the following assumption.

Assumption 1. *The kernels κ_X, κ_Y and κ_Z satisfies $E[\kappa_X(X, X)] < \infty$, $E[\kappa_Y(Y, Y)] < \infty$ and $E[\kappa_Z(Z, Z)] < \infty$.*

With the reproducing kernel κ_X specified, we define the mean element of X in the RKHS $\mathcal{G}_X$. Under Assumption 1, there is a unique element $\mu_X \in \mathcal{G}_X$ satisfying

$$\langle \mu_X, f \rangle_{\mathcal{G}_X} = E[f(X)], \qquad \forall f \in \mathcal{G}_X.$$

This element is defined as the mean element of X. We denote this element by $E[\kappa_X(\cdot, X)]$, so that the identity

$$\langle E[\kappa_X(\cdot, X)], f \rangle_{\mathcal{G}_X} = E\langle \kappa_X(\cdot, X), f \rangle_{\mathcal{G}_X} = E[f(X)]$$

is satisfied. We define the mean elements $\mu_Y = E[\kappa_Y(\cdot, Y)]$ and $\mu_Z = E[\kappa_Z(\cdot, Z)]$ in the same way.

We also need the definition of characteristic kernels, also known as probability-determining kernels (see Fukumizu et al. (2004); Zhang et al. (2011)).

Definition 1 (characteristic kernel)**.** *Let $(A, \mathcal{F}_A)$ be a measurable space, and $\kappa : A \times A \to \mathbb{R}$ be a reproducing kernel. Let $X : \Omega \to A$ and $Y : \Omega \to A$ be random elements that are measurable with respect to $\mathcal{F}/\mathcal{F}_A$. We say that κ is characteristic if*

$$E[\kappa(\cdot, X)] = E[\kappa(\cdot, Y)] \;\Rightarrow\; P \circ X^{-1} = P \circ Y^{-1}.$$

According to Proposition 2 of Zhang et al. (2024), the Gaussian kernel and Laplace kernel in $L^2(T)$, defined by

$$\kappa_1(\cdot, z) = \exp(-\gamma_1 \left\| \cdot - z \right\|^2), \quad \gamma_1 > 0, \quad \text{and} \quad \kappa_2(\cdot, z) = \exp\left(-\gamma_2 \left\| \cdot - z \right\|\right), \quad \gamma_2 > 0,$$

respectively, are both characteristic kernels on their corresponding RKHS's.

We then define the covariance operator as follows. Under Assumption 1, there is a unique bounded linear operator $\Sigma_{XX} : \mathcal{G}_X \to \mathcal{G}_X$ satisfying

$$\langle f, \Sigma_{XX} g \rangle_{\mathcal{G}_X} = \operatorname{cov}\left[f(X), g(X)\right]$$

for all $f, g \in \mathcal{G}_X$. This operator is called the covariance operator of X. It can be equivalently expressed as

$$\Sigma_{XX} = E\left[(\kappa_X(\cdot, X) - \mu_X) \otimes (\kappa_X(\cdot, X) - \mu_X)\right]. \tag{3}$$

It can be shown that Σ_{XX} is a Hilbert-Schmidt operator on $\mathcal{G}_X$ under Assumption 1; see, for example, Fukumizu et al. (2007a). Linear operators Σ_{YY} and Σ_{ZZ} are defined similarly. Letting $\tilde{\kappa}$ be centralized kernel $\tilde{\kappa}_X(\cdot, x) = \kappa_X(\cdot, x) - \mu_X$, we can represent (3) as

$$\Sigma_{XX} = E\left[\tilde{\kappa}_X(\cdot, X) \otimes \tilde{\kappa}_X(\cdot, X)\right]. \tag{4}$$

Similarly, we can rewrite Σ_{YY} and Σ_{ZZ} as

$$\Sigma_{YY} = E\left[\tilde{\kappa}_Y(\cdot, Y) \otimes \tilde{\kappa}_Y(\cdot, Y)\right], \quad \text{and} \quad \Sigma_{ZZ} = E\left[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)\right].$$

We further define the (cross-)covariance operator between (X, Y), $\Sigma_{XY} : \mathcal{G}_Y \to \mathcal{G}_X$, as

$$\Sigma_{XY} = E\left[(\kappa_X(\cdot, X) - \mu_X) \otimes (\kappa_Y(\cdot, Y) - \mu_Y)\right] = E\left[\tilde{\kappa}_X(\cdot, X) \otimes \tilde{\kappa}_Y(\cdot, Y)\right]. \tag{5}$$

Equivalently, for all $f \in \mathcal{G}_X$ and $g \in \mathcal{G}_Y$, we have

$$\langle f, \Sigma_{XY} g \rangle = \operatorname{cov}\left[f(X), g(Y)\right].$$

A well-known fact is that, if κ_X and κ_Y are characteristic, then X and Y are independent if and only if $\Sigma_{XY} = 0$. See, for example, Fukumizu et al. (2007b). This fact is often used to construct tests for independence (Gretton et al., 2005; Sejdinovic et al., 2013). We define Σ_{XZ}, Σ_{YZ}, Σ_{YX}, Σ_{ZX} and Σ_{ZY} similarly.

2.3 Conditional Covariance Operator (CCO)

Based on the definitions of the mean element and the covariance operator, we define the conditional covariance operator (CCO) of (X, Y) given Z, denoted by $\Sigma_{XY|Z} : \mathcal{G}_Y \to \mathcal{G}_X$, as follows:

$$\Sigma_{XY|Z} = \Sigma_{XY} - \Sigma_{XZ}\Sigma_{ZZ}^{\dagger}\Sigma_{ZY}, \tag{6}$$

where $\Sigma_{ZZ}^{\dagger}$ is the Moore-Penrose inverse of Σ_{ZZ} (see, for example, Li (2018a); Hsing and Eubank (2015)). Further define

$$B_{X|Z} = \Sigma_{XZ}\Sigma_{ZZ}^{\dagger}, \qquad B_{Y|Z} = \Sigma_{YZ}\Sigma_{ZZ}^{\dagger}.$$

Note that $B_{X|Z}$ and $B_{Y|Z}$ are precisely the adjoint operators of the regression operators introduced in Chapter 13 of Li (2018b) and in Li (2018a). We adopt a different notation here (than, say, R_{ZX}^* and R_{ZY}^*) for convenience in subsequent developments.

To establish an equivalent representation of $\Sigma_{YX|Z}$, we first present the following lemma and corollary, which connect the operators $B_{X|Z}$ and $B_{Y|Z}$ to the regression between two centralized kernels. We begin by stating an additional technical assumption.

Assumption 2.

(i) For all $g \in \mathcal{G}_X$, the function $E[g(X)|Z = \cdot]$ is an element of $\mathcal{G}_Z$;

(ii) For all $g \in \mathcal{G}_Y$, the function $E[g(Y)|Z = \cdot]$ is an element of $\mathcal{G}_Z$.

Lemma 1. *If Assumptions 1 and 2 are satisfied, then, for all $f \in \ker(\Sigma_{ZZ})^{\perp} \subset \mathcal{G}_Z$, $g \in \mathcal{G}_X$ and $h \in \mathcal{G}_Y$, we have*

$$\langle B_{X|Z}f, g\rangle_{\mathcal{H}_X} = \langle f, E[g(X)|Z = \cdot]\rangle_{\mathcal{H}_Z}, \quad \langle B_{Y|Z}f, h\rangle_{\mathcal{H}_Y} = \langle f, E[h(Y)|Z = \cdot]\rangle_{\mathcal{H}_Z}. \tag{7}$$

Proof. We will only show the first part of (7), as the second part is the same statement with X replaced by Y and g replaced by h. By the definition of an adjoint operator,

$$\langle B_{X|Z}f, g\rangle_{\mathcal{H}_X} = \langle \Sigma_{XZ}\Sigma_{ZZ}^{\dagger}f, g\rangle_{\mathcal{H}_X} = \langle f, \Sigma_{ZZ}^{\dagger}\Sigma_{ZX}g\rangle_{\mathcal{H}_Z}.$$

The rest of the proof follows from Theorem 2 and Corollary 3 of Fukumizu et al. (2004). □

Corollary 1. *Under Assumptions 1 and 2, we have*

$$E[\tilde{\kappa}_X(\cdot, X)|Z = z] = B_{X|Z}\tilde{\kappa}_Z(\cdot, z), \quad E[\tilde{\kappa}_Y(\cdot, Y)|Z = z] = B_{Y|Z}\tilde{\kappa}_Z(\cdot, z). \tag{8}$$

The proof of Corollary 1 is given in Appendix B.1. As we can see from (8), $B_{X|Z}$ and $B_{Y|Z}$ can be viewed as the regression coefficients of $\tilde{\kappa}_X(\cdot, X)$ and $\tilde{\kappa}_Y(\cdot, Y)$ on $\tilde{\kappa}_Z(\cdot, Z)$, which is the motivation for choosing $X|Z$ and $Y|Z$ as the subscripts of B. Also, from Corollary 1, we derive the equivalent representation of $\Sigma_{XY|Z}$.

Proposition 1. *Let $\Sigma_{XY|Z}$ be defined by (6). Under Assumptions 1 and 2,*

$$\Sigma_{XY|Z} = E\left[\left(\tilde{\kappa}_X(\cdot, X) - E\left[\tilde{\kappa}_X(\cdot, X)|Z\right]\right) \otimes \left(\tilde{\kappa}_Y(\cdot, Y) - E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right]\right)\right]. \tag{9}$$

The proof of Proposition 1 is given in Appendix B.2. Note that the μ_X and μ_Y terms in $\tilde{\kappa}_X(\cdot, x)$ and $\tilde{\kappa}_Y(\cdot, y)$ are nonrandom. Two direct corollaries of Proposition 1 are as follows.

Corollary 2. *Under Assumptions 1 and 2,*

$$\Sigma_{XY|Z} = E\left[(\kappa_X(\cdot, X) - E\left[\kappa_X(\cdot, X)|Z\right]) \otimes (\kappa_Y(\cdot, Y) - E\left[\kappa_Y(\cdot, Y)|Z\right])\right]. \tag{10}$$

Corollary 3. *Under Assumptions 1 and 2, for any $f \in \mathcal{G}_X$ and $g \in \mathcal{G}_Y$, we have*

$$\langle f, \Sigma_{XY|Z} g\rangle_{\mathcal{G}_X} = E\left\{\mathrm{cov}[f(X), g(Y)|Z]\right\}.$$

For notational convenience, define the conditional centered kernel as

$$\tilde{\kappa}_{X|Z}(\cdot, x|z) = \tilde{\kappa}_X(\cdot, x) - E[\tilde{\kappa}_X(\cdot, X)|Z = z] = \tilde{\kappa}_X(\cdot, x) - B_{X|Z}\tilde{\kappa}_Z(\cdot, z). \tag{11}$$

By construction, $\tilde{\kappa}_{X|Z}(\cdot, x|z) \in \mathcal{H}_X$. Analogously, we define $\tilde{\kappa}_{Y|Z}(\cdot, y|z) \in \mathcal{H}_Y$. With these notations, the representation in (9) can be written more compactly as

$$\Sigma_{XY|Z} = E\big[\tilde{\kappa}_{X|Z}(\cdot, X|Z) \otimes \tilde{\kappa}_{Y|Z}(\cdot, Y|Z)\big]. \tag{12}$$

3 Test Construction for Conditional Independence

3.1 Construction of Test Statistic

The conditional covariance operator $\Sigma_{XY|Z}$ is inadequate for testing conditional independence because, by Corollary 3, $\Sigma_{XY|Z} = 0$ if and only if, for all $f \in \mathcal{G}_X$ and $g \in \mathcal{G}_Y$,

$$E\{\mathrm{cov}[f(X), g(Y)|Z]\} = 0.$$

Clearly, this does not imply conditional independence between X and Y given Z. Instead, conditional independence is implied by

$$\mathrm{cov}[f(X), g(Y)|Z] = 0. \tag{13}$$

For this reason, we need to make a modification of $\Sigma_{XY|Z}$ so that the modified version corresponding to (13).

Let $\ddot{X} = (X, Z)$ and $\kappa_{\ddot{X}} = \kappa_X\kappa_Z$, and let $\mathcal{G}_{\ddot{X}}$ be the RKHS generated by $\kappa_{\ddot{X}}$. We define the modified conditional covariance operator as

$$\Sigma_{\ddot{X}Y|Z} = \Sigma_{\ddot{X}Y} - \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\Sigma_{ZY}, \tag{14}$$

This operator was first introduced by Fukumizu et al. (2008). To distinguish this operator from $\Sigma_{XY|Z}$ (without the dots), we refer to the operator in (14) as the conjoined conditional covariance operator and abbreviate it by CCCO (see Li and Kim (2024)). The next theorem shows that the CCCO can characterize conditional independence. Before stating the theorem, we need to present a modified version of Assumptions 1 and 2 using $\ddot{X}$.

Assumption 3. *The kernels $\kappa_{\ddot{X}}$, κ_Y and κ_Z satisfies $E[\kappa_{\ddot{X}}(\ddot{X}, \ddot{X})] < \infty$, $E[\kappa_Y(Y, Y)] < \infty$ and $E[\kappa_Z(Z, Z)] < \infty$.*

Assumption 4. *The following two parts hold:*

(i) For all $g \in \mathcal{G}_{\ddot{X}}$, the function $E[g(\ddot{X})|Z = \cdot]$ is an element of $\mathcal{G}_Z$;

(ii) For all $g \in \mathcal{G}_Y$, the function $E[g(Y)|Z = \cdot]$ is an element of $\mathcal{G}_Z$.

Theorem 1. *If Assumptions 3 and 4 hold and $\kappa_{\ddot{X}} \otimes \kappa_Y$ is characteristic, then,*

$$\Sigma_{\ddot{X}Y|Z} = 0 \quad \Leftrightarrow \quad X \perp\!\!\!\perp Y|Z. \tag{15}$$

This theorem was first established in Theorem 8 and Corollary 9 of Fukumizu et al. (2004), where X, Y, Z are random vectors. On the other hand, when X, Y, Z are random functions, although the proof is similar, the definitions of their corresponding Borel σ-fields $\mathcal{F}_X$, $\mathcal{F}_Y$, $\mathcal{F}_Z$ are more complicated, and we refer to Sections 1.1-1.2 of Bosq (2012) and Section 2.2, items 5-8 of Shiryaev (2016). Because of the similarity, we omit the proof of this result in the functional setting.

By (15), testing the hypothesis $H_0 : X \perp\!\!\!\perp Y|Z$ is equivalent to testing $H_0 : \Sigma_{\ddot{X}Y|Z} = 0$. This motivates us to use the statistic

$$T_n = n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}},$$

as the test statistic, where $\hat{\Sigma}_{\ddot{X}Y|X}$ is the sample estimate of $\Sigma_{\ddot{X}Y|Z}$, which will be constructed in the next subsection, and $\|\cdot\|_{\text{HS}}$ is the Hilbert-Schmidt norm. Under H_0, we expect T_n to be small, whereas under the alternative hypothesis $H_1 : \Sigma_{\ddot{X}Y|Z} \neq 0$, T_n should tend to be large. We therefore reject H_0 for sufficiently large values of T_n. In this paper, we will focus on the test statistic of the form $n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}}$.

3.2 Estimating CCCO and its Hilbert-Schmidt norm

We now construct an estimator of $\Sigma_{\ddot{X}Y|Z}$. The construction proceeds in two parts. In the first part, we derive the estimator $\hat{\Sigma}_{\ddot{X}Y|Z}$ and therefrom the corresponding test statistic $T_n = n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}}$. In the second part, we develop an implementable numerical procedure based on coordinate representations.

Henceforth, we will use $E_n(\cdots)$ to denote the sample average. For example, for a generic kernel κ and a sample of random variables $X_1, \ldots, X_n$, we use $E_n[\kappa(\cdot, X)]$ is the function $n^{-1}\sum_{i=1}^n \kappa(\cdot, X_i)$. We let $\hat{\kappa}(\cdot, x)$ denote the centered version

$$\hat{\kappa}(\cdot, x) = \kappa(\cdot, x) - E_n[\kappa(\cdot, X)].$$

Recall that we have used $\tilde{\kappa}(\cdot, x)$ to represent the version centered by the true mean $E[\kappa(\cdot, X)]$. Thus, $\hat{\kappa}$ can be regarded as the estimate of $\tilde{\kappa}$.

We begin the construction by estimating $B_{\ddot{X}|Z} = \Sigma_{\ddot{X}Z}\Sigma^{\dagger}_{ZZ}$. Each covariance operator can be estimated by replacing the population expectations with sample means. Specifically, the empirical estimators of $\Sigma_{\ddot{X}Z}$ and Σ_{ZZ} are given by

$$\hat{\Sigma}_{\ddot{X}Z} = E_n[\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) \otimes \hat{\kappa}_Z(\cdot, Z)], \quad \hat{\Sigma}_{ZZ} = E_n[\hat{\kappa}_Z(\cdot, Z) \otimes \hat{\kappa}_Z(\cdot, Z)]$$

Since $\hat{\Sigma}_{ZZ}$ is not invertible, we apply Tikhonov regularization when estimating $\hat{B}_{X|Z}$. That is, we can estimate $B_{\ddot{X}|Z}$ by

$$\hat{B}_{\ddot{X}|Z} = \hat{\Sigma}_{\ddot{X}Z}(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1},$$

where $\epsilon_n > 0$ is a tuning constant. We further estimate $\tilde{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{x}|z)$ by

$$\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{x}|z) = \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{x}) - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, z). \tag{16}$$

We perform exactly the same sequence of operations to obtain the estimates of $B_{Y|Z}$ and $\tilde{\kappa}_{Y|Z}(\cdot, y|z)$ as

$$\hat{B}_{Y|Z} = \hat{\Sigma}_{Y|Z}(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}, \quad \hat{\kappa}_{Y|Z}(\cdot, y|z) = \hat{\kappa}_Y(\cdot, y) - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, z).$$

We construct an estimator of $\Sigma_{\ddot{X}Y|Z}$ by simply replacing the expectation in (12) by sample mean and replacing $\tilde{\kappa}_{\ddot{X}|Z}$ and $\tilde{\kappa}_{Y|Z}$ by $\hat{\kappa}_{\ddot{X}|Z}$ and $\hat{\kappa}_{Y|Z}$. That is,

$$\hat{\Sigma}_{\ddot{X}Y|Z} = E_n[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}|Z) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y|Z)]. \tag{17}$$

Substituting (16) into (17), we have

$$\begin{aligned}
\hat{\Sigma}_{\ddot{X}Y|Z} =& E_n\{[\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z)] \otimes [\hat{\kappa}_Y(\cdot, Y) - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z)]\} \\
=& E_n[\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) \otimes \hat{\kappa}_Y(\cdot, Y)] - E_n[\hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z) \otimes \hat{\kappa}_Y(\cdot, Y)] \\
& - E_n[\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) \otimes \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z)] + E_n[\hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z) \otimes \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z)] \\
=& \hat{\Sigma}_{\ddot{X}Y} - \hat{B}_{\ddot{X}|Z}\hat{\Sigma}_{ZY} - \hat{\Sigma}_{\ddot{X}Z}\hat{B}^*_{Y|Z} + \hat{B}_{\ddot{X}|Z}\hat{\Sigma}_{ZZ}\hat{B}^*_{Y|Z},
\end{aligned} \tag{18}$$

where, for a linear operator A, A^* stands for the adjoint operator of A. The squared Hilbert-Schmidt norm of $\hat{\Sigma}_{\ddot{X}Y|Z}$ is computed as follows:

$$\begin{aligned}
\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}} =& \left\| E_n\left\{[\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z)] \otimes [\hat{\kappa}_Y(\cdot, Y) - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z)]\right\}\right\|^2_{\text{HS}} \\
=& E_n\Big[\langle \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z), \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}') - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z')\rangle_{\mathcal{G}_{\ddot{X}}} \times \\
& \qquad \langle \hat{\kappa}_Y(\cdot, Y) - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z), \hat{\kappa}_Y(\cdot, Y') - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z')\rangle_{\mathcal{G}_Y}\Big] \\
=& \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n \Big[\langle \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_i) - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z_i), \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_j) - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot, Z_j)\rangle_{\mathcal{G}_{\ddot{X}}} \times \\
& \qquad \langle \hat{\kappa}_Y(\cdot, Y_i) - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z_i), \hat{\kappa}_Y(\cdot, Y_j) - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot, Z_j)\rangle_{\mathcal{G}_Y}\Big].
\end{aligned}$$

The above expression is at the operator level. In order to implement it numerically we need to turn operators to matrices via coordinate mapping (see, for example, Section 12.3 of Li (2018b)). The subspace $\text{ran}(\hat{\Sigma}_{\ddot{X}\ddot{X}})$ is spanned by

$$\mathcal{B}_{\ddot{X}} = \{\kappa_{\ddot{X}}(\cdot, \ddot{X}_i) - E_n[\kappa_{\ddot{X}}(\cdot, \ddot{X})] : i \in \{1, \ldots, n\}\} = \{\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_i) : i \in \{1, \ldots, n\}\}.$$

Similarly, $\text{ran}(\hat{\Sigma}_{YY})$ and $\text{ran}(\hat{\Sigma}_{ZZ})$ are spanned by

$$\mathcal{B}_Y = \{\hat{\kappa}_Y(\cdot, Y_i) : i \in \{1, \ldots, n\}\}, \quad \mathcal{B}_Z = \{\hat{\kappa}_Z(\cdot, Z_i) : i \in \{1, \ldots, n\}\},$$

respectively. Let $K_X \in \mathbb{R}^{n\times n}$ be the Gram matrix of κ_X acting on $X_1, \ldots, X_n$; that is, $(K_X)_{ij} = \kappa_X(X_i, X_j)$. Similarly, we define K_Y and K_Z as the Gram matrices for κ_Y and κ_Z. Since $\kappa_{\ddot{X}} = \kappa_X\kappa_Z$, the Gram matrix $K_{\ddot{X}}$ of $\kappa_{\ddot{X}}$ can be represented as

$$K_{\ddot{X}} = K_X \odot K_Z, \tag{19}$$

where $\odot$ represents the Hadamard product between matrices; that is, for example, $(K_{\ddot{X}})_{ij} = (K_X)_{ij}(K_Z)_{ij}$ for $i, j = 1, \ldots, n$. Furthermore, let $\tilde{K}_{\ddot{X}} = HK_{\ddot{X}}H$ be the centralized Gram matrix where $H = I_n - \frac{1}{n}1_n1_n^T$, where I_n is the $n \times n$ identity matrix, and 1_n is the n-dimensional vector whose entries are 1. Equivalently, $(\tilde{K}_{\ddot{X}})_{ij} = \langle\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_i), \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_j)\rangle_{\mathcal{G}_{\ddot{X}}}$. Similarly, $\tilde{K}_Y = HK_YH$ and $\tilde{K}_Z = HK_ZH$. We now give the coordinate mappings for some related operators as follows (in the notations described in, for example, Li (2018b), Li and Solea (2018)):

$$\begin{aligned}
&{}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}\ddot{X}}]_{\mathcal{B}_{\ddot{X}}} = n^{-1}\tilde{K}_{\ddot{X}}, \quad {}_{\mathcal{B}_Y}[\hat{\Sigma}_{YY}]_{\mathcal{B}_Y} = n^{-1}\tilde{K}_Y, \quad {}_{\mathcal{B}_Z}[\hat{\Sigma}_{ZZ}]_{\mathcal{B}_{ZZ}} = n^{-1}\tilde{K}_Z,\\
&{}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Y}]_{\mathcal{B}_Y} = n^{-1}\tilde{K}_Y, \quad {}_{\mathcal{B}_Y}[\hat{\Sigma}_{Y\ddot{X}}]_{\mathcal{B}_{\ddot{X}}} = n^{-1}\tilde{K}_{\ddot{X}}, \quad {}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Z}]_{\mathcal{B}_Z} = n^{-1}\tilde{K}_Z,\\
&{}_{\mathcal{B}_Z}[\hat{\Sigma}_{Z\ddot{X}}]_{\mathcal{B}_{\ddot{X}}} = n^{-1}\tilde{K}_{\ddot{X}}, \quad {}_{\mathcal{B}_Y}[\hat{\Sigma}_{YZ}]_{\mathcal{B}_Z} = n^{-1}\tilde{K}_Z, \quad {}_{\mathcal{B}_Z}[\hat{\Sigma}_{ZY}]_{\mathcal{B}_Y} = n^{-1}\tilde{K}_Y.
\end{aligned}$$

Using the above results, and applying the rules for coordinate mapping described in Li and Solea (2018), we derive the coordinate mappings of $\hat{B}_{\ddot{X}|Z}$ and $\hat{B}^*_{Y|Z}$ as follows:

$$\begin{aligned}
{}_{\mathcal{B}_{\ddot{X}}}[\hat{B}_{\ddot{X}|Z}]_{\mathcal{B}_Z} =& \ {}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Z}]_{\mathcal{B}_Z}\ {}_{\mathcal{B}_Z}[(\hat{\Sigma}_{ZZ} + \epsilon_nI)^{-1}]_{\mathcal{B}_Z} = n^{-1}\tilde{K}_Z(n^{-1}\tilde{K}_Z + \epsilon_nI)^{-1},\\
{}_{\mathcal{B}_Z}[\hat{B}^*_{Y|Z}]_{\mathcal{B}_Y} =& \ {}_{\mathcal{B}_Z}[(\hat{\Sigma}_{ZZ} + \epsilon_nI)^{-1}]_{\mathcal{B}_Z}\ {}_{\mathcal{B}_Z}[\hat{\Sigma}_{ZY}]_{\mathcal{B}_Y} = (n^{-1}\tilde{K}_Z + \epsilon_nI)^{-1}n^{-1}\tilde{K}_Y.
\end{aligned}$$

Therefore, the coordinate mapping of $\hat{\Sigma}_{\ddot{X}Y|Z}$ is

$$\begin{aligned}
{}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Y|Z}]_{\mathcal{B}_Y} =& \ {}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Y} - \hat{B}_{\ddot{X}|Z}\hat{\Sigma}_{ZY} - \hat{\Sigma}_{\ddot{X}Z}\hat{B}^*_{Y|Z} + \hat{B}_{\ddot{X}|Z}\hat{\Sigma}_{ZZ}\hat{B}^*_{Y|Z}]_{\mathcal{B}_Y}\\
=& \ {}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Y}]_{\mathcal{B}_Y} - {}_{\mathcal{B}_{\ddot{X}}}[\hat{B}_{\ddot{X}|Z}]_{\mathcal{B}_Z}\ {}_{\mathcal{B}_Z}[\hat{\Sigma}_{ZY}]_{\mathcal{B}_Y} - {}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Z}]_{\mathcal{B}_Z}\ {}_{\mathcal{B}_Z}[\hat{B}^*_{Y|Z}]_{\mathcal{B}_Y}\\
&+ {}_{\mathcal{B}_{\ddot{X}}}[\hat{B}_{\ddot{X}|Z}]_{\mathcal{B}_Z}\ {}_{\mathcal{B}_Z}[\hat{\Sigma}_{ZZ}]_{\mathcal{B}_Z}\ {}_{\mathcal{B}_Z}[\hat{B}^*_{Y|Z}]_{\mathcal{B}_Y}\\
=& n^{-1}\tilde{K}_Y - \tilde{K}_Z(\tilde{K}_Z + n\epsilon_nI)^{-1}n^{-1}\tilde{K}_Y - n^{-1}\tilde{K}_Z(\tilde{K}_Z + n\epsilon_nI)^{-1}\tilde{K}_Y\\
&+ \tilde{K}_Z(\tilde{K}_Z + n\epsilon_nI)^{-1}n^{-1}\tilde{K}_Z(\tilde{K}_Z + n\epsilon_nI)^{-1}\tilde{K}_Y\\
=& n^{-1}[I - \tilde{K}_Z(\tilde{K}_Z + n\epsilon_nI)^{-1}]^2\tilde{K}_Y.
\end{aligned}$$

Next, we derive the expression for $\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}}$. Let

$$R_Z = I - \tilde{K}_Z(\tilde{K}_Z + n\epsilon_nI)^{-1} = n\epsilon_n(\tilde{K}_Z + n\epsilon_nI)^{-1}. \tag{20}$$

Then ${}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Y|Z}]_{\mathcal{B}_Y}$ can be abbreviated as $n^{-1}R_Z^2\tilde{K}_Y$. By Theorem 8 (part 4) of Li and Solea (2018),

$$\begin{aligned}
\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}} =& \left\|\tilde{K}_{\ddot{X}}^{1/2}\ {}_{\mathcal{B}_{\ddot{X}}}[\hat{\Sigma}_{\ddot{X}Y|Z}]_{\mathcal{B}_Y}\tilde{K}_Y^{-1/2}\right\|_F^2 = \left\|n^{-1}\tilde{K}_{\ddot{X}}^{1/2}R_Z^2\tilde{K}_Y^{1/2}\right\|_F^2\\
=& n^{-2}\text{tr}\left(\tilde{K}_{\ddot{X}}^{1/2}R_Z^2\tilde{K}_Y^{1/2}\tilde{K}_Y^{1/2}R_Z^2\tilde{K}_{\ddot{X}}^{1/2}\right) = n^{-2}\text{tr}\left(R_Z\tilde{K}_{\ddot{X}}R_ZR_Z\tilde{K}_YR_Z\right),
\end{aligned}$$

where tr denotes the trace of a matrix. Let

$$\tilde{K}_{\ddot{X}|Z} = R_Z \tilde{K}_{\ddot{X}} R_Z, \quad \tilde{K}_{Y|Z} = R_Z \tilde{K}_Y R_Z. \tag{21}$$

Then, $\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}}$ can be further abbreviated as

$$\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}} = n^{-2}\text{tr}(\tilde{K}_{\ddot{X}|Z}\tilde{K}_{Y|Z}). \tag{22}$$

The test statistic is then expressed as $T_n = n^{-1}\text{tr}(\tilde{K}_{\ddot{X}|Z}\tilde{K}_{Y|Z})$, which coincides the form given in equation (13) of Zhang et al. (2011).

4 Asymptotic Distribution of T_n

4.1 Asymptotic properties under null distribution

In this section we derive the asymptotic distribution of T_n. We make the following additional assumptions.

Assumption 5. *The kernels κ_X and κ_Y are bounded, and the kernel κ_Z satisfies $E[\kappa_Z^2(Z,Z)] < \infty$.*

This assumption is satisfied, for example, by the Gaussian radial basis function (RBF) kernel and the Laplace kernel. Before stating the next assumption, we define an intermediate operator

$$\Lambda_{\ddot{X}Z} = E[\tilde{\kappa}_X(\cdot, X)\kappa_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)].$$

Note that this operator is different from $\Sigma_{\ddot{X}Z}$, which can be represented by

$$\Sigma_{\ddot{X}Z} = E[\kappa_X(\cdot, X)\kappa_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)].$$

Clearly, we have

$$\Sigma_{\ddot{X}Z} = E\{[\mu_X + \tilde{\kappa}_X(\cdot, X)]\kappa_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)\} = \mu_X\Sigma_{ZZ} + \Lambda_{\ddot{X}Z}, \tag{23}$$

where $\mu_X\Sigma_{ZZ}$ is a linear operator from $\mathcal{G}_Z$ to $\mathcal{G}_{\ddot{X}}$ defined by

$$(\mu_X\Sigma_{ZZ})f = \mu_X(\Sigma_{ZZ}f) \in \mathcal{G}_{\ddot{X}}.$$

The following smoothness assumption is imposed in terms of $\Lambda_{\ddot{X}Z}$ and Σ_{YZ}.

Assumption 6. *There exists some $\beta > 0$ such that $\Lambda_{\ddot{X}Z} = S_{\ddot{X}Z}\Sigma_{ZZ}^{1+\beta}$ and $\Sigma_{YZ} = S_{YZ}\Sigma_{ZZ}^{1+\beta}$ for some bounded linear operators $S_{\ddot{X}Z} : \mathcal{G}_Z \to \mathcal{G}_{\ddot{X}}$ and $S_{YZ} : \mathcal{G}_Z \to \mathcal{G}_Y$.*

Furthermore, let

$$\Sigma_{ZZ} = \sum_{k=1}^{\infty} \gamma_k(\phi_k \otimes \phi_k), \quad \Sigma_{XX} = \sum_{k=1}^{\infty} \eta_k(\chi_k \otimes \chi_k), \quad \Sigma_{YY} = \sum_{k=1}^{\infty} \mu_k(\psi_k \otimes \psi_k) \tag{24}$$

be the spectral decompositions of Σ_{ZZ}, Σ_{XX} and Σ_{YY}, where, for example, $\gamma_1 \geq \gamma_2 \geq \cdots$ are the eigenvalues of Σ_{ZZ}, and $\phi_1, \phi_2, \ldots$ are the corresponding eigenfunctions. The Karhunen-Loéve expansions of $\kappa_Z(\cdot, Z)$, $\kappa_X(\cdot, X)$ and $\kappa_Y(\cdot, Y)$ are

$$\begin{aligned}
\kappa_Z(\cdot, Z) &= \mu_Z + \sum_{k=1}^{\infty} \sqrt{\gamma_k}\xi_k\phi_k, \quad \xi_k = \frac{\langle \kappa_Z(\cdot, Z) - \mu_Z, \phi_k\rangle_{\mathcal{G}_Z}}{\sqrt{\gamma_k}}, \\
\kappa_X(\cdot, X) &= \mu_X + \sum_{k=1}^{\infty} \sqrt{\eta_k}\omega_k\chi_k, \quad \omega_k = \frac{\langle \kappa_X(\cdot, X) - \mu_X, \chi_k\rangle_{\mathcal{G}_X}}{\sqrt{\eta_k}}, \\
\kappa_Y(\cdot, Y) &= \mu_Y + \sum_{k=1}^{\infty} \sqrt{\mu_k}\theta_k\psi_k, \quad \theta_k = \frac{\langle \kappa_Y(\cdot, Y) - \mu_Y, \psi_k\rangle_{\mathcal{G}_Y}}{\sqrt{\mu_k}}.
\end{aligned} \tag{25}$$

Note that ξ_k's are uncorrelated random variables with mean 0 and variance 1, and same for ω_k's and θ_k's. We make an additional assumption on the decay rate of eigenvalues of Σ_{ZZ} in (24). Henceforth, for two sequences of positive real numbers (a_j) and (b_j), we write $a_j \preceq b_j$ if there exists a constant C such that $a_j \leq Cb_j$ for all j, and write $a_j \asymp b_j$ if $a_j \preceq b_j$ and $b_j \preceq a_j$.

Assumption 7. *$\gamma_j \preceq j^{-\alpha}$ for some $\alpha > 1$, and for all $j \in \{1, 2, \ldots\}$.*

We now give some equivalent conditions and concrete examples for Assumptions 6 and 7.

Proposition 2. *Suppose β is any positive constant, and* $\operatorname{rank}(\Sigma_{ZZ}) = K$ *for some $K \in \mathbb{N} \cup \{\infty\}$.*

(i) $\Sigma_{YZ} = S_{YZ}\Sigma_{ZZ}^{1+\beta}$ holds for some bounded linear operator $S_{YZ} : \mathcal{G}_Z \to \mathcal{G}_Y$ if and only if

$$\sup_{j,k\in\mathbb{N}, k\leq K} \mu_j^{1/2}\gamma_k^{-1/2-\beta}|E(\theta_j\xi_k)| < \infty. \tag{26}$$

(ii) $\Lambda_{\ddot{X}Z} = S_{\ddot{X}Z}\Sigma_{ZZ}^{1+\beta}$ holds for some bounded linear operator $S_{\ddot{X}Z} : \mathcal{G}_Z \to \mathcal{G}_{\ddot{X}}$ if and only if

$$\sup_{j,k,l\in\mathbb{N}, k,l\leq K} \eta_j^{1/2}\gamma_k^{1/2}\gamma_l^{-1/2-\beta}|E(\omega_j\xi_k\xi_l)| < \infty, \quad \sup_{j,k\in\mathbb{N}, k\leq K} \eta_j^{1/2}\gamma_k^{-1/2-\beta}|E(\omega_j\xi_k)| < \infty. \tag{27}$$

The proof of Proposition 2 is given in Appendix B.3. The equivalences in Proposition 2 show that Assumption 6 in effect imposes a certain degree of smoothness between $\kappa_Z(\cdot, Z)$ and $\kappa_X(\cdot, X)$ and between $\kappa_Z(\cdot, Z)$ and $\kappa_Y(\cdot, Y)$. That is, the dependence between them occurs mostly in the low-frequency parts of the spectra, whereas high-frequency dependence is ignorable. This point was also discussed in Li (2018b) and Li (2018a). To provide more insights into the nature of Assumption 6, we give two illustrative examples where it is satisfied in Appendix A.

In particular, the Σ_{YZ} part of Assumption 6 is commonly used in RKHS regression or dimension reduction literature, such as Caponnetto and De Vito (2007); Li and Song (2017).

However, the similar condition is only imposed on $\Lambda_{\ddot{X}Z}$ rather than $\Sigma_{\ddot{X}Z}$ because their difference term, given by (23), $\mu_X\Sigma_{ZZ}$ cannot be as smooth as $\Sigma_{ZZ}^{1+\beta}$ unless the eigenvalues of Σ_{ZZ} do not decay, which contradicts Assumption 7. Therefore, when deriving the asymptotic properties, the $\mu_X\Sigma_{ZZ}$ term will be handled separately.

Assumption 7 regulates the behavior of the distribution of the random function Z. Instead of assuming $\gamma_j \asymp j^{-\alpha}$, which is common in functional data analysis (Cai and Hall, 2006; Sang and Li, 2026), our Assumption 7 only requires a weaker condition that γ_j can be either faster than or of the same order as $j^{-\alpha}$. Based on our Assumption 7, we prove a similar lemma as Lemma 8 of Sang and Li (2026) as follows.

Lemma 2. *Under Assumption 7, if $\epsilon_n \to 0$, then we have*

$$\sum_{j=1}^{\infty} \frac{\gamma_j}{(\gamma_j+\epsilon_n)^2} = O\left(\epsilon_n^{-(\alpha+1)/\alpha}\right), \qquad \sum_{j=1}^{\infty} \frac{\gamma_j^2}{(\gamma_j+\epsilon_n)^2} = O\left(\epsilon_n^{-1/\alpha}\right).$$

The proof of Lemma 2 is presented in Appendix B.4. The next Proposition gives a concrete example to illustrate Assumption 7, and its proof is given in Appendix B.5.

Proposition 3 (Brownian bridge with Gaussian kernel)**.** *Let $\mathcal{H}_Z = L^2[0,1]$, Z be a Brownian bridge on $[0,1]$, and κ_Z be the Gaussian kernel*

$$\kappa_Z(z,z') = \exp\{-\eta\|z-z'\|_{L^2}^2\}, \qquad z,z' \in \mathcal{H}_Z.$$

If $0 < \eta \le \pi^2/8$, then Assumption 7 is satisfied for every $\alpha \in (1,2)$. Furthermore, if Z is replaced by any Gaussian process whose eigenvalues of covariance operator, $\rho_1 \ge \rho_2 \ge \cdots$ decays at the rate $\rho_m \preceq m^{-r}$ for some $r > 1$, then Assumption 7 is satisfied for every $\alpha \in (1,r)$.

We introduce the following operator, which will contribute to the bias term in $\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z}$:

$$R = \mu_X\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \mu_X. \tag{28}$$

Similar to the notation used in (23), here, $\mu_X\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}$ stands for the operator from $\mathcal{G}_Z$ to $\mathcal{G}_{\ddot{X}}$ defined by

$$[\mu_X\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}]f = \mu_X[\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}f].$$

Henceforth, we use $\|\cdot\|_{\mathrm{OP}}$ to denote the operator norm. The next lemma develops some properties of R which will be used later, and its proof is given in Appendix B.6.

Lemma 3. *Under Assumption 5, we have*

(i) $\|R\|_{\mathrm{OP}} \le \|\mu_X\|_{\mathcal{G}_X} < \infty$, i.e., R is bounded;

(ii) for any $\theta > 0$, we have

$$\|R\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} = O_P(n^{-1/2}\epsilon_n^{\theta\wedge 1-1} + \epsilon_n^{\theta\wedge 1}); \tag{29}$$

(iii) in particular,

$$\|R\Sigma_{ZZ}\|_{\mathrm{HS}} = O_P(n^{-1/2} + \epsilon_n^{1-1/(2\alpha)}). \tag{30}$$

We next state a lemma regarding the convergence rates of $\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R$ and $\hat{B}_{Y|Z} - B_{Y|Z}$, which were established in a recent work by Choi et al. (2026).

Lemma 4. *Suppose Assumptions 4–7 are satisfied for some $\beta > 0$ and $\alpha > 1$. If $\epsilon_n \to 0$ as $n \to \infty$, then*

$$\|\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R\|_{\mathrm{OP}} = O_P(\epsilon_n^{\beta\wedge 1} + n^{-1/2}\epsilon_n^{\beta\wedge 1-1} + n^{-1}\epsilon_n^{-(3\alpha+1)/(2\alpha)} + n^{-1/2}\epsilon_n^{-(\alpha+1)/(2\alpha)})$$

and

$$\|\hat{B}_{Y|Z} - B_{Y|Z}\|_{\mathrm{OP}} = O_P(\epsilon_n^{\beta\wedge 1} + n^{-1/2}\epsilon_n^{\beta\wedge 1-1} + n^{-1}\epsilon_n^{-(3\alpha+1)/(2\alpha)} + n^{-1/2}\epsilon_n^{-(\alpha+1)/(2\alpha)}).$$

The proof of Lemma 4 mimics that of Theorem 3.4 of Choi et al. (2026) and Theorem 9 of Sang and Li (2026), and we place the full proof in our setting in Appendix B for completeness.

Furthermore, Theorem 3.7 of Choi et al. (2026) provides the optimal choice of ϵ_n, which yields the convergence rates for $\|\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R\|_{\mathrm{OP}}$ and $\|\hat{B}_{Y|Z} - B_{Y|Z}\|_{\mathrm{OP}}$ under the optimal choice of the tuning parameter. We restate that result below in the form of a corollary tailored to our setting.

Corollary 4. *Suppose the conditions in Lemma 4 hold and $\epsilon_n \asymp n^{-\delta}$ for some $\delta > 0$.*

(i) If $\beta > \frac{\alpha-1}{2\alpha}$, then the optimal choice of δ is $\frac{\alpha}{2\alpha(\beta\wedge 1)+\alpha+1}$, and the convergence rates of $\|\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R\|_{\mathrm{OP}}$ and $\|\hat{B}_{Y|Z} - B_{Y|Z}\|_{\mathrm{OP}}$ under the optimal choice of the tuning parameter are $O_P(n^{-\frac{\alpha(\beta\wedge 1)}{2\alpha(\beta\wedge 1)+\alpha+1}})$;

(ii) If $\beta \le \frac{\alpha-1}{2\alpha}$, then the optimal choice of δ is $\frac{1}{2}$, and the convergence rates of $\|\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R\|_{\mathrm{OP}}$ and $\|\hat{B}_{Y|Z} - B_{Y|Z}\|_{\mathrm{OP}}$ under the optimal choice of the tuning parameter are $O_P(n^{-\frac{\beta}{2}})$.

We now state the asymptotic normality of $\hat{\Sigma}_{\ddot{X}Y|Z}$ based on the convergence rate of $\hat{B}_{\ddot{X}|Z}$ and $\hat{B}_{Y|Z}$ under the optimal choice of the tuning parameter in Corollary 4.

Theorem 2. *Suppose that Assumptions 2–7 hold for some $\beta > 0$ and $\alpha > 2$. Further suppose that $\beta > \frac{\alpha-1}{2\alpha}$ and $\frac{\alpha(\beta\wedge 1)}{2\alpha(\beta\wedge 1)+\alpha+1} > \frac{1}{4}$. Let $\epsilon_n \asymp n^{-\frac{\alpha(\beta\wedge 1)}{2\alpha(\beta\wedge 1)+\alpha+1}}$. Then, we have*

$$\sqrt{n}(\hat{\Sigma}_{\ddot{X}Y|Z} - \Sigma_{\ddot{X}Y|Z}) \xrightarrow{\mathcal{D}} N(0, \Gamma_{\ddot{X}Y|Z}), \tag{31}$$

where $\Gamma_{\ddot{X}Y|Z} : \mathcal{G}_{\ddot{X}} \otimes \mathcal{G}_Y \to \mathcal{G}_{\ddot{X}} \otimes \mathcal{G}_Y$ is the linear operator defined by

$$\Gamma_{\ddot{X}Y|Z} = E\left\{[\tilde{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}|Z) \otimes \tilde{\kappa}_{Y|Z}(\cdot, Y|Z) - \Sigma_{\ddot{X}Y|Z}] \otimes [\tilde{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}|Z) \otimes \tilde{\kappa}_{Y|Z}(\cdot, Y|Z) - \Sigma_{\ddot{X}Y|Z}]\right\}. \tag{32}$$

A key step in the proof of Theorem 2 is to show that the three terms in

$$\sqrt{n}(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})E_n[\tilde{\kappa}_Z(\cdot, Z) \otimes \epsilon_Y] + \sqrt{n}E_n[\epsilon_{\ddot{X}} \otimes \tilde{\kappa}_Z(\cdot, Z)](B_{Y|Z} - \hat{B}_{Y|Z})^* \\ + \sqrt{n}(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})E_n[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)](B_{Y|Z} - \hat{B}_{Y|Z})^*$$

are all $o_P(1)$, which, as mentioned in the Introduction, is ensured by applying Corollary 4 and Lemma 3. Without these conditions — that is, if we only use the optimal rate implied by (1) — the above quantity would have the same order of magnitude as the leading term

$$\sqrt{n}\left(E_n\left\{[\tilde{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) - B_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot, Z)] \otimes [\tilde{\kappa}_Y(\cdot, Y) - B_{Y|Z}\tilde{\kappa}_Z(\cdot, Z)]\right\} - \Sigma_{\ddot{X}Y|Z}\right),$$

and consequently the asymptotic distribution in (31) and (32) would fail to hold. The full proof of Theorem 2 is placed in Appendix B.8.

By Theorem 2 and continuous mapping theorem, we have the following asymptotic distribution of our test statistic.

Corollary 5. *Suppose that $\Gamma_{\ddot{X}Y|Z}$ defined in (32) is trace class. Under all assumptions in Theorem 2, when $\Sigma_{\ddot{X}Y|Z} = 0$, we have*

$$n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\mathrm{HS}} \xrightarrow{\mathcal{D}} \sum_{k=1}^{\infty} \lambda_k Z_k^2, \tag{33}$$

where $\lambda_1 \geq \lambda_2 \geq \cdots$ are eigenvalues of $\Gamma_{\ddot{X}Y|Z}$ in (32) without the $\Sigma_{\ddot{X}Y|Z}$ term, and Z_k's are independent standard normal random variables.

4.2 Welch-Satterthwaite approximation

Following the idea of Kokoszka et al. (2017) and Zhang (2013), we use the Welch-Satterwaite approximation to the asymptotic distribution in (33). Let $V = \sum_{i=1}^{\infty} \lambda_i Z_i^2$, where λ_i's and Z_i's are as defined in Corollary 5. Our goal is to approximate the distribution of V using

$$\tilde{V} \sim \zeta\chi^2(\nu), \tag{34}$$

where ζ and ν are calibrated to capture the first two moments of V. Since $Z_1, Z_2, \ldots$ are i.i.d. $N(0, 1)$, we have

$$\mu_V = E(V) = \sum_{i=1}^{\infty} \lambda_i E(Z_i^2) = \sum_{i=1}^{\infty} \lambda_i = \mathrm{tr}(\Gamma_{\ddot{X}Y|Z})$$

and

$$\sigma_V^2 = \mathrm{var}(V) = \sum_{i=1}^{\infty} \lambda_i^2 \mathrm{var}(Z_i^2) = 2\sum_{i=1}^{\infty} \lambda_i^2 = \mathrm{tr}(\Gamma^2_{\ddot{X}Y|Z}).$$

By matching the first two moments of $\zeta\chi(\nu)$ and V, as was done in equation (14) of Kokoszka et al. (2017), we have

$$\zeta = \frac{\sigma_V^2}{2\mu_V} = \frac{\mathrm{tr}(\Gamma^2_{\ddot{X}Y|Z})}{2\,\mathrm{tr}(\Gamma_{\ddot{X}Y|Z})}, \qquad \nu = \frac{2\mu_V^2}{\sigma_V^2} = \frac{2\,[\mathrm{tr}(\Gamma_{\ddot{X}Y|Z})]^2}{\mathrm{tr}(\Gamma^2_{\ddot{X}Y|Z})}. \tag{35}$$

4.3 Local power analysis

In this subsection, we derive the asymptotic distribution under the local alternative distribution using the technique in Theorem 3 of Tang and Li (2024) and Theorem 13 of Gretton et al. (2012), with the detailed proof placed in Appendix B.9.

Theorem 3. *Suppose all assumptions in Theorem 2 are satisfied and suppose*

(i) $\Gamma_{\ddot{X}Y|Z}$ *has spectral decomposition* $\sum_{j=1}^{\infty}\lambda_j(v_j\otimes v_j)$*, where* $\lambda_1\geq\lambda_2\geq\ldots$ *are eigenvalues of* $\Gamma_{\ddot{X}Y|Z}$*, and* $v_1, v_2, \ldots$ *are eigenfunctions of* $\Gamma_{\ddot{X}Y|Z}$*, which form an orthonormal basis in* $\mathcal{G}_{\ddot{X}}\otimes\mathcal{G}_Y$*;*

(ii) Σ_1 *is a fixed linear operator in* $\mathcal{G}_{\ddot{X}}\otimes\mathcal{G}_Y$ *with expansion* $\sum_{j=1}^{\infty}\sigma_j v_j$ *and* $\|\Sigma_1\|_{\text{HS}}=c>0$*.*

Then, under the local alternative hypothesis $H_1^{(n)}:\Sigma_{\ddot{X}Y|Z}=n^{-1/2}\Sigma_1$*, the asymptotic distribution of* $\|\hat{\Sigma}_{\ddot{X}Y|Z}\|_{\text{HS}}$ *is*

$$n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|_{\text{HS}}^2\xrightarrow{\mathcal{D}}\sum_{j=1}^{\infty}\lambda_j\tilde{Z}_j^2$$

where $\tilde{Z}_j$ *are independent* $N(\sigma_j/\sqrt{\lambda_j},1)$ *random variables. Thus, the local power of the test is*

$$P\left(n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|_{\text{HS}}^2>s\right)\to P\left(\sum_{j=1}^{\infty}\lambda_j\tilde{Z}_j^2>s\right).$$

5 Approximation for Random Functions

5.1 Approximating trajectories by reproducing-kernel-based smoother

In practice, the processes $X(t)$, $Y(t)$, and $Z(t)$ are not observed continuously over T; instead, they are observed only at a finite collection of discrete time points. The underlying curves must therefore be reconstructed from these discrete observations. In the following, we describe the procedure for estimating these curves, using $X(t)$ as an illustration. Let $X_1,\ldots,X_n$ be i.i.d. samples from X. Assume, for sample i, we can only observe $X_i(t)$ at time points in $T_i=\{t_{i1},\ldots,t_{im_i}\}\subseteq T$. We use an RKHS smoother to approximate the functions (see, for example, Section 6.2 of Li and Song (2017) or Section 7.2 of Kokoszka and Reimherr (2017)). Let $\kappa_T:T\times T\to\mathbb{R}$ be a positive definite kernel, and let $\mathcal{H}$ be a Hilbert space spanned by $\mathcal{L}=\{\kappa_T(\cdot,t):t\in T\}$.

For subject i, we use the subset $\mathcal{L}_i=\{\kappa_T(\cdot,t):t\in T_i\}$ to construct an approximation of the function $X_i(t)$, and denote the resulting approximation by $\hat{X}_i(t)$. Let $K_T^{(i)}$ be the kernel matrix generated by κ_T evaluated on T_i, that is,

$$(K_T^{(i)})_{rs}=\kappa_T(t_{ir},t_{is}),\quad r,s\in\{1,\ldots,m_i\}. \tag{36}$$

Furthermore, let $X_i(T_i)=\{X_i(t_{ir}):r\in\{1,\ldots,m_i\}\}$ denote the observed values of $X_i(t)$ at the time points in T_i.

We now represent $X_i(t)$ in the coordinate system induced by $\mathcal{L}_i$, and denote the corresponding coordinate vector by $[X_i]_{\mathcal{L}_i}$. By the definition of the coordinate mapping, for $s \in \{1, \ldots, m_i\}$,

$$X_i(t_{is}) = \sum_{r=1}^{m_i} ([X_i]_{\mathcal{L}_i})_r \, \kappa_T(t_{is}, t_{ir}).$$

Stacking the observations $X_i(t_{is})$ together yields

$$X_i(T_i) = K_T^{(i)} [X_i]_{\mathcal{L}_i}.$$

To solve for $[X_i]_{\mathcal{L}_i}$, we account for the possible singularity of $K_T^{(i)}$ by applying Tikhonov regularization. The regularized coordinate mapping is given by

$$[\hat{X}_i]_{\mathcal{L}_i} = (K_T^{(i)} + \delta_n I)^{-1} X_i(T_i), \tag{37}$$

where $\delta_n > 0$ is a regularization tuning parameter. We use the notation $[\hat{X}_i]_{\mathcal{L}_i}$ instead of $[X_i]_{\mathcal{L}_i}$ to emphasize that this is the coordinate of the estimated curve $\hat{X}_i$. Let

$$\kappa_T(\cdot, T_i) = \{\kappa_T(\cdot, t_{ir}) : r \in \{1, \ldots, m_i\}\}.$$

The coordinate vector $[\hat{X}_i]_{\mathcal{L}_i}$ then yields the following representation of the reconstructed curve:

$$\hat{X}_i(\cdot) = [\hat{X}_i]_{\mathcal{L}_i}^{\mathsf{T}} \kappa_T(\cdot, T_i).$$

5.2 Approximating inner products and distances

To construct the second layer kernel κ_X (again, using X as an illustration), we need to approximate the inner products $\langle X_i, X_j \rangle$ and the distances $\|X_i - X_j\|$, both taken in the space $L^2(T)$. To estimate the inner product $\langle X_i, X_j \rangle = \int_T X_i(t) X_j(t) \, dt$, we approximate X_i and X_j by their reconstructions $\hat{X}_i$ and $\hat{X}_j$, respectively, and evaluate the integral numerically. Specifically, we evaluate $\hat{X}_i$ and $\hat{X}_j$ at $\ell + 1$ equally-spaced points in T, denoted by $U = \{u_0, u_1, \ldots, u_\ell\}$. For simplicity, we assume that ℓ is even and that u_0 and u_ℓ are the left and right endpoints of T. Applying Simpson's rule to approximate the integral, we obtain

$$\int_{u_0}^{u_\ell} \hat{X}_i(t) \hat{X}_j(t) \, dt \approx \hat{X}_i(U)^{\mathsf{T}} D \hat{X}_j(U), \tag{38}$$

where $\hat{X}_i(U)$ is the vector $\{\hat{X}_i(u_r) : r \in \{0, 1, \ldots, \ell\}\}$ and same for $\hat{X}_j(U)$,

$$D = (h/3)\mathrm{diag}(1, 4, 2, 4, \ldots, 2, 4, 1),$$

and $h = (u_\ell - u_0)/\ell$. Using the coordinate mappings, we have

$$\langle X_i, X_j \rangle \approx \langle \hat{X}_i, \hat{X}_j \rangle = \int_{u_0}^{u_\ell} \hat{X}_i(t) \hat{X}_j(t) dt \approx [\hat{X}_i]_{\mathcal{L}_i}^{\mathsf{T}} \kappa_T(T_i, U) \, D \, \kappa_T(U, T_j) [\hat{X}_j]_{\mathcal{L}_j} \tag{39}$$

where $\kappa_T(T_i, U)$ is the $m_i \times \ell$ matrix $\{\kappa_T(t_{ir}, u_s) : r \in \{1, \ldots, m_i\}, s \in \{0, \ldots, \ell\}\}$, and $\kappa_T(U, T_j)$ is the ℓ by m_j matrix $\{\kappa_T(u_r, t_{js}) : r \in \{0, \ldots, \ell\}, s \in \{1, \ldots, m_j\}\}$. The distance $\|X_i - X_j\|$ can then be estimated by

$$\|\hat{X}_i - \hat{X}_j\| = \sqrt{\langle \hat{X}_i, \hat{X}_i \rangle - 2\langle \hat{X}_i, \hat{X}_j \rangle + \langle \hat{X}_j, \hat{X}_j \rangle}, \tag{40}$$

where the inner products on the right-hand side are given the right-hand side of (39).

5.3 Simplification under balanced observation schedule

An important special case of the observation schedule is where all subjects are observed according to the same set of equally-spaced time points, which is known as the balanced observation scheme. In symbols, $T_1 = \cdots = T_n = U$, where U is the common set of time points. In this case we can simplify the estimation procedure by directly applying Simpson's rule (38) to $X_i(U)$ and $X_j(U)$ as an approximation of $\langle X_i, X_j\rangle$. That is,

$$\langle X_i, X_j\rangle = \int_{u_0}^{u_l} X_i(t)X_j(t)dt \approx X_i(U)^{\mathsf{T}} D X_j(U). \tag{41}$$

Based on the inner product, we can calculate the inner product matrix and the pairwise distance matrix needed for constructing the second layer kernel κ_X (as well as κ_Y and κ_Z), Note that we do not need to involve κ_T and the tuning parameter, and the calculation is significantly simplified.

6 Sample-Level Implementation

6.1 Estimation of eigenvalues

The linear operator $\Gamma_{\ddot{X}Y|Z}$ defined by (32) depends on the unknown population distribution of (X, Y, Z). So, to apply Theorem 2 and Corollary 5 in hypothesis testing of conditional independence we must first approximate this linear operator.

Since $\Gamma_{\ddot{X}Y|Z}$ is a self-adjoint linear operator on $\mathcal{G}_{\ddot{X}\otimes Y} = \mathcal{G}_{\ddot{X}} \otimes \mathcal{G}_Y$, we begin by characterizing the tensor product space $\mathcal{G}_{\ddot{X}\otimes Y}$ at the sample level. By Theorem 2.7.13 of Hsing and Eubank (2015), $\kappa_{\ddot{X}}\kappa_Y$ is a reproducing kernel of $\mathcal{G}_{\ddot{X}\otimes Y}$.

To estimate the eigenvalues of $\Gamma_{\ddot{X}Y|Z}$, we perform coordinate mapping on its empirical version $\hat{\Gamma}_{\ddot{X}Y|Z}$ under the following basis:

$$\mathcal{B}_{\ddot{X}Y} = \left\{\hat{\kappa}_{\ddot{X}Y}(\cdot, \ddot{X}_i, \star, Y_j) : i \in \{1, \ldots, n\}; j \in \{1, \ldots, n\}\right\},$$

where $\hat{\kappa}_{\ddot{X}Y}(\cdot, \ddot{x}, \star, y) = \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{x})\hat{\kappa}_Y(\star, y)$.

In the operator level, $\hat{\Gamma}_{\ddot{X}Y|Z}$ is estimated by replacing all expectations in (32) by the corresponding sample means, as follows:

$$\hat{\Gamma}_{\ddot{X}Y|Z} = \frac{1}{n}\sum_{k=1}^{n}[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)] \otimes [\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)].$$

We then turn the linear operators in the right-hand above into $n\times n$ matrices by coordinating mapping. Applying the rules of coordinate mapping described in Lemma 12.3 of Li (2018b), the coordinate of the linear operator $\hat{\Gamma}_{\ddot{X}Y|Z}$ with respect to the spanning system $\mathcal{B}_{\ddot{X}Y}$ is

$$\begin{aligned}
{}_{\mathcal{B}_{\ddot{X}Y}}[\hat{\Gamma}_{\ddot{X}Y|Z}]_{\mathcal{B}_{\ddot{X}Y}} =& \frac{1}{n}\sum_{k=1}^{n} {}_{\mathcal{B}_{\ddot{X}Y}}[[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)] \otimes [\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)]]_{\mathcal{B}_{\ddot{X}Y}} \\
=& \frac{1}{n}\sum_{k=1}^{n}[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)]_{\mathcal{B}_{\ddot{X}Y}}[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)]_{\mathcal{B}_{\ddot{X}Y}}^{\mathsf{T}} G_{\mathcal{B}_{\ddot{X}Y}},
\end{aligned} \tag{42}$$

where $G_{\mathcal{B}_{\ddot{X}Y}}$ is the Gram matrix of the set $\mathcal{B}_{\ddot{X}Y}$; that is, the $((i,j),(i',j'))$-th entry of $G_{\mathcal{B}_{\ddot{X}Y}}$ is

$$\begin{aligned}(G_{\mathcal{B}_{\ddot{X}Y}})_{(ij)(i'j')} =& \langle \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_i) \otimes \hat{\kappa}_Y(\cdot, Y_j), \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_{i'}) \otimes \hat{\kappa}_Y(\cdot, Y_{j'}) \rangle_{\mathcal{H}_{\ddot{X}\otimes Y}} \\ =& \langle \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_i), \hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_{i'}) \rangle_{\mathcal{H}_{\ddot{X}}} \langle \hat{\kappa}_Y(\cdot, Y_j), \hat{\kappa}_Y(\cdot, Y_{j'}) \rangle_{\mathcal{H}_Y} \\ =& \hat{\kappa}_{\ddot{X}}(\ddot{X}_i, \ddot{X}_{i'}) \hat{\kappa}_Y(Y_j, Y_{j'}) = (\tilde{K}_{\ddot{X}})_{ii'} (\tilde{K}_Y)_{jj'}.\end{aligned}$$

Hence, we have

$$G_{\mathcal{B}_{\ddot{X}Y}} = \tilde{K}_{\ddot{X}} \otimes \tilde{K}_Y,$$

where $\otimes$ denotes the Kronecker product between two matrices. Also, on the right-hand side of (42),

$$[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)]_{\mathcal{B}_{\ddot{X}Y}} = [\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k)]_{\mathcal{B}_{\ddot{X}}} \otimes [\hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)]_{\mathcal{B}_Y},$$

where the first term is

$$\begin{aligned}[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k)]_{\mathcal{B}_{\ddot{X}}} &= [\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_k) - \hat{B}_{\ddot{X}|Z} \hat{\kappa}_Z(\cdot, Z_k)]_{\mathcal{B}_{\ddot{X}}} = [\hat{\kappa}_{\ddot{X}}(\cdot, \ddot{X}_k)]_{\mathcal{B}_{\ddot{X}}} - {}_{\mathcal{B}_{\ddot{X}}}[\hat{B}_{\ddot{X}|Z}]_{\mathcal{B}_Z} [\hat{\kappa}_Z(\cdot, Z_k)]_{\mathcal{B}_Z} \\ &= e_k - \tilde{K}_Z (\tilde{K}_Z + n\epsilon_n I)^{-1} e_k = [I - \tilde{K}_Z (\tilde{K}_Z + n\epsilon_n I)^{-1}] e_k = n\epsilon_n (\tilde{K}_Z + n\epsilon_n I)^{-1} e_k = R_Z e_k,\end{aligned}$$

and same arguments apply to the second term, which gives

$$[\hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)]_{\mathcal{B}_Y} = R_Z e_k,$$

where R_Z is defined by (20). Therefore, we have

$$[\hat{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}_k|Z_k) \otimes \hat{\kappa}_{Y|Z}(\cdot, Y_k|Z_k)]_{\mathcal{B}_{\ddot{X}Y}} = (R_Z e_k) \otimes (R_Z e_k).$$

We plug this result into the coordinate mapping of $\hat{\Gamma}_{\ddot{X}Y|Z}$, and we have

$${}_{\mathcal{B}_{\ddot{X}Y}}[\hat{\Gamma}_{\ddot{X}Y|Z}]_{\mathcal{B}_{\ddot{X}Y}} = \frac{1}{n} \sum_{k=1}^{n} [(R_Z e_k) \otimes (R_Z e_k)][(R_Z e_k) \otimes (R_Z e_k)]^{\mathsf{T}} (\tilde{K}_{\ddot{X}} \otimes \tilde{K}_Y). \tag{43}$$

Since the rank of the matrix ${}_{\mathcal{B}_{\ddot{X}Y}}[\hat{\Gamma}_{\ddot{X}Y|Z}]_{\mathcal{B}_{\ddot{X}Y}}$ is at most n, we compute its first n eigenvalues $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_n$ and use them as estimates of the eigenvalues λ_k's in (33).

6.2 Acceleration of computation

Since the coordinate mapping ${}_{\mathcal{B}_{\ddot{X}Y}}[\hat{\Gamma}_{\ddot{X}Y|Z}]_{\mathcal{B}_{\ddot{X}Y}}$ in (43), is an $n^2 \times n^2$ matrix, the direct computational cost for its eigenvalues is very high even for moderate sample sizes. However, since the rank of this matrix is only n, the cost can be substantially reduced by a carefully designed algorithm. In the following we adapt the idea in Section 6.3 of Tang and Li (2024) to accelerate the computation.

Specifically, since $\tilde{K}_{\ddot{X}}$ and $\tilde{K}_Y$ are symmetric, (43) can be rewritten as

$$\begin{aligned}
\mathcal{B}_{\ddot{X}Y}[\hat{\Gamma}_{\ddot{X}Y|Z}]\mathcal{B}_{\ddot{X}Y} =& \frac{1}{n}\sum_{k=1}^{n}(\tilde{K}_{\ddot{X}}^{1/2}\otimes\tilde{K}_Y^{1/2})[(R_Ze_k)\otimes(R_Ze_k)][(R_Ze_k)\otimes(R_Ze_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}^{1/2}\otimes\tilde{K}_Y^{1/2}) \\
=& \sum_{k=1}^{n}\left[(\tilde{K}_{\ddot{X}}^{1/2}R_Ze_k)\otimes(\tilde{K}_Y^{1/2}R_Ze_k)/\sqrt{n}\right]\left[(\tilde{K}_{\ddot{X}}^{1/2}R_Ze_k)\otimes(\tilde{K}_Y^{1/2}R_Ze_k)/\sqrt{n}\right]^{\mathsf{T}} \\
=& \sum_{k=1}^{n}\left[(L_{\ddot{X}}e_k)\otimes(L_Ye_k)/\sqrt{n}\right]\left[(L_{\ddot{X}}e_k)\otimes(L_Ye_k)/\sqrt{n}\right]^{\mathsf{T}},
\end{aligned}$$

where $L_{\ddot{X}} = \tilde{K}_{\ddot{X}}^{1/2}R_Z$ and $L_Y = \tilde{K}_Y^{1/2}R_Z$. Let $L = (L_1, \ldots, L_n)$ be the $n^2 \times n$ matrix, whose kth column is $L_k = (L_{\ddot{X}}e_k)\otimes(L_Ye_k)/\sqrt{n}$. Then,

$$\mathcal{B}_{\ddot{X}Y}[\hat{\Gamma}_{\ddot{X}Y|Z}]\mathcal{B}_{\ddot{X}Y} = \sum_{k=1}^{n} L_kL_k^{\mathsf{T}} = LL^{\mathsf{T}}.$$

Since the nonzero eigenvalues of LL^{T} are same as those of $L^{\mathsf{T}}L$, we only need to calculate the eigenvalues of the $n \times n$ matrix $L^{\mathsf{T}}L$, substantially reducing the computing time. The computer memory needed is also substantially reduced, because we never need to save the $n^2 \times n^2$ matrix $\mathcal{B}_{\ddot{X}Y}[\hat{\Gamma}_{\ddot{X}Y|Z}]\mathcal{B}_{\ddot{X}Y}$, and the largest matrix we need to save is the $n^2 \times n$ matrix L.

6.3 Implementation of Welch-Satterthwaite approximation

To implement the Welch-Satterthwaite approximation in subsection 4.2 at the sample level, we simply substitute $\hat{\Gamma}_{\ddot{X}Y|Z}$ for $\Gamma_{\ddot{X}Y|Z}$ to estimate $\hat{\mu}_V$ and $\hat{\sigma}_V^2$, as follows:

$$\hat{\mu}_V = \mathrm{tr}(\hat{\Gamma}_{\ddot{X}Y|Z}) = \mathrm{tr}(\,\mathcal{B}_{\ddot{X}Y}[\hat{\Gamma}_{\ddot{X}Y|Z}]\mathcal{B}_{\ddot{X}Y}), \qquad \hat{\sigma}_V^2 = \mathrm{tr}(\hat{\Gamma}_{\ddot{X}Y|Z}^2) = \mathrm{tr}(\,\mathcal{B}_{\ddot{X}Y}[\hat{\Gamma}_{\ddot{X}Y|Z}]_{\mathcal{B}_{\ddot{X}Y}}^2). \tag{44}$$

By (21), we have the following identity: for any $k, \ell \in \{1, \ldots, n\}$,

$$\begin{aligned}
[(R_Ze_k)\otimes(R_Ze_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)[(R_Ze_\ell)\otimes(R_Ze_\ell)] &= (e_k^{\mathsf{T}}R_Z\tilde{K}_{\ddot{X}}R_Ze_\ell)(e_k^{\mathsf{T}}R_Z\tilde{K}_YR_Ze_\ell) \\
&= (e_k^{\mathsf{T}}\tilde{K}_{\ddot{X}|Z}e_\ell)(e_k^{\mathsf{T}}\tilde{K}_{Y|Z}e_\ell) = (\tilde{K}_{\ddot{X}|Z})_{k\ell}(\tilde{K}_{Y|Z})_{k\ell} = (\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z})_{k\ell}.
\end{aligned} \tag{45}$$

Substituting (43) into the first relation in (44) and then invoking the above identity, we have

$$\begin{aligned}
\hat{\mu}_V &= \mathrm{tr}\left\{\frac{1}{n}\sum_{k=1}^{n}[(R_Ze_k)\otimes(R_Ze_k)][(R_Ze_k)\otimes(R_Ze_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)\right\} \\
&= \frac{1}{n}\sum_{k=1}^{n}\mathrm{tr}\left\{[(R_Ze_k)\otimes(R_Ze_k)][(R_Ze_k)\otimes(R_Ze_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)\right\} \\
&= \frac{1}{n}\sum_{k=1}^{n}[(R_Ze_k)\otimes(R_Ze_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)[(R_Ze_k)\otimes(R_Ze_k)]
\end{aligned}$$

$$= \frac{1}{n}\sum_{k=1}^{n}(\tilde{K}_{\ddot{X}|Z} \odot \tilde{K}_{Y|Z})_{kk} = \frac{1}{n}\mathrm{tr}(\tilde{K}_{\ddot{X}|Z} \odot \tilde{K}_{Y|Z}).$$

Similarly, substituting (43) into the second relation in (44) and then invoking identity (45), we have

$$\begin{aligned}
\hat{\sigma}_V^2 &= \mathrm{tr}\left(\left\{\frac{1}{n}\sum_{k=1}^{n}[(R_Z e_k)\otimes(R_Z e_k)][(R_Z e_k)\otimes(R_Z e_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)\right\}^2\right) \\
&= \frac{1}{n^2}\sum_{k=1}^{n}\sum_{\ell=1}^{n}\mathrm{tr}\Big\{[(R_Z e_k)\otimes(R_Z e_k)][(R_Z e_k)\otimes(R_Z e_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y) \\
&\qquad\qquad \times[(R_Z e_\ell)\otimes(R_Z e_\ell)][(R_Z e_\ell)\otimes(R_Z e_\ell)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)\Big\} \\
&= \frac{1}{n^2}\sum_{k=1}^{n}\sum_{\ell=1}^{n}[(R_Z e_k)\otimes(R_Z e_k)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)[(R_Z e_\ell)\otimes(R_Z e_\ell)] \\
&\qquad\qquad \times[(R_Z e_\ell)\otimes(R_Z e_\ell)]^{\mathsf{T}}(\tilde{K}_{\ddot{X}}\otimes\tilde{K}_Y)[(R_Z e_k)\otimes(R_Z e_k)] \\
&= \frac{1}{n^2}\sum_{k=1}^{n}\sum_{\ell=1}^{n}(\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z})_{k\ell}(\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z})_{\ell k} \\
&= \frac{1}{n^2}\sum_{k=1}^{n}\sum_{\ell=1}^{n}(\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z})_{k\ell}^2 = \frac{1}{n^2}\|\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z}\|_{\mathrm{F}}^2,
\end{aligned}$$

where $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm of a matrix. Then, we can estimate ζ and ν in (35) by

$$\hat{\zeta} = \frac{\hat{\sigma}_V^2}{2\hat{\mu}_V} = \frac{\|\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z}\|_{\mathrm{F}}^2}{2n\,\mathrm{tr}(\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z})}, \qquad \hat{\nu} = \frac{2\hat{\mu}_V^2}{\hat{\sigma}_V^2} = \frac{2[\mathrm{tr}(\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z})]^2}{\|\tilde{K}_{\ddot{X}|Z}\odot\tilde{K}_{Y|Z}\|_{\mathrm{F}}^2}. \tag{46}$$

6.4 Tuning

In this section, we describe the procedure for selecting tuning parameters. Three types of tuning parameters are involved: the kernel bandwidths, the regularization parameter δ_n associated with K_T, and the regularization parameter ϵ_n associated with K_Z. We propose a step-by-step procedure for selecting these parameters sequentially.

6.4.1 Kernel bandwidths

For all kernels, we use Gaussian radial basis functions (RBFs) as reproducing kernels. Each kernel involves a tuning parameter controlling the bandwidth. Specifically, the kernel on T is defined as

$$\kappa_T : T\times T\to\mathbb{R}, \qquad (t_1,t_2)\mapsto\exp\{-\gamma_T(t_1-t_2)^2\}, \tag{47}$$

where $\gamma_T>0$ is a bandwidth parameter. Similarly, the kernel on $\mathcal{G}_X$ is

$$\kappa_X : \mathcal{G}_X\times\mathcal{G}_X\to\mathbb{R}, \qquad (x_1,x_2)\mapsto\exp\{-\gamma_X\|x_1-x_2\|^2\}, \tag{48}$$

where $\gamma_X > 0$ is the corresponding bandwidth parameter. The kernels κ_Y and κ_Z are defined analogously with tuning parameters γ_Y and γ_Z.

Following the criterion in Section 6.4 of Li and Solea (2018), we choose the bandwidth parameters according to the average pairwise distances. Specifically,

$$\frac{1}{\sqrt{\gamma_T}} = \binom{\ell+1}{2}^{-1} \sum_{k=0}^{\ell-1} \sum_{r=k+1}^{\ell} |u_k - u_r|, \tag{49}$$

where $U = \{u_0, u_1, \ldots, u_\ell\}$ are the $\ell+1$ equally-spaced points on T as defined in Section 5.2, and

$$\frac{1}{\sqrt{\gamma_X}} = \binom{n}{2}^{-1} \sum_{a=1}^{n-1} \sum_{b=a+1}^{n} \|X_a - X_b\|. \tag{50}$$

The parameters γ_Y and γ_Z are chosen in the same way.

6.4.2 Tikhonov regularization constant δ_n

To achieve appropriate scaling, we reset δ_n to $\delta_n \lambda_{\max}(K_T^{(i)})$, where $\lambda_{\max}(K_T^{(i)})$ is the largest eigenvalue of $K_T^{(i)}$. Inspired by the approach in Section 6.4 of Li and Solea (2018), we employ the generalized cross-validation (GCV) criterion

$$\begin{aligned}
\mathrm{GCV}_T(\delta_n) = &\sum_{i=1}^{n} \frac{\|X_i(T_i) - K_T^{(i)}(K_T^{(i)} + \delta_n \lambda_{max}(K_T^{(i)}) I_{m_i})^{-1} X_i(T_i)\|_F^2}{\{1 - \mathrm{tr}[K_T^{(i)}(K_T^{(i)} + \delta_n \lambda_{max}(K_T^{(i)}) I_{m_i})^{-1}]/m_i\}^2} \\
&+ \sum_{i=1}^{n} \frac{\|Y_i(T_i) - K_T^{(i)}(K_T^{(i)} + \delta_n \lambda_{max}(K_T^{(i)}) I_{m_i})^{-1} Y_i(T_i)\|_F^2}{\{1 - \mathrm{tr}[K_T^{(i)}(K_T^{(i)} + \delta_n \lambda_{max}(K_T^{(i)}) I_{m_i})^{-1}]/m_i\}^2} \\
&+ \sum_{i=1}^{n} \frac{\|Z_i(T_i) - K_T^{(i)}(K_T^{(i)} + \delta_n \lambda_{max}(K_T^{(i)}) I_{m_i})^{-1} Z_i(T_i)\|_F^2}{\{1 - \mathrm{tr}[K_T^{(i)}(K_T^{(i)} + \delta_n \lambda_{max}(K_T^{(i)}) I_{m_i})^{-1}]/m_i\}^2}.
\end{aligned} \tag{51}$$

In practice, we minimize $GCV_T(\delta_n)$ over a prespecified range I_T to determine the optimal tuning parameter δ_n^*:

$$\delta_n^* = \arg\min_{\delta_n \in I_T} \mathrm{GCV}_T(\delta_n). \tag{52}$$

6.4.3 Tikhonov regularization constant ϵ_n

Similarly, let κ_{XY} be a reproducing kernel on the tensor-product space $\mathcal{G}_{X\otimes Y} = \mathcal{G}_X \otimes \mathcal{G}_Y$, defined by $\kappa_{XY}(\cdot, x; \star, y) = \kappa_X(\cdot, x)\, \kappa_Y(\star, y)$. The corresponding Gram matrix for $\mathcal{H}_{X\otimes Y}$ is $K_{XY} = K_X \odot K_Y$, where $\odot$ denotes the Hadamard product. The centered Gram matrix is therefore $\tilde{K}_{XY} = H(K_X \odot K_Y)H$. Let $\mathcal{R}_Z = \mathrm{ran}(\hat{\Sigma}_{ZZ}) = \mathrm{span}(\mathcal{B}_Z)$ and $\mathcal{R}_{XY} = \mathrm{span}(\mathcal{B}_{XY})$, where

$$\begin{aligned}
\mathcal{B}_{XY} &= \{\hat{\kappa}_{XY}(\cdot, X_i; \star, Y_i) : i \in \{1, \ldots, n\}\} \\
&= \{\kappa_{XY}(\cdot, X_i; \star, Y_i) - E_n[\kappa_{XY}(\cdot, X; \star, Y)] : i \in \{1, \ldots, n\}\}.
\end{aligned}$$

We predict functions in $\mathcal{R}_{XY}$ using functions in $\mathcal{R}_Z$, which amounts to projecting $\hat{\kappa}_{XY}(\cdot, X_i, \star, Y_i)$ onto $\mathcal{R}_Z$. The empirical projection is $\hat{\Sigma}_{ZZ}^{-1}\hat{\Sigma}_{Z(XY)}\hat{\kappa}_{XY}(\cdot, X_i, \star, Y_i)$, which minimizes the empirical variance

$$\mathrm{var}_n\{\hat{\kappa}_{XY}(X, X_i, Y, Y_i) - [A\hat{\kappa}_{XY}(\cdot, X_i, \star, Y_i)](Z)\}. \tag{53}$$

among all bounded linear operators $A : \mathcal{G}_{X\otimes Y} \to \mathcal{G}_Z$. Under Tikhonov regularization, the coordinate representation of $\hat{\Sigma}_{ZZ}^{-1}\hat{\Sigma}_{Z(XY)}$ with respect to the bases $\mathcal{B}_Z$ and $\mathcal{B}_{XY}$ is

$$_{\mathcal{B}_Z}[\hat{\Sigma}_{ZZ}^{-1}\hat{\Sigma}_{Z(XY)}]_{\mathcal{B}_{XY}} = (\tilde{K}_Z + n\epsilon_n I)^{-1}\tilde{K}_{XY}.$$

Hence the empirical error in (53) becomes

$$\|\tilde{K}_{XY} - \tilde{K}_Z(\tilde{K}_Z + n\epsilon_n I)^{-1}\tilde{K}_{XY}\|_{\mathrm{F}}^2,$$

where $\|\cdot\|_{\mathrm{F}}$ stands for the Frobenius matrix norm.

As with δ_n, we replace ϵ_n by $\epsilon_n\lambda_{\max}(\tilde{K}_Z)$ to ensure appropriate scaling. The generalized cross-validation (GCV) criterion is

$$\mathrm{GCV}_Z(\epsilon_n) = \frac{\|\tilde{K}_{XY} - \tilde{K}_Z(\tilde{K}_Z + n\epsilon_n\lambda_{\max}(\tilde{K}_Z)I)^{-1}\tilde{K}_{XY}\|_{\mathrm{F}}^2}{\{1 - \mathrm{tr}[\tilde{K}_Z(\tilde{K}_Z + n\epsilon_n\lambda_{\max}(\tilde{K}_Z)I)^{-1}]/n\}^2}. \tag{54}$$

The optimal parameter is chosen as

$$\epsilon_n^* = \arg\min_{\epsilon_n \in I_Z} \mathrm{GCV}_Z(\epsilon_n). \tag{55}$$

for some prespecified range I_Z.

6.5 Algorithm

We summarize the procedures developed before as the following algorithm.

1. Choose the kernel κ_T for $\mathcal{H}$, along with its corresponding bandwidth. For example, if κ_T is the Gaussian kernel (47), then set the bandwidth by (49).

2. Compute the matrices $K_T^{(i)}$ based on κ_T, $i \in \{1, \ldots, n\}$, by (36).

3. Choose the regularization parameter δ_n by minimizing $GCV_T(\delta_n)$ in (51) over a prespecified range.

4. Based on the observations $X_i(T_i)$, $Y_i(T_i)$, $Z_i(T_i)$, $i \in \{1, \ldots, n\}$, compute the coordinate mapping $[\hat{X}_i]_{\mathcal{L}_i}$, $[\hat{Y}_i]_{\mathcal{L}_i}$, $[\hat{Z}_i]_{\mathcal{L}_i}$ by (37), $i \in \{1, \ldots, n\}$.

5. Set the grid points U and compute the matrix $\kappa_T(T_i, U)$, $i \in \{1, \ldots, n\}$. Compute the inner products $\langle \hat{X}_i, \hat{X}_j\rangle$, $\langle \hat{Y}_i, \hat{Y}_j\rangle$ and $\langle Z_i, Z_j\rangle$ by (39), and further compute $\|\hat{X}_i - \hat{X}_j\|$, $\|\hat{Y}_i - \hat{Y}_j\|$ and $\|\hat{Z}_i - \hat{Z}_j\|$ by (40).

6. Choose the bandwidths for the Gaussian kernels κ_X, κ_Y, κ_Z by (50).

7. Compute the Gram matrices K_X, K_Y, K_Z based on the kernels κ_X, κ_Y, κ_Z, and further compute $K_{\ddot{X}}$ based on (19).

8. Choose the regularization parameters δ_n and ϵ_n by minimizing $\mathrm{GCV}_T(\delta_n)$ in (51) and $\mathrm{GCV}_Z(\epsilon_n)$ in (54), respectively, over their prespecified ranges.

9. Compute R_Z using (20), and further compute $\tilde{K}_{\ddot{X}|Z}$ and $\tilde{K}_{Y|Z}$ using (21).

10. Compute the test statistic $n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\mathrm{HS}}$ via (22).

11. Compute the eigenvalues $\hat{\lambda}_1, \ldots, \hat{\lambda}_n$ of $L^\intercal L$ as defined in Section 6.2.

12. Perform the test using the asymptotic distribution in (33).

One side note is that the first five steps can be simplified under some special cases, where all observations are on the same set of equally-spaced grid points. The simplified steps are to be carried out according to Section 5.3. Moreover, if Welch-Satterthwaite approximation is applied, the last two steps are modified as follows:

11'. Compute $\hat{\zeta}$ and $\hat{\nu}$ as given in (46).

12'. Perform the test using (34) as an approximation of the asymptotic distribution in (33).

7 Simulation

We first generate the random functions $\epsilon_i(t)$ for $i \in \{1, 2, 3\}$ by

$$\epsilon_i(t) = \sum_{k=1}^{r} \xi_k \kappa_T(t, t_k),$$

where $\kappa_T(t, s) = \min(t, s)$ is the Brownian motion kernel, $\xi_1, \ldots, \xi_r$ are i.i.d. $N(0, 1)$, $t_1, \ldots, t_r$ are i.i.d. $U(0, 1)$. We generate $\epsilon_1, \epsilon_2, \epsilon_3$ independently according to the above model, where we choose $r = 50$. Note that, even though we have used the Brownian motion kernel to generate the functional random errors, we use the Gaussian kernel in our estimation process, as mentioned in Section 6.4.1.

For all models, the random functions X and Z are generated from

$$Z(t) = \epsilon_1(t), \qquad X(t) = 2Z(t) + \epsilon_2(t).$$

The dependence structures are introduced through different representations of $Y(t)$ in the five models described below. The models are summarized in Table 1.

In Table 1, only Model 1 corresponds to the case where $X(t)$ and $Y(t)$ are independent given $Z(t)$. In Models 2–4, $Y(t)$ depends on both $Z(t)$ and $X(t)$ through additive structures with different transformations applied to $X(t)$. In Model 5, we consider a multiplicative interaction between $Z(t)$ and $X(t)$.

In the simulation study, the sample sizes are set to be $n \in \{50, 100, 200, 500\}$, and the number of simulation replications is set to $n_{\mathrm{sim}} = 100$. We examine both balanced and

Table 1: Data-generating models used in the simulations. For all models, $Z(t) = \epsilon_1(t)$ and $X(t) = 2Z(t) + \epsilon_2(t)$, where $\epsilon_1, \epsilon_2, \epsilon_3$ are mutually independent functional errors. Model 1 satisfies $X \perp\!\!\!\perp Y \mid Z$; Models 2–5 represent quadratic, linear, logarithmic, and multiplicative alternatives, respectively.

Model	Representation of $Y(t)$
1	$Y(t) = Z(t) + \epsilon_3(t)$
2	$Y(t) = Z(t) + X^2(t) + \epsilon_3(t)$
3	$Y(t) = Z(t) + X(t) + \epsilon_3(t)$
4	$Y(t) = Z(t) + 4\log(\lvert X(t)\rvert + 1) + \epsilon_3(t)$
5	$Y(t) = Z(t)X(t) + \epsilon_3(t)$

unbalanced observation schemes with $m = 200$. In the balanced case, the observation times $t_{i0}, \ldots, t_{im}$ are fixed as $t_{ij} = j/200$, for $j \in \{0, 1, \ldots, 200\}$. In the unbalanced case, for the ith subject, the observation times $t_{i1}, \ldots, t_{im}$ are obtained by randomly sampling 200 points without replacement from $U = \{j/2000 : j \in \{1, \ldots, 2000\}\}$, and we set $t_{i0} = 0$. We then use (49) with grid points $\{0\} \cup U$ to calculate γ_T, and use the GCV criterion (52) to determine the tuning parameter δ_n, where the prespecified search range I_T is given by

$$I_T = \{0.001, 0.002, \ldots, 0.009, 0.01, 0.02, \ldots, 0.09, 0.1\}.$$

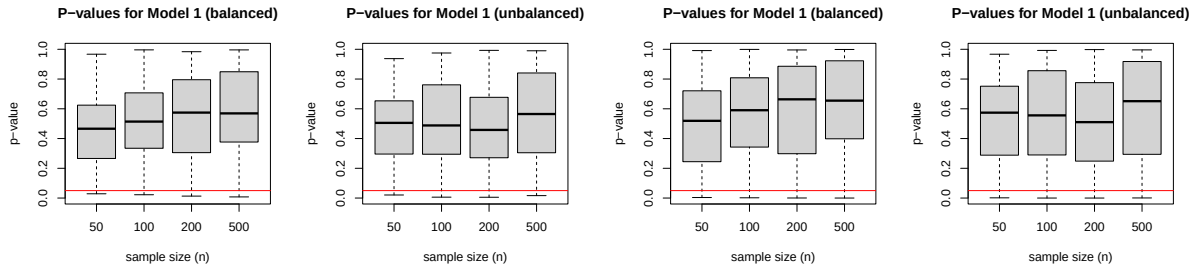
Furthermore, for selecting the regularization parameter ϵ_n, we adopt the GCV criterion (55) with the prespecified search range I_Z given by

$$I_Z = \{0.001/n, 0.002/n, \ldots, 0.009/n, 0.01/n, 0.02/n, \ldots, 0.09/n, 0.1/n\}.$$

The resulting p-values for all models under the balanced and unbalanced settings are displayed in Figures 1–5, where the test is performed either using the original weighted sum of $\chi^2_{(1)}$ distributions in (33) or using the Welch-Satterthwaite approximation (34).

Figs. 1–5 show that the performances under the balanced and unbalanced settings are similar for all models. Also, the results of tests using Welch-Satterthwaite approximation is similar to those via weighted sum of $\chi^2_{(1)}$ distributions, indicating that the Welch-Satterthwaite method approximates the asymptotic distribution (33) well. In particular, Figure 1 indicates that most of the p-values are much larger than 0.05, suggesting that X and Y are independent given Z, which agrees with the conditional-independence null hypothesis. For the remaining models, nearly all p-values fall below 0.05, which demonstrate the strong power of our CCCO-based test. The only exception occurs for the small sample size ($n = 50$) in Model 4, as shown in Figure 4. This behavior is reasonable because the model involves the transformation $\log(|X(t)| + 1)$, which weakens the influence of $X(t)$ on $Y(t)$ and makes conditional dependence more difficult to detect when the sample size is small.

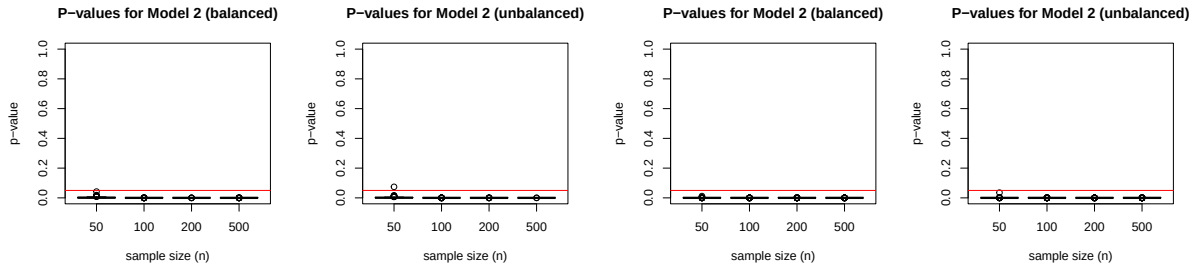
We also examined the simulation running times, as reported in Table 2. In the table, Stage 1 refers to the computation of the test statistic T_n, which is required regardless of whether the Welch-Satterthwaite approximation is used. Stage 2.1 refers to performing the test using the Welch-Satterthwaite approximation, whereas Stage 2.2 refers to performing the test using the original weighted sum of $\chi^2_{(1)}$ distributions. From Table 2 we see that



(a) Results via weighted sum of $\chi^2_{(1)}$ distributions.

(b) Results with Welch-Satterthwaite approximation.

Figure 1: Simulation p-values for Model 1, where $Y(t) = Z(t) + \epsilon_3(t)$. Each boxplot summarizes 100 simulation replications. The sample sizes are $n \in \{50, 100, 200, 500\}$. The left and right plots within each panel show results for the balanced and unbalanced observation schedules. Panel (a) uses the weighted sum of $\chi^2_{(1)}$ distributions, and panel (b) uses the Welch–Satterthwaite approximation. The red line marks the significance level $\alpha = 0.05$. Model 1 satisfies $X \perp\!\!\!\perp Y \mid Z$, and most of the tests correctly do not reject this null hypothesis.

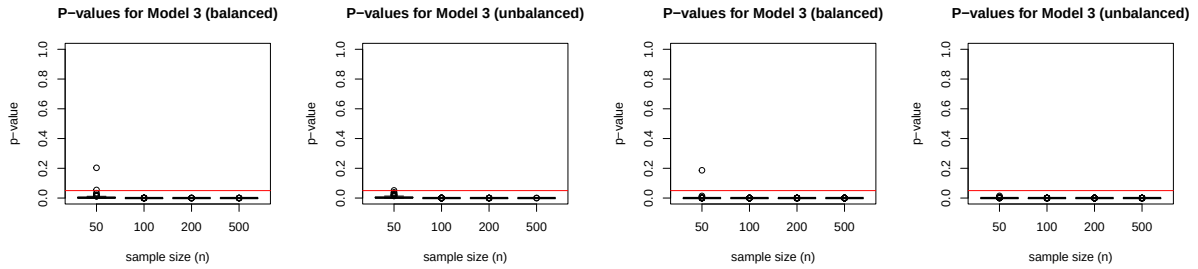


(a) Results via weighted sum of $\chi^2_{(1)}$ distributions.

(b) Results with Welch-Satterthwaite approximation.

Figure 2: Simulation p-values for Model 2, where $Y(t) = Z(t) + X^2(t) + \epsilon_3(t)$. The panel arrangement, sample sizes, observation schedules, methods for computing the p-values, and significance reference line are the same as in Fig. 1. Both methods successfully detect this conditional dependence.

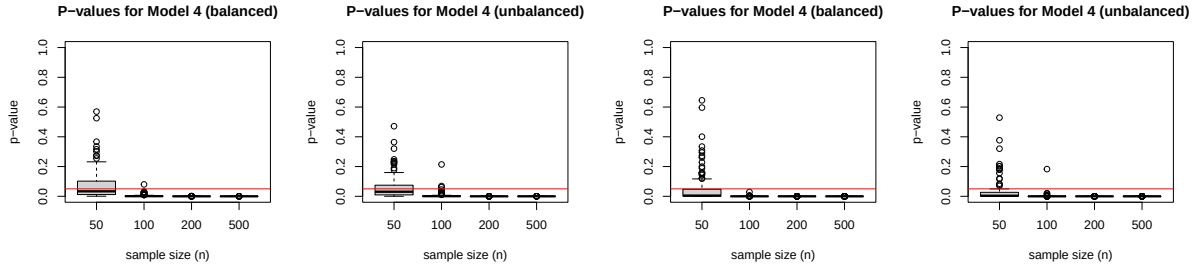
Stage 2.1 is significantly faster than Stage 2.2, especially when the sample size is large. In fact, Stage 2.2 requires calculating the eigenvalues of an $n \times n$ matrix $L^\mathsf{T} L$ where L is an $n^2 \times n$ matrix, which is very time consuming for large n. In contrast, the Welch-Satterthwaite approximation only requires the trace and Frobenius norm of the $n \times n$ matrix $\tilde{K}_{\dot{X}|Z} \odot \tilde{K}_{Y|Z}$, which had already been calculated in Stage 1. Consequently, very little additional calculation is needed. We also note that the time for computing the test statistic is substantially shorter under the balanced observation schedule, since the inner products of the random functions can be approximated directly by (41), without the smoothing step required to estimate the entire functions.



(a) Results via weighted sum of $\chi^2_{(1)}$ distributions.

(b) Results with Welch-Satterthwaite approximation.

Figure 3: Simulation p-values for Model 3, where $Y(t) = Z(t) + X(t) + \epsilon_3(t)$. The panel arrangement, sample sizes, observation schedules, methods for computing the p-values, and significance reference line are the same as in Fig. 1. Both methods successfully detect this conditional dependence.



(a) Results via weighted sum of $\chi^2_{(1)}$ distributions.

(b) Results with Welch-Satterthwaite approximation.

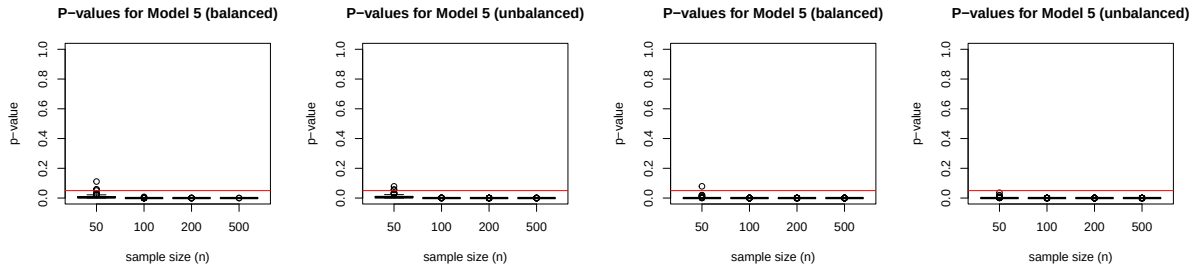
Figure 4: Simulation p-values for Model 4, where $Y(t) = Z(t) + 4\log(|X(t)| + 1) + \epsilon_3(t)$. The panel arrangement, sample sizes, observation schedules, methods for computing the p-values, and significance reference line are the same as in Fig. 1. This dependence is hard to detect at $n = 50$, but when sample size increases, both methods successfully detect this conditional dependence.

8 Application on WISDM Dataset

We apply our method to the Wireless Sensor Data Mining (WISDM) dataset (Kwapisz et al., 2010).[1] This dataset contains labeled accelerometer measurements collected from multiple users under controlled laboratory conditions by the WISDM Lab. For each subject, triaxial accelerometer signals corresponding to three spatial directions are recorded during various physical activities.

Thirty-six users participate in the experiment, during which accelerometer measurements are collected while they perform various activities, including Walking, Jogging, Upstairs, Downstairs, Sitting, and Standing. The predictor consists of multivariate functional data,

[1]The dataset is available at `https://www.cis.fordham.edu/wisdm/dataset.php`. We use the file `WISDM_ar_v1.1_raw.txt` from the Activity Prediction Dataset (v1.1).



(a) Results via weighted sum of $\chi^2_{(1)}$ distributions.

(b) Results with Welch-Satterthwaite approximation.

Figure 5: Simulation p-values for Model 5, where $Y(t) = Z(t)X(t) + \epsilon_3(t)$. The panel arrangement, sample sizes, observation schedules, methods for computing the p-values, and significance reference line are the same as in Fig. 1. Both methods successfully detect this conditional dependence.

Table 2: Computation times, in seconds, for different sample sizes under balanced and unbalanced observation schedules. Stage 1 calculates the test statistic T_n. Stages 2.1 and 2.2 compute the p-value using the Welch–Satterthwaite approximation and the weighted sum of $\chi^2_{(1)}$ distributions, respectively. Stage 1 takes longer under the unbalanced schedule, while Stage 2.1 is substantially faster than Stage 2.2 for large sample sizes.

(a) Elapsed time (in seconds) for balanced cases.

n	Stage 1	Stage 2.1	Stage 2.2
50	0.3763	0.0373	0.1121
100	0.6971	0.0393	0.3808
200	1.9652	0.0273	2.2478
500	11.8508	0.0101	27.5814

(b) Elapsed time (in seconds) for unbalanced cases.

n	Stage 1	Stage 2.1	Stage 2.2
50	5.1011	0.0074	0.1049
100	12.9999	0.0051	0.3918
200	38.8418	0.0050	2.3504
500	201.1896	0.0062	26.7144

with time series representing accelerations along the x-, y-, and z-directions. For each user-activity pair, the accelerations are recorded at a sampling rate of 20 Hz (i.e., one sample every 50 ms). For each user performing a given activity, we use the observations from the first 10, 15, and 20 seconds, which correspond to $m = 200$, 300, and 400 time points, respectively. Taking $m = 200$, for example, the curves of each activity are given by Figures S.1–S.6 in Appendix C, where $X(t)$, $Y(t)$, and $Z(t)$ represent the acceleration curves along the x-, y-, and z-directions, respectively.

We plot the first 15 eigenvalues of $\hat{\Sigma}_{ZZ}$ for the six activities in Figure 6. As shown in the figure, the decaying patterns of the eigenvalues for all six activities suggest that Assumption 7 is justifiable.

To justify Assumption 6, by Proposition 2, it suffices to justify the boundedness of the three quantities in (26) and (27). Specifically, we consider the quantities

$$s^{(1)}_{jk} = \hat{\mu}_j^{1/2}\hat{\gamma}_k^{-1/2-\beta}|\hat{E}(\hat{\theta}_j\hat{\xi}_k)|, \qquad s^{(2)}_{jk} = \hat{\eta}_j^{1/2}\hat{\gamma}_k^{-1/2-\beta}|\hat{E}(\hat{\omega}_j\hat{\xi}_k)|, \qquad j,k \in \{1,2,\ldots,15\},$$
$$s^{(3)}_{jk\ell} = \hat{\eta}_j^{1/2}\hat{\gamma}_k^{1/2}\hat{\gamma}_\ell^{-1/2-\beta}|\hat{E}(\hat{\omega}_j\hat{\xi}_k\hat{\xi}_\ell)|, \qquad j,k,\ell \in \{1,2,\ldots,15\},$$

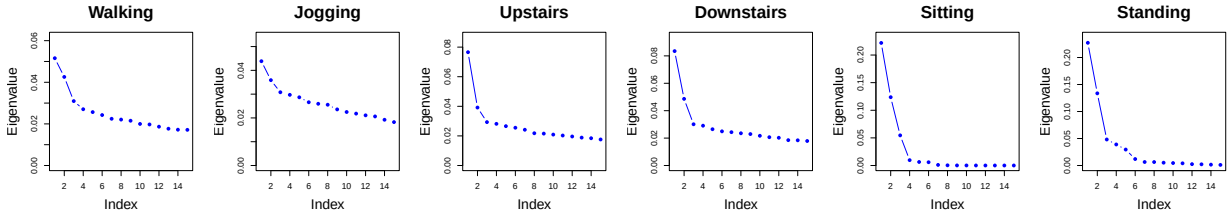


Figure 6: The first 15 eigenvalues of the estimated covariance operator $\hat{\Sigma}_{ZZ}$ for the six WISDM activities. The eigenvalues are ordered from largest to smallest within each panel. The vertical scales differ across activities to show the decay patterns clearly. The observed decay supports Assumption 7.

where all the hats refer to the empirical estimators of the corresponding quantities. We pick $\beta = 0.25$. The heatmaps for $s_{jk}^{(1)}$ and $s_{jk}^{(2)}$ are presented in Figs. 7 and 8, respectively. For $s_{jk\ell}^{(3)}$, we fix k in each heatmap and present the plots for $k \in \{1, \ldots, 5\}$ in Fig. 9 and Fig. S.7 in the Supplementary Material. We also report the maximum values of $s_{jk\ell}^{(3)}$, for $k \in \{1, \ldots, 15\}$ and for all six activities in Table S.1 in the Supplementary Material. The heatmaps for $k \in \{6, \ldots, 15\}$ are omitted because all values are less than 0.01 according to Table S.1. Note that the ranges of Figs. 7 and 8 are set to be $[0, 0.8]$ while the ranges of Fig. 9 and Fig. S.7 are set to be $[0, 0.3]$. We also give the maximum values of $s_{jk}^{(1)}$, $s_{jk}^{(2)}$ and $s_{jk\ell}^{(3)}$ over their corresponding index ranges as Table 3. As we can see from the figures, most values of the quantities are very small. Relatively large values only appear in Sitting and Standing with small indices (j, k) or (j, k, ℓ), while with the increase of the indices, all the quantities have a decaying trend. This justifies the boundedness of the three quantities in (26) and (27), which further justifies that Assumption 6 is plausible for this dataset.

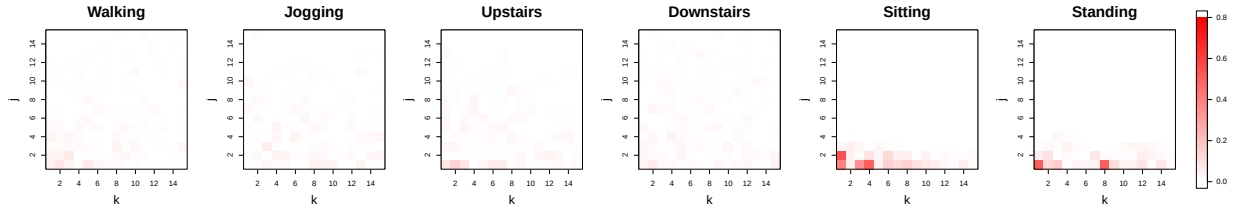


Figure 7: Heatmaps of $s_{jk}^{(1)}$ for $j, k \in \{1, \ldots, 15\}$, which measures a scaled association between the j-th component on the Y side and the k-th component of Z. All panels use the same color scale from 0 to 0.8, with darker colors indicating larger values. Larger values are concentrated in the low-order components for Sitting and Standing.

Our goal is to test, for each activity, whether the accelerations in the x- and y-directions are conditionally independent given the acceleration in the z-direction. Note that the sample size n varies across activities, because some users do not perform all activities in the dataset. For each activity, we perform the CCCO-based test under each sample size and time window. The resulting p-values are summarized in Table 4, where panel (a) is obtained from the weighted sum of $\chi^2_{(1)}$ distributions, and panel (b) is obtained from the Welch-Satterthwaite

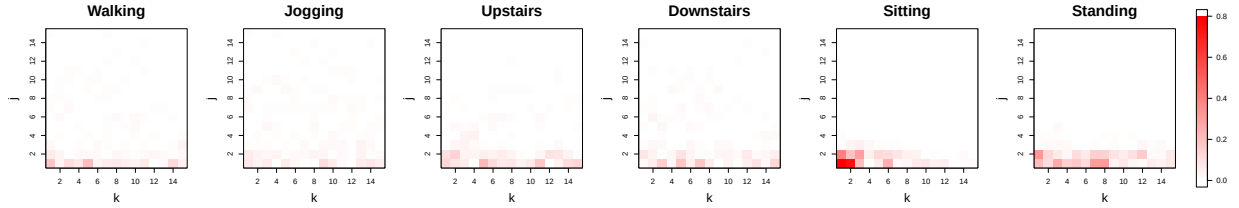


Figure 8: Heatmaps of $s^{(2)}_{jk}$ for $j, k \in \{1, \ldots, 15\}$, which measures a scaled association between the j-th component on the $\ddot{X}$ side and the k-th component of Z. All panels use the same color scale from 0 to 0.8, with darker colors indicating larger values. Larger values are concentrated in the low-order components for Sitting.

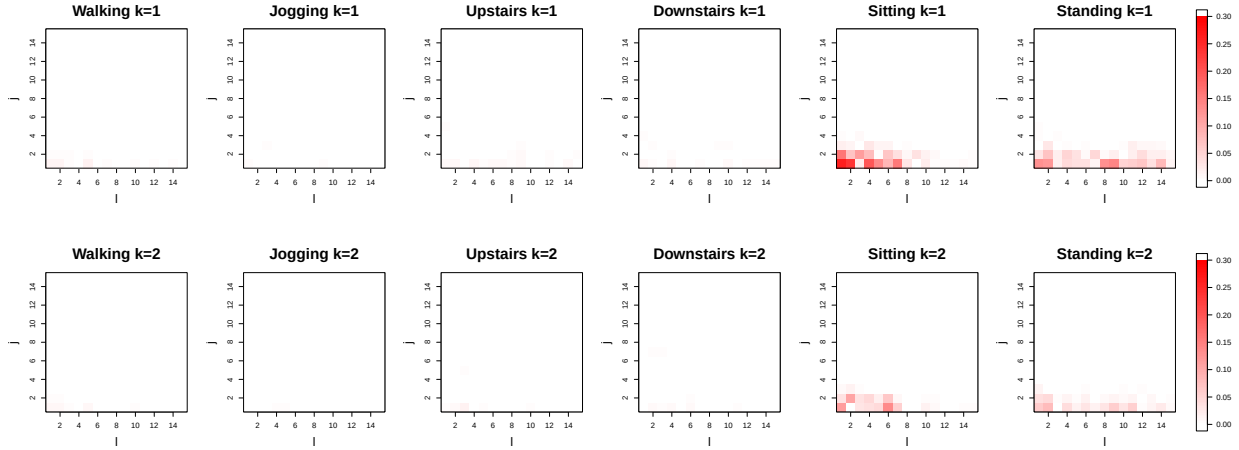


Figure 9: Heatmaps of $s^{(3)}_{jk\ell}$ for $j, \ell \in \{1, \ldots, 15\}$, which measures a scaled third-order association involving the j-th component on the $\ddot{X}$ side and the k-th and ℓ-th components of Z. The upper and lower rows correspond to $k = 1$ and $k = 2$, respectively. All panels use the same color scale from 0 to 0.3, with darker colors indicating larger values. Larger values are concentrated in the low-order components for Sitting and Standing.

approximation.

From Table 4a we see that, at the significance level $\alpha = 0.05$, for all activities except sitting, the accelerations in the x- and y-directions are conditionally independent given the acceleration in the z-direction. For the sitting activity, however, the accelerations in the x- and y-directions are conditionally dependent given the acceleration in the z-direction, regardless of the time window used. The results from the Welch-Satterthwaite approximation in Table 4b are slightly different as conditional dependence is significant for the standing activity as well. This is reasonable since, with only $n = 24$ samples, the Welch-Satterthwaite approximation might not be very accurate, and the p-values are close to $\alpha = 0.05$.

Our statistical results are consistent with the physical analysis (Figure 2) of Kwapisz et al. (2010). For the first four activities (walking, jogging, upstairs, and downstairs), there are clear periodic patterns in all three directions, and the oscillations across directions exhibit noticeable synchronization. As a result, the accelerations in the z-direction largely

Table 3: Maximum values of the three empirical quantities over the first 15 components. The rows report $\max_{j,k} s_{jk}^{(1)}$, $\max_{j,k} s_{jk}^{(2)}$, and $\max_{j,k,\ell} s_{jk\ell}^{(3)}$. Sitting has the largest maximum for all three quantities.

	Walking	Jogging	Upstairs	Downstairs	Sitting	Standing
$\max s_{jk}^{(1)}$	0.0759	0.0589	0.1370	0.0565	0.5495	0.5054
$\max s_{jk}^{(2)}$	0.2024	0.1046	0.2231	0.1788	0.7947	0.3230
$\max s_{jk\ell}^{(3)}$	0.0154	0.0042	0.0172	0.0096	0.2636	0.1413

Table 4: CCCO p-values for the six WISDM activities. For each activity, the test examines whether the acceleration functions in the x- and y-directions are conditionally independent given the acceleration function in the z-direction. The column n gives the number of users included in the analysis. The values $m = 200$, 300, and 400 represent the first 10, 15, and 20 seconds of data recorded at 20 Hz. Panels (a) and (b) use the weighted sum of $\chi^2_{(1)}$ distributions and the Welch-Satterthwaite approximation, respectively. At the 0.05 significance level, Sitting is significant in both panels, while Standing is significant only in panel (b).

(a) Results via weighted sum of $\chi^2_{(1)}$ distributions.

Activity	n	$m = 200$	$m = 300$	$m = 400$
Walking	36	0.1610	0.2016	0.2221
Jogging	32	0.2121	0.2435	0.2519
Upstairs	32	0.1393	0.1731	0.1974
Downstairs	32	0.1951	0.2072	0.2053
Sitting	23	0.0094	0.0097	0.0102
Standing	24	0.0789	0.0836	0.0796

(b) Results with Welch-Satterthwaite approximation.

Activity	n	$m = 200$	$m = 300$	$m = 400$
Walking	36	0.1407	0.1977	0.2376
Jogging	32	0.1702	0.2090	0.2197
Upstairs	32	0.0870	0.1285	0.1576
Downstairs	32	0.1650	0.1816	0.1740
Sitting	23	0.0003	0.0003	0.0004
Standing	24	0.0340	0.0395	0.0359

capture the relationship between those in the x- and y-directions, making the conditional dependence between accelerations in the x- and y-directions given the z-direction very weak.

In contrast, for sitting and standing, the accelerations in the z-direction are nearly constant, as shown in Figure 2(e)-(f) of Kwapisz et al. (2010), and therefore provide little information about the x- and y-directions. However, Figure 2(e) indicates that during sitting the oscillations in the x- and y-directions remain highly synchronized, suggesting a strong dependency between them. By comparison, the pattern for standing in Figure 2(f) appears much weaker, indicating a substantially weaker dependence between the two directions.

A Illustrative Examples for Assumption 6

Example 1. In this example we show that if $X \perp\!\!\!\perp Z$ and $Y \perp\!\!\!\perp Z$, then Assumption 6 is satisfied for any $\beta > 0$. Since $Y \perp\!\!\!\perp Z$, we have $\theta_j \perp\!\!\!\perp \xi_k$ for all j, k. Hence $E(\theta_j \xi_k) = E(\theta_j)E(\xi_k) = 0$ for all j, k. So (26) is satisfied. Since $X \perp\!\!\!\perp Z$, we have $\omega_j \perp\!\!\!\perp \xi_k$ for all j, k, which implies that $E(\omega_j \xi_k) = E(\omega_j)E(\xi_k) = 0$ and $E(\omega_j \xi_k \xi_\ell) = E(\omega_j)E(\xi_k \xi_\ell) = 0$ for all j, k, ℓ. So (27) is satisfied.

Example 2. In this example we show that, if there exists an $n_0 \in \mathbb{N}$ such that

$$\{\omega_j : j > n_0\} \perp\!\!\!\perp Z, \quad X \perp\!\!\!\perp \{\xi_k : k > n_0\}, \quad \text{and} \quad Y \perp\!\!\!\perp \{\xi_k : k > n_0\}, \tag{S.1}$$

and $\sup_{j\in\mathbb{N}} E|\omega_j|^3 < \infty$, $\sup_{j\in\mathbb{N}} E|\xi_j|^3 < \infty$, then Assumption 6 is satisfied for any $\beta > 0$. The first two independence conditions in (S.1) mean that the dependence between X and Z only happens in both low-frequency components, and the last condition in (S.1) means that Y only depends on the low-frequency components in Z.

Since $Y \perp\!\!\!\perp \{\xi_k : k > n_0\}$, we have $E(\theta_j \xi_k) = E(\theta_j)E(\xi_k) = 0$ for all $k > n_0$ and all j. Also note that for all $k \le n_0$ and all j, we have $|E(\theta_j \xi_k)| \le \sqrt{E(\theta_j^2)E(\xi_k^2)} = 1$, so the supremum in (26) is

$$\sup_{j,k\in\mathbb{N}} \mu_j^{1/2} \gamma_k^{-1/2-\beta} |E(\theta_j \xi_k)| = \sup_{j,k\in\mathbb{N}, k\le n_0} \mu_j^{1/2} \gamma_k^{-1/2-\beta} |E(\theta_j \xi_k)| \le \mu_1^{1/2} \gamma_{n_0}^{-1/2-\beta} < \infty.$$

Since $X \perp\!\!\!\perp \{\xi_k : k > n_0\}$, we have $E(\omega_j \xi_k) = E(\omega_j)E(\xi_k) = 0$ for all $k > n_0$ and all j. Also note that for all $k \le n_0$ and all j, we have $|E(\omega_j \xi_k)| \le \sqrt{E(\omega_j^2)E(\xi_k^2)} = 1$, so the second supremum in (27) is

$$\sup_{j,k\in\mathbb{N}} \eta_j^{1/2} \gamma_k^{-1/2-\beta} |E(\omega_j \xi_k)| = \sup_{j,k\in\mathbb{N}, k\le n_0} \eta_j^{1/2} \gamma_k^{-1/2-\beta} |E(\omega_j \xi_k)| \le \eta_1^{1/2} \gamma_{n_0}^{-1/2-\beta} < \infty.$$

Furthermore, since $\{\omega_j : j > n_0\} \perp\!\!\!\perp Z$, we have $E(\omega_j \xi_k \xi_\ell) = E(\omega_j)E(\xi_k \xi_\ell) = 0$ for all $j > n_0$ and all k, ℓ. Also note that for all $j \le n_0$ and all k, ℓ, we have $|E(\omega_j \xi_k \xi_l)| \le \sqrt[3]{E|\omega_j|^3 E|\xi_k|^3 E|\xi_l|^3} < \infty$, so the first supremum in (27) is

$$\begin{aligned} \sup_{j,k,\ell\in\mathbb{N}} \eta_j^{1/2} \gamma_k^{1/2} \gamma_\ell^{-1/2-\beta} |E(\omega_j \xi_k \xi_\ell)| &= \sup_{j,k,\ell\in\mathbb{N}, j\le n_0} \eta_j^{1/2} \gamma_k^{1/2} \gamma_\ell^{-1/2-\beta} |E(\omega_j \xi_k \xi_\ell)| \\ &\preceq \eta_1^{1/2} \gamma_1^{1/2} \gamma_{n_0}^{-1/2-\beta} < \infty, \end{aligned}$$

as desired.

B Technical Proofs

B.1 Proof of Corollary 1

Proof. Again, we will only show the first part of (8). Take $f = \tilde{\kappa}_Z(\cdot, z)$ and $g = \kappa_X(x, \cdot)$ in the first equation in (7), and we have

$$\langle B_{X|Z} \tilde{\kappa}_Z(\cdot, z), \kappa_X(x, \cdot) \rangle_{\mathcal{H}_X} = \langle \tilde{\kappa}_Z(\cdot, z), E[\kappa_X(x, X)|Z = \cdot] \rangle_{\mathcal{H}_Z}. \tag{S.2}$$

The right-hand side of (S.2) is

$$\begin{aligned}\langle \tilde{\kappa}_Z(\cdot, z), E[\kappa_X(x, X)|Z = \cdot]\rangle_{\mathcal{H}_Z} &= E[\kappa_X(x, X)|Z = z] - E\{E[\kappa_X(x, X)|Z]\} \\ &= E[\kappa_X(x, X)|Z = z] - E[\kappa_X(x, X)] \\ &= E[\kappa_X(x, X) - \mu_X(x)|Z = z] \\ &= E[\tilde{\kappa}_X(x, X)|Z = z].\end{aligned}$$

The left-hand side of (S.2) is

$$\langle B_{X|Z}\tilde{\kappa}_Z(\cdot, z), \kappa_X(x, \cdot)\rangle_{\mathcal{H}_X} = \langle B_{X|Z}\tilde{\kappa}_Z(\cdot, z), \kappa_X(\cdot, x)\rangle_{\mathcal{H}_X} = [B_{X|Z}\tilde{\kappa}_Z(\cdot, z)](x).$$

Equating the two sides of (S.2) gives the desired result (8). It remains to check that $f = \tilde{\kappa}_Z(\cdot, z) \in \ker(\Sigma_{ZZ})^{\perp}$. For any $h \in \ker(\Sigma_{ZZ})$, we have $\Sigma_{ZZ}h = 0$, so

$$0 = \langle h, \Sigma_{ZZ}h\rangle_{\mathcal{H}_Z} = \mathrm{var}[h(Z)],$$

which implies that h is a constant function almost surely. Therefore,

$$\langle \tilde{\kappa}_Z(\cdot, z), h\rangle_{\mathcal{H}_Z} = h(z) - E[h(Z)] = 0$$

almost surely. This implies that $\tilde{\kappa}_Z(\cdot, z) \in \ker(\Sigma_{ZZ})^{\perp}$. □

B.2 Proof of Proposition 1

Proof. We expand the right-hand side of (9) as follows.

$$\begin{aligned}&E\left[(\tilde{\kappa}_X(\cdot, X) - E\left[\tilde{\kappa}_X(\cdot, X)|Z\right]) \otimes (\tilde{\kappa}_Y(\cdot, Y) - E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right])\right] \\ =&E\left[\tilde{\kappa}_X(\cdot, X) \otimes \tilde{\kappa}_Y(\cdot, Y)\right] - E\left[E\left[\tilde{\kappa}_X(\cdot, X)|Z\right] \otimes \tilde{\kappa}_Y(\cdot, Y)\right] \\ &- E\left[\tilde{\kappa}_X(\cdot, X) \otimes E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right]\right] + E\left[E\left[\tilde{\kappa}_X(\cdot, X)|Z\right] \otimes E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right]\right]\end{aligned} \tag{S.3}$$

By (5), the first term on the right-hand side is Σ_{XY}. We now consider the other three terms separately. By Corollary 1, we have

$$E\left[\tilde{\kappa}_X(\cdot, X)|Z\right] = B_{X|Z}\tilde{\kappa}_Z(\cdot, Z) = \Sigma_{XZ}\Sigma_{ZZ}^{\dagger}, \quad E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right] = B_{Y|Z}\tilde{\kappa}_Z(\cdot, Z) = \Sigma_{YZ}\Sigma_{ZZ}^{\dagger}.$$

Hence the second term on the right-hand side of (S.3) is

$$\begin{aligned}E\left[E\left[\tilde{\kappa}_X(\cdot, X)|Z\right] \otimes \tilde{\kappa}_Y(\cdot, Y)\right] =&E\left[B_{X|Z}\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Y(\cdot, Y)\right] = B_{X|Z}E\left[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Y(\cdot, Y)\right] \\ =&\Sigma_{XZ}\Sigma_{ZZ}^{\dagger}\Sigma_{ZY}.\end{aligned}$$

By the law of total expectation, we have

$$\begin{aligned}E\left[E\left[\tilde{\kappa}_X(\cdot, X)|Z\right] \otimes \tilde{\kappa}_Y(\cdot, Y)\right] =&E\left[E\left[\tilde{\kappa}_X(\cdot, X)|Z\right] \otimes E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right]\right], \\ E\left[\tilde{\kappa}_X(\cdot, X) \otimes E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right]\right] =&E\left[E\left[\tilde{\kappa}_X(\cdot, X)|Z\right] \otimes E\left[\tilde{\kappa}_Y(\cdot, Y)|Z\right]\right].\end{aligned}$$

Consequently, the four terms on the right-hand of (S.3) are simply

$$\Sigma_{XY} - \Sigma_{XZ}\Sigma_{ZZ}^{\dagger}\Sigma_{ZY} - \Sigma_{XZ}\Sigma_{ZZ}^{\dagger}\Sigma_{ZY} + \Sigma_{XZ}\Sigma_{ZZ}^{\dagger}\Sigma_{ZY} = \Sigma_{XY|Z},$$

as desired. □

B.3 Proof of Proposition 2

Proof. 1. If $\text{rank}(\Sigma_{ZZ}) = \infty$, by (24), we have

$$\Sigma_{ZZ}^{-1-\beta} = \sum_{k=1}^{\infty} \gamma_k^{-1-\beta} (\phi_k \otimes \phi_k).$$

By the Karhunen-Loéve expansions of $\kappa_Z(\cdot, Z)$ and $\kappa_Y(\cdot, Y)$,

$$\Sigma_{YZ} = E[\tilde{\kappa}_Y(\cdot, Y) \otimes \tilde{\kappa}_Z(\cdot, Z)] = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \sqrt{\mu_j \gamma_k} E(\theta_j \xi_k)(\psi_j \otimes \phi_k).$$

Hence, $\Sigma_{YZ} = S_{YZ} \Sigma_{ZZ}^{1+\beta}$ is equivalent to

$$\begin{aligned} S_{YZ} = \Sigma_{YZ} \Sigma_{ZZ}^{-1-\beta} &= \left[\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \mu_j^{1/2} \gamma_k^{1/2} E(\theta_j \xi_k)(\psi_j \otimes \phi_k) \right] \left[\sum_{\ell=1}^{\infty} \gamma_\ell^{-1-\beta} (\phi_\ell \otimes \phi_\ell) \right] \\ &= \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} \mu_j^{1/2} \gamma_k^{1/2} \gamma_\ell^{-1-\beta} E(\theta_j \xi_k)(\psi_j \otimes \phi_k)(\phi_\ell \otimes \phi_\ell) \\ &= \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \mu_j^{1/2} \gamma_k^{-1/2-\beta} E(\theta_j \xi_k)(\psi_j \otimes \phi_k). \end{aligned}$$

Therefore, S_{YZ} is bounded if and only if (26) holds.

If $\text{rank}(\Sigma_{ZZ}) = K < \infty$, then $\gamma_k = 0$ for $k > K$. Let $\{\phi_{K+1}, \phi_{K+2}, \dots\}$ be a completion of $\{\phi_0, \phi_1, \dots, \phi_K\}$ as an orthonormal basis of $\mathcal{G}_Z$, where $\phi_0 = \mu_Z / \|\mu_Z\|_{\mathcal{G}_Z}$. Suppose that

$$S_{YZ} = \sum_{j=1}^{\infty} \sum_{k=0}^{\infty} c_{jk} (\psi_j \otimes \phi_k)$$

is a bounded operator such that $\Sigma_{YZ} = S_{YZ} \Sigma_{ZZ}^{1+\beta}$. Then, we have

$$\begin{aligned} \sum_{j=1}^{\infty} \sum_{k=1}^{K} \mu_j^{1/2} \gamma_k^{1/2} E(\theta_j \xi_k)(\psi_j \otimes \phi_k) &= \left[\sum_{j=1}^{\infty} \sum_{k=0}^{\infty} c_{jk} (\psi_j \otimes \phi_k) \right] \left[\sum_{k=1}^{K} \gamma_k^{1+\beta} (\phi_k \otimes \phi_k) \right] \\ &= \sum_{j=1}^{\infty} \sum_{k=0}^{\infty} \sum_{\ell=1}^{K} c_{jk} \gamma_\ell^{1+\beta} (\psi_j \otimes \phi_k)(\phi_\ell \otimes \phi_\ell) \\ &= \sum_{j=1}^{\infty} \sum_{k=1}^{K} c_{jk} \gamma_k^{1+\beta} (\psi_j \otimes \phi_k), \end{aligned}$$

which indicates that

$$c_{jk} = \mu_j^{1/2} \gamma_k^{-1/2-\beta} E(\theta_j \xi_k), \qquad \text{for all } j = 1, 2, \dots, \quad k = 1, 2, \dots, K.$$

Thus, the boundedness of S_{YZ} implies $\sup_{j\in\mathbb{N},k\in\mathbb{N}_0}|c_{jk}|<\infty$ ($\mathbb{N}_0$ being the set of nonnegative integers), which further implies (26). Conversely, if (26) holds, we can set

$$S_{YZ}=\sum_{j=1}^{\infty}\sum_{k=1}^{K}\mu_j^{1/2}\gamma_k^{-1/2-\beta}E(\theta_j\xi_k)(\psi_j\otimes\phi_k),$$

which is clearly bounded due to (26), so that

$$\begin{aligned}S_{YZ}\Sigma_{ZZ}^{1+\beta}&=\left[\sum_{j=1}^{\infty}\sum_{k=1}^{K}\mu_j^{1/2}\gamma_k^{-1/2-\beta}E(\theta_j\xi_k)(\psi_j\otimes\phi_k)\right]\left[\sum_{k=1}^{K}\gamma_k^{1+\beta}(\phi_k\otimes\phi_k)\right]\\&=\sum_{j=1}^{\infty}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\mu_j^{1/2}\gamma_k^{-1/2-\beta}\gamma_\ell^{1+\beta}E(\theta_j\xi_k)(\psi_j\otimes\phi_k)(\phi_\ell\otimes\phi_\ell)\\&=\sum_{j=1}^{\infty}\sum_{k=1}^{K}\mu_j^{1/2}\gamma_k^{1/2}E(\theta_j\xi_k)(\psi_j\otimes\phi_k)=\Sigma_{YZ}.\end{aligned}$$

2. If $\operatorname{rank}(\Sigma_{ZZ})=\infty$, by the Karhunen-Loéve expansions of $\kappa_Z(\cdot,Z)$ and $\kappa_X(\cdot,X)$,

$$\begin{aligned}\Lambda_{\ddot{X}Z}&=E[\tilde{\kappa}_X(\cdot,X)\kappa_Z(\cdot,Z)\otimes\tilde{\kappa}_Z(\cdot,Z)]\\&=E[\tilde{\kappa}_X(\cdot,X)\tilde{\kappa}_Z(\cdot,Z)\otimes\tilde{\kappa}_Z(\cdot,Z)]+E[\tilde{\kappa}_X(\cdot,X)\mu_Z\otimes\tilde{\kappa}_Z(\cdot,Z)]\\&=\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\sum_{\ell=1}^{\infty}\sqrt{\eta_j\gamma_k\gamma_\ell}E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_l]+\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\sqrt{\eta_j\gamma_k}E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k].\end{aligned}$$

Hence, $\Lambda_{\ddot{X}Z}=S_{\ddot{X}Z}\Sigma_{ZZ}^{1+\beta}$ is equivalent to

$$\begin{aligned}S_{\ddot{X}Z}&=\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{-1-\beta}\\&=\left[\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\sum_{\ell=1}^{\infty}\eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{1/2}E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell]\right]\left[\sum_{r=1}^{\infty}\gamma_r^{-1-\beta}(\phi_r\otimes\phi_r)\right]\\&\quad+\left[\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\eta_j^{1/2}\gamma_k^{1/2}E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k]\right]\left[\sum_{\ell=1}^{\infty}\gamma_\ell^{-1-\beta}(\phi_\ell\otimes\phi_\ell)\right]\\&=\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\sum_{\ell=1}^{\infty}\sum_{r=1}^{\infty}\eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{1/2}\gamma_r^{-1-\beta}E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell](\phi_r\otimes\phi_r)\\&\quad+\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\sum_{\ell=1}^{\infty}\eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{-1-\beta}E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k](\phi_\ell\otimes\phi_\ell)\\&=\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\sum_{\ell=1}^{\infty}\eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{-1/2-\beta}E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell]\\&\quad+\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\eta_j^{1/2}\gamma_k^{-1/2-\beta}E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k].\end{aligned}$$

Therefore, $S_{\ddot{X}Z}$ is bounded if and only if (27) holds.

If $\mathrm{rank}(\Sigma_{ZZ}) = K < \infty$, then $\gamma_k = 0$ for $k > K$. Similarly, let $\{\phi_{K+1}, \phi_{K+2}, \dots\}$ be a completion of $\{\phi_0, \phi_1, \dots, \phi_K\}$ as an orthonormal basis of $\mathcal{G}_Z$, where $\phi_0 = \mu_Z / \|\mu_Z\|_{\mathcal{G}_Z}$. Suppose that

$$S_{\ddot{X}Z} = \sum_{j=1}^{\infty}\sum_{k=0}^{\infty}\sum_{\ell=0}^{\infty} c_{jk\ell}[(\chi_j\phi_k)\otimes\phi_\ell]$$

is a bounded operator such that $\Lambda_{\ddot{X}Z} = S_{\ddot{X}Z}\Sigma_{ZZ}^{1+\beta}$. Then, we have

$$\begin{aligned}
&\sum_{j=1}^{\infty}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{1/2}E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell] + \sum_{j=1}^{\infty}\sum_{k=1}^{K}\eta_j^{1/2}\gamma_k^{1/2}E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k] \\
&= \left[\sum_{j=1}^{\infty}\sum_{k=0}^{\infty}\sum_{\ell=0}^{\infty} c_{jk\ell}[(\chi_j\phi_k)\otimes\phi_\ell]\right]\left[\sum_{k=1}^{K}\gamma_k^{1+\beta}(\phi_k\otimes\phi_k)\right] \\
&= \sum_{j=1}^{\infty}\sum_{k=0}^{\infty}\sum_{\ell=0}^{\infty}\sum_{r=1}^{K} c_{jk\ell}\gamma_r^{1+\beta}[(\chi_j\phi_k)\otimes\phi_\ell](\phi_r\otimes\phi_r) \\
&= \sum_{j=1}^{\infty}\sum_{k=0}^{\infty}\sum_{\ell=1}^{K} c_{jk\ell}\gamma_l^{1+\beta}[(\chi_j\phi_k)\otimes\phi_\ell] \\
&= \sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\sum_{\ell=1}^{K} c_{jk\ell}\gamma_l^{1+\beta}[(\chi_j\phi_k)\otimes\phi_\ell] + \sum_{j=1}^{\infty}\sum_{k=1}^{K} c_{j0k}\|\mu_Z\|_{\mathcal{G}_Z}^{-1}\gamma_l^{1+\beta}[(\chi_j\mu_Z)\otimes\phi_k]
\end{aligned}$$

which indicates that

$$\begin{aligned}
c_{jk\ell} &= \eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{-1/2-\beta}E(\omega_j\xi_k\xi_\ell), \qquad \text{for all } j = 1, 2, \dots, \quad k, \ell = 1, 2, \dots, K, \quad \text{and} \\
c_{j0k} &= \eta_j^{1/2}\gamma_k^{-1/2-\beta}\|\mu_Z\|_{\mathcal{G}_Z}E(\omega_j\xi_k), \qquad \text{for all } j = 1, 2, \dots, \quad k = 1, 2, \dots, K.
\end{aligned}$$

Thus, the boundedness of $S_{\ddot{X}Z}$ implies $\sup_{j\in\mathbb{N},k,l\in\mathbb{N}_0}|c_{jk\ell}| < \infty$, which further implies (27). Conversely, if (27) holds, we can set

$$\begin{aligned}
S_{\ddot{X}Z} = &\sum_{j=1}^{\infty}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{-1/2-\beta}E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell] \\
&+ \sum_{j=1}^{\infty}\sum_{k=1}^{K}\eta_j^{1/2}\gamma_k^{-1/2-\beta}E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k],
\end{aligned}$$

which is clearly bounded due to (27), so that

$$\begin{aligned}
S_{\ddot{X}Z}\Sigma_{ZZ}^{1+\beta} = &\left[\sum_{j=1}^{\infty}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{-1/2-\beta}E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell]\right]\left[\sum_{k=1}^{K}\gamma_k^{1+\beta}(\phi_k\otimes\phi_k)\right] \\
&+ \left[\sum_{j=1}^{\infty}\sum_{k=1}^{K}\eta_j^{1/2}\gamma_k^{-1/2-\beta}E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k]\right]\left[\sum_{k=1}^{K}\gamma_k^{1+\beta}(\phi_k\otimes\phi_k)\right]
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{j=1}^{\infty}\sum_{k=1}^{K}\sum_{\ell=1}^{K}\sum_{r=1}^{K} \eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{-1/2-\beta}\gamma_r^{1+\beta} E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell](\phi_r\otimes\phi_r) \\
&\quad + \sum_{j=1}^{\infty}\sum_{k=1}^{K}\sum_{\ell=1}^{K} \eta_j^{1/2}\gamma_k^{-1/2-\beta}\gamma_\ell^{1+\beta} E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k](\phi_\ell\otimes\phi_\ell) \\
&= \sum_{j=1}^{\infty}\sum_{k=1}^{K}\sum_{\ell=1}^{K} \eta_j^{1/2}\gamma_k^{1/2}\gamma_\ell^{1/2} E(\omega_j\xi_k\xi_\ell)[(\chi_j\phi_k)\otimes\phi_\ell] \\
&\quad + \sum_{j=1}^{\infty}\sum_{k=1}^{K} \eta_j^{1/2}\gamma_k^{1/2} E(\omega_j\xi_k)[(\chi_j\mu_Z)\otimes\phi_k] \\
&= \Lambda_{\ddot{X}Z}.
\end{aligned}
$$

□

B.4 Proof of Lemma 2

Proof. Under Assumption 7, there exists a constant C_α such that $\gamma_j \le C_\alpha j^{-\alpha}$. Let $J_\epsilon = \lceil (C_\alpha/\epsilon_n)^{1/\alpha} \rceil$. Then

$$
\sum_{j=1}^{\infty} \frac{\gamma_j}{(\gamma_j+\epsilon_n)^2} = \sum_{j\le J_\epsilon} \frac{\gamma_j}{(\gamma_j+\epsilon_n)^2} + \sum_{j>J_\epsilon} \frac{\gamma_j}{(\gamma_j+\epsilon_n)^2}. \tag{S.4}
$$

For the first term on the right-hand side of (S.4), we have

$$
\sum_{j\le J_\epsilon} \frac{\gamma_j}{(\gamma_j+\epsilon_n)^2} \le \sum_{j\le J_\epsilon} \frac{1}{\epsilon_n} = \frac{J_\epsilon}{\epsilon_n} = O\left(\epsilon_n^{-1-1/\alpha}\right) = O\left(\epsilon_n^{-(\alpha+1)/\alpha}\right).
$$

For the second term on the right-hand side of (S.4), since

$$
\sum_{j>J_\epsilon} j^{-\alpha} \le \int_{J_\epsilon}^{\infty} x^{-\alpha} dx = \frac{J_\epsilon^{1-\alpha}}{\alpha-1} = O\left(J_\epsilon^{1-\alpha}\right),
$$

we have

$$
\sum_{j>J_\epsilon} \frac{\gamma_j}{(\gamma_j+\epsilon_n)^2} \le \epsilon_n^{-2} \sum_{j>J_\epsilon} \gamma_j \le C_\alpha \epsilon_n^{-2} \sum_{j>J_\epsilon} j^{-\alpha} = O\left(\epsilon_n^{-2} J_\epsilon^{1-\alpha}\right) = O\left(\epsilon_n^{-1-1/\alpha}\right) = O\left(\epsilon_n^{-(\alpha+1)/\alpha}\right).
$$

Combining the two terms above gives the first rate.

For the second summation, we apply the same decomposition

$$
\sum_{j=1}^{\infty} \frac{\gamma_j^2}{(\gamma_j+\epsilon_n)^2} = \sum_{j\le J_\epsilon} \frac{\gamma_j^2}{(\gamma_j+\epsilon_n)^2} + \sum_{j>J_\epsilon} \frac{\gamma_j^2}{(\gamma_j+\epsilon_n)^2}. \tag{S.5}
$$

For the first term on the right-hand side of (S.5), we have

$$
\sum_{j\le J_\epsilon} \frac{\gamma_j^2}{(\gamma_j+\epsilon_n)^2} \le \sum_{j\le J_\epsilon} 1 = J_\epsilon = O\left(\epsilon_n^{-1/\alpha}\right).
$$

For the second term on the right-hand side of (S.5), since

$$\sum_{j>J_\epsilon} j^{-2\alpha} \leq \int_{J_\epsilon}^{\infty} x^{-2\alpha} dx = \frac{J_\epsilon^{1-2\alpha}}{2\alpha - 1} = O\left(J_\epsilon^{1-2\alpha}\right),$$

we have

$$\sum_{j>J_\epsilon} \frac{\gamma_j^2}{(\gamma_j + \epsilon_n)^2} \leq \epsilon_n^{-2} \sum_{j>J_\epsilon} \gamma_j^2 \leq C_\alpha^2 \epsilon_n^{-2} \sum_{j>J_\epsilon} j^{-2\alpha} = O\left(\epsilon_n^{-2} J_\epsilon^{1-2\alpha}\right) = O\left(\epsilon_n^{-1/\alpha}\right).$$

Combining the two terms above gives the second rate. □

B.5 Proof of Proposition 3

Proof. For a Brownian bridge Z on $[0,1]$, the covariance function is $c(s,t) = \min(s,t) - st$ for all $s,t \in [0,1]$, and its Karhunen-Loève expansion can be written as

$$Z(t) = \sum_{m=1}^{\infty} \sqrt{\rho_m} Z_m e_m(t), \tag{S.6}$$

where

$$\rho_m = \frac{1}{\pi^2 m^2}, \quad e_m(t) = \sqrt{2}\sin(m\pi t), \quad m = 1, 2, \ldots,$$

and $Z_1, Z_2, \ldots$ are i.i.d. standard normal random variables. See, for example, Deheuvels (2006); Chigansky et al. (2020) for detailed discussions of the Karhunen-Loève expansions of the Brownian bridge. The rest of the proof will be based on the map of Z to its Karhunen-Loève coefficient sequence $\tilde{Z} = (Z_1, Z_2, \ldots)$.

Step 1: Construction of space of $\tilde{Z}$ and its relationship to $\mathcal{H}_Z$. We first define the space where the Karhunen-Loève coefficient sequence $\tilde{Z}$ resides. Note that $\{e_m : m = 1, 2, \ldots\}$ forms a complete orthonormal basis of $L^2[0,1]$ (see, for example, Section 5.4 of Vretblad (2003)). Let $\mathcal{H}_Z = L^2[0,1]$ and define another Hilbert space

$$\tilde{\mathcal{H}}_Z = \left\{ \tilde{z} = (z_1, z_2, \ldots) \in \mathbb{R}^\infty : \sum_{m=1}^{\infty} \rho_m z_m^2 < \infty \right\},$$

where the inner product in $\tilde{\mathcal{H}}_Z$ is defined as

$$\langle \tilde{z}, \tilde{z}' \rangle_{\tilde{\mathcal{H}}_Z} = \sum_{m=1}^{\infty} \rho_m z_m z_m', \quad \text{for all } \tilde{z} = (z_1, z_2, \ldots),\ \tilde{z}' = (z_1', z_2', \ldots) \in \tilde{\mathcal{H}}_Z.$$

Define $T : \mathcal{H}_Z \to \tilde{\mathcal{H}}_Z$ by $Tz = (T_1 z, T_2 z, \ldots)$, where $T_m : \mathcal{H}_Z \to \mathbb{R}$ is the mapping

$$T_m z = \frac{\langle z, e_m \rangle_{\mathcal{H}_Z}}{\sqrt{\rho_m}}.$$

Clearly, T is linear and bijective, and

$$\langle z, z' \rangle_{\mathcal{H}_Z} = \sum_{m=1}^{\infty} \rho_m (T_m z)(T_m z') = \langle Tz, Tz' \rangle_{\tilde{\mathcal{H}}_Z}.$$

Therefore, $\mathcal{H}_Z$ and $\tilde{\mathcal{H}}_Z$ are isomorphic, and T is an isomorphism.

Step 2: Constructing RKHS's on $\mathcal{H}_Z$ and $\tilde{H}_Z$ and developing their relation. Define another positive-definite kernel function $\kappa : \tilde{H}_Z \times \tilde{H}_Z \to \mathbb{R}$ by

$$\kappa(\tilde{z}, \tilde{z}') = \exp\{-\eta \|\tilde{z} - \tilde{z}'\|^2_{\tilde{\mathcal{H}}_Z}\}, \quad \text{for all } \tilde{z}, \tilde{z}' \in \tilde{\mathcal{H}}_Z.$$

Then, we have

$$\kappa_Z(z, z') = \exp\{-\eta \|z - z'\|^2_{\mathcal{H}_Z}\} = \exp\{-\eta \|Tz - Tz'\|^2_{\tilde{\mathcal{H}}_Z}\} = \kappa(Tz, Tz').$$

Let $\mathcal{G}_Z$ be the RKHS of real-valued functions on $\mathcal{H}_Z$ generated by $\kappa_Z(z, z')$ and, similarly, let $\tilde{\mathcal{G}}_Z$ be the RKHS on $\tilde{H}_Z$ generated $\kappa(\tilde{z}, \tilde{z}')$. Let $U : \tilde{\mathcal{G}}_Z \to \mathcal{G}_Z$ be the map defined by $U(\tilde{f}) = \tilde{f} \circ T$. Clearly, U is linear and bijective. Furthermore, for all $z \in \mathcal{H}_Z$, we have

$$U\kappa(\cdot, Tz) = \kappa(T\cdot, Tz) = \kappa_Z(\cdot, z).$$

Therefore, for all $z, z' \in \mathcal{H}_Z$, using the reproducing properties of $\mathcal{G}_Z$ and $\tilde{\mathcal{G}}_Z$, we have

$$\begin{aligned}\langle U\kappa(\cdot, Tz), U\kappa(\cdot, Tz')\rangle_{\mathcal{G}_Z} &= \langle \kappa_Z(\cdot, z), \kappa_Z(\cdot, z')\rangle_{\mathcal{G}_Z} = \kappa_Z(z, z') \\ &= \kappa(Tz, Tz') = \langle \kappa(\cdot, Tz), \kappa(\cdot, Tz')\rangle_{\tilde{\mathcal{G}}_Z}.\end{aligned}$$

Based on the definitions of $\mathcal{G}_Z$ and $\tilde{\mathcal{G}}_Z$, we know that for all $\tilde{f}, \tilde{f}' \in \tilde{\mathcal{G}}_Z$, we have

$$\langle U\tilde{f}, U\tilde{f}'\rangle_{\mathcal{G}_Z} = \langle \tilde{f}, \tilde{f}'\rangle_{\tilde{\mathcal{G}}_Z}.$$

Therefore, $\mathcal{G}_Z$ and $\tilde{\mathcal{G}}_Z$ are isomorphic, and U is an isomorphism.

Step 3: Second moment operators in two RKHS's. Let $\tilde{M}_{ZZ} : \tilde{\mathcal{G}}_Z \to \tilde{\mathcal{G}}_Z$ and $M_{ZZ} : \mathcal{G}_Z \to \mathcal{G}_Z$ be the second moment operators

$$\tilde{M}_{ZZ} = E_{\tilde{Z}\sim\mu}[\kappa(\cdot, \tilde{Z}) \otimes \kappa(\cdot, \tilde{Z})], \qquad M_{ZZ} = E_{Z\sim P_Z}[\kappa_Z(\cdot, Z) \otimes \kappa_Z(\cdot, Z)],$$

where $\mu = P \circ \tilde{Z}^{-1}$ is the distribution of a sequence of i.i.d. standard normal variables. Note that $Z \sim P_Z$ (i.e., Z is a Brownian bridge) if and only if $TZ \sim \mu$ (Deheuvels, 2006; Chigansky et al., 2020).

By construction, we know that, for all $f \in \mathcal{G}_Z$ and $\tilde{f} \in \tilde{\mathcal{G}}_Z$, we have

$$(M_{ZZ}f)(\cdot) = E_{Z\sim P_Z}[\kappa_Z(\cdot, Z)f(Z)], \qquad (\tilde{M}_{ZZ}\tilde{f})(\cdot) = E_{\tilde{Z}\sim\mu}[\kappa(\cdot, \tilde{Z})\tilde{f}(\tilde{Z})].$$

Thus, for all $\tilde{f} \in \tilde{\mathcal{G}}_Z$, for all $z \in \mathcal{H}_Z$, we have

$$\begin{aligned}(M_{ZZ}U\tilde{f})(z) &= E_{Z\sim P_Z}[\kappa_Z(z, Z)(U\tilde{f})(Z)] = E_{Z\sim P_Z}[\kappa(Tz, TZ)\tilde{f}(TZ)] \\ &= E_{\tilde{Z}\sim\mu}[\kappa(Tz, \tilde{Z})\tilde{f}(\tilde{Z})] = (\tilde{M}_{ZZ}\tilde{f})(Tz) = (U\tilde{M}_{ZZ}\tilde{f})(z).\end{aligned}$$

Therefore, we have $\tilde{M}_{ZZ} = U^{-1}M_{ZZ}U$, and since U is a Hilbert space homomorphism, by Theorem 8.3 of Conway (2019), $\tilde{M}_{ZZ}$ and M_{ZZ} have the same eigenvalues. Hence, to find the properties of eigenvalues of M_{ZZ}, it suffices to study the eigenvalues of $\tilde{M}_{ZZ}$.

Step 4: Mercer decomposition of $\kappa(\cdot,\cdot)$ under $L^2(\mu\times\mu)$. Since, for all $\tilde{f}\in\tilde{\mathcal{G}}_Z$, we have

$$\tilde{M}_{ZZ}\tilde{f}(\cdot)=\int\kappa(\cdot,\tilde{z}')\tilde{f}(\tilde{z}')d\mu(\tilde{z}'),$$

we can view $\tilde{M}_{ZZ}$ as the integral operator on $\tilde{\mathcal{G}}_Z$ with $\kappa(\cdot,\cdot)$ as its kernel function. By Mercer's Theorem (Steinwart and Scovel, 2012), to find the eigenvalues and eigenfunctions of $\tilde{M}_{ZZ}$, it suffices to find the Mercer decomposition of $\kappa(\cdot,\cdot)$.

Clearly, for all $\tilde{z}=(z_1,z_2,\dots)\in\tilde{\mathcal{H}}_Z$ and $\tilde{z}'=(z_1',z_2',\dots)\in\tilde{\mathcal{H}}_Z$, we have

$$\kappa(\tilde{z},\tilde{z}')=\exp\{-\eta\|\tilde{z}-\tilde{z}'\|^2_{\tilde{\mathcal{H}}_Z}\}=\exp\left\{-\eta\sum_{m=1}^{\infty}\rho_m(z_m-z_m')^2\right\}=\prod_{m=1}^{\infty}\kappa_m(z_m,z_m')$$

where

$$\kappa_m(z_m,z_m')=\exp\{-\eta\rho_m(z_m-z_m')^2\},\quad m=1,2,\dots.$$

Since $\sum_{m=1}^{\infty}\rho_m<\infty$, the product above is well-defined almost surely $\mu\times\mu$. Let $\mathcal{H}_m$ be the RKHS with reproducing kernel κ_m. Then, equation (2.4) of Karvonen and Särkkä (2019) gives the Mercer decomposition of the Gaussian kernel as

$$\kappa_m(z_m,z_m')=\sum_{j=1}^{\infty}\lambda_{m,j}\psi_{m,j}(z_m)\psi_{m,j}(z_m')\quad\text{in}\quad L^2(\Phi\times\Phi).\tag{S.7}$$

where Φ is the standard normal distributions and the eigenvalues have the form

$$\lambda_{m,j}=a_m b_m^j,\qquad j=0,1,\dots,$$

with,

$$a_m=\left(\frac{1}{2(1+8\eta\rho_m)^{1/2}-1+2\eta\rho_m}\right)^{1/2},\qquad b_m=\frac{2\eta\rho_m}{2(1+8\eta\rho_m)^{1/2}-1+2\eta\rho_m},\tag{S.8}$$

The forms of the eigenfunctions $\psi_{m,j}$'s are omitted, and can be written in terms of the Hermite polynomials according to equation (2.5) of Karvonen and Särkkä (2019).

Next, for any $M\geq 1$, let

$$\kappa^{(M)}(\tilde{z},\tilde{z}')=\prod_{m=1}^{M}\kappa_m(z_m,z_m').\tag{S.9}$$

By (S.7) and (S.9), for each M, we have

$$\kappa^{(M)}(\tilde{z},\tilde{z}')=\sum_{\nu_1,\dots,\nu_M\geq 0}\left(\prod_{m=1}^{M}\lambda_{m,\nu_m}\right)\prod_{m=1}^{M}\psi_{m,\nu_m}(z_m)\psi_{m,\nu_m}(z_m'),$$

with convergence in $L^2(\mu_M \times \mu_M)$ where $\mu_M = \Phi \times \cdots \times \Phi$ is the m-fold product measure. Moreover, since $0 \le \kappa^{(M)} \le 1$ and $\kappa^{(M)}(\tilde{z}, \tilde{z}') \to \kappa(\tilde{z}, \tilde{z}')$ almost surely $\mu \times \mu$, using dominated convergence theorem, we have

$$\|\kappa^{(M)} - \kappa\|_{L^2(\mu \otimes \mu)} \to 0.$$

Consequently, the Mercer's decomposition of $\kappa(\tilde{z}, \tilde{z}')$ is

$$\kappa(\tilde{z}, \tilde{z}') = \sum_{\nu \in \mathcal{I}} \lambda_\nu^* \Psi_\nu^*(\tilde{z}) \Psi_\nu^*(\tilde{z}') \quad \text{in } L^2(\mu \times \mu), \tag{S.10}$$

where $\mathcal{I}$ is the collection of all sequences, whose entries are nonnegative integers, and with at most finitely many nonzero entries, and

$$\lambda_\nu^* = \prod_{m=1}^{\infty} \lambda_{m,\nu_m}, \qquad \Psi_\nu^*(\tilde{z}) = \prod_{m=1}^{\infty} \psi_{m,\nu_m}(z_m).$$

Since $\{\psi_{m,j} : j = 1, 2, \dots\}$ is an orthonormal basis of $L^2(\Phi)$, we know that $\{\Psi_\nu^* : \nu \in \mathcal{I}\}$ is an orthonormal basis of $L^2(\mu)$. Therefore, λ_ν for $\nu \in \mathcal{I}$ are all the eigenvalues of $\tilde{M}_{ZZ}$.

Step 5: Order of magnitude of the eigenvalues of $\tilde{M}_{ZZ}$. We first analyze the order of magnitude of the quantities a_m and b_m in (S.8). By

$$(1 + 8\eta\rho_m)^{1/2} \le 1 + 4\eta\rho_m,$$

we have

$$a_m \ge \left(\frac{1}{1 + 10\eta\rho_m}\right)^{1/2}.$$

By

$$\sum_{m=1}^{\infty} \rho_m = \frac{1}{\pi^2} \sum_{m=1}^{\infty} \frac{1}{m^2} < \infty,$$

we have

$$\sum_{m=1}^{\infty} |\log a_m| \le \frac{1}{2} \sum_{m=1}^{\infty} \log(1 + 10\eta\rho_m) \le 5\eta \sum_{m=1}^{\infty} \rho_m < \infty.$$

Hence,

$$0 < \prod_{m=1}^{\infty} a_m < \infty.$$

Also, it is obvious that $2(1 + 8\eta\rho_m)^{1/2} - 1 + 2\eta\rho_m \ge 1$, so

$$b_m \le 2\eta\rho_m.$$

Applying this, and invoking $\eta \le \pi^2/8$, we have

$$\lambda_{m,j} \le a_m(2\eta\rho_m)^j = a_m(2\eta/\pi^2)^j m^{-2j} \le a_m(m+1)^{-2j}, \tag{S.11}$$

and consequently, for $\nu \in \mathcal{I}$,

$$\lambda_\nu^* \le C \prod_{m=1}^{\infty} (m+1)^{-2\nu_m}$$

where $C = \prod_{m=1}^{\infty} a_m$.

Fix an $s > 1/2$. Then

$$\sum_{\nu\in\mathcal{I}} (\lambda_\nu^*)^s \le C^s \sum_{\nu\in\mathcal{I}} \prod_{m=1}^{\infty} (m+1)^{-2s\nu_m}.$$

Note that, if $\nu \in \mathbb{N}_0^\infty \setminus \mathcal{I}$, then for any $m_0 > 0$, there exists $r > m_0$ such that $\nu_r \ge 1$, which implies that the product $\prod_{m=1}^{\infty}(m+1)^{-2s\nu_m} \le (r+1)^{-2s\nu_r} \le m_0^{-2s}$. By taking $\lim_{m_0\to\infty}$ on the right, we see that $\prod_{m=1}^{\infty}(m+1)^{-2s\nu_m} = 0$. Therefore,

$$\begin{aligned}\sum_{\nu\in\mathcal{I}} \prod_{m=1}^{\infty} (m+1)^{-2s\nu_m} &= \sum_{\nu\in\mathbb{N}_0^\infty} \prod_{m=1}^{\infty} (m+1)^{-2s\nu_m} \\ &= \prod_{m=1}^{\infty} \left\{ \sum_{r=0}^{\infty} (m+1)^{-2sr} \right\} = \prod_{m=1}^{\infty} \frac{1}{1-(m+1)^{-2s}}.\end{aligned}$$

The product on the right-hand side is finite because, when $s > 1/2$, we have $\sum_{m=1}^{\infty}(m+1)^{-2s} < \infty$ and $0 < (m+1)^{-2sr} < 1$ for all m and r, so by Corollary 42.5 of Agarwal et al. (2011), $\prod_{m=1}^{\infty}[1-(m+1)^{-2s}] > 0$. Thus

$$\sum_{\nu\in\mathcal{I}} (\lambda_\nu^*)^s < \infty. \tag{S.12}$$

Let $\lambda_1 \ge \lambda_2 \ge \cdots$ be the decreasing rearrangement of $\{\lambda_\nu^* : \nu \in \mathcal{I}\}$. Then

$$j\lambda_j^s \le \sum_{\ell=1}^{j} \lambda_\ell^s \le \sum_{\ell=1}^{\infty} \lambda_\ell^s < \infty,$$

so

$$\lambda_j \le C_{1/s} j^{-1/s}, \qquad \text{where } C_{1/s} = \left(\sum_{\ell=1}^{\infty} \lambda_\ell^s \right)^{1/s}.$$

Since $s > 1/2$ is arbitrary, this gives

$$\lambda_j \le C_\alpha j^{-\alpha} \qquad \text{for every } 1 < \alpha < 2, \qquad j = 1, 2, \ldots.$$

Step 6: Relationship between γ_j and λ_j. By Corollary 12.3 of Schmüdgen (2012), since $\Sigma_{ZZ} \preceq M_{ZZ}$ in the operator sense (i.e., $\langle f, \Sigma_{ZZ} f\rangle_{\mathcal{H}_Z} \leq \langle f, M_{ZZ} f\rangle_{\mathcal{H}_Z}$ for all $f \in \mathcal{H}_Z$), for all $j = 1, 2, \ldots$, we have $\gamma_j \leq \lambda_j$, where γ_j, $j = 1, 2, \ldots$, are eigenvalues of Σ_{ZZ}. Thus, we have

$$\gamma_j \leq \lambda_j \leq C_\alpha j^{-\alpha} \qquad \text{for every } 1 < \alpha < 2, \qquad j = 1, 2, \ldots,$$

ad desired.

General case. Regarding the general case where $\rho_m \preceq m^{-r}$ for some $r > 1$, we only need to make some changes in Step 5. Specifically, the exponent 2 should be replaced by r starting from (S.11). For example, (S.11) becomes

$$\lambda_{m,j} \leq a_m (2\eta\rho_m)^j = a_m (2\eta/\pi^2)^j m^{-rj} \leq a_m (m+1)^{-rj}.$$

In the subsequent analysis, the condition $s > 1/2$ can be replaced by $s > 1/r$, and the essential convergence result (S.12) still holds. The rest of the argument is unchanged, which gives $\lambda_j \preceq j^{-1/s}$, and for any $1 < \alpha < r$, choosing $1/r < s < 1/\alpha$ gives $\lambda_j \preceq j^{-\alpha}$. □

B.6 Proof of Lemma 3

Proof. By definition (28), we have

$$R = \mu_X \hat{\Sigma}_{ZZ} (\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - \mu_X = \epsilon_n \mu_X (\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}.$$

Therefore,

$$\begin{aligned} \|R\|_{\text{OP}} &= \epsilon_n \|\mu_X (\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}} \leq \epsilon_n \|\mu_X\|_{\mathcal{G}_X} \|(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}} \\ &\leq \epsilon_n \|\mu_X\|_{\mathcal{G}_X} \epsilon_n^{-1} = \|\mu_X\|_{\mathcal{G}_X}, \end{aligned}$$

which proves assertion 1.

To prove assertion 2, we decompose R into $R_1 + R_2$, where

$$R_1 = \epsilon_n \mu_X [(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - (\Sigma_{ZZ} + \epsilon_n I)^{-1}], \qquad R_2 = \epsilon_n \mu_X (\Sigma_{ZZ} + \epsilon_n I)^{-1}. \tag{S.13}$$

For any $\theta > 0$, we have

$$\begin{aligned} R_1 \Sigma_{ZZ}^\theta &= \epsilon_n \mu_X [(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - (\Sigma_{ZZ} + \epsilon_n I)^{-1}] \Sigma_{ZZ}^\theta \\ &= \epsilon_n \mu_X (\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} (\Sigma_{ZZ} - \hat{\Sigma}_{ZZ}) (\Sigma_{ZZ} + \epsilon_n I)^{-1} \Sigma_{ZZ}^\theta. \end{aligned}$$

Thus,

$$\|R_1 \Sigma_{ZZ}^\theta\|_{\text{OP}} \leq \epsilon_n \|\mu_X (\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}} \|\Sigma_{ZZ} - \hat{\Sigma}_{ZZ}\|_{\text{OP}} \|(\Sigma_{ZZ} + \epsilon_n I)^{-1} \Sigma_{ZZ}^\theta\|_{\text{OP}}. \tag{S.14}$$

In the right-hand side above, $\|\hat{\Sigma}_{ZZ} - \Sigma_{ZZ}\|_{\text{OP}} = O_P(n^{-1/2})$ by Hilbert space central limit theorem,

$$\|\mu_X (\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}} \leq \epsilon_n^{-1} \|\mu_X I\|_{\text{OP}} = \epsilon_n^{-1} \|\mu_X\|_{\mathcal{H}_X} = O(\epsilon_n^{-1}),$$

and

$$\|(\Sigma_{ZZ}+\epsilon_n I)^{-1}\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} \le \|(\Sigma_{ZZ}+\epsilon_n I)^{-1+\theta}\|_{\mathrm{OP}} = O(\epsilon_n^{\theta\wedge 1-1}). \tag{S.15}$$

Therefore,

$$\|R_1\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} = \epsilon_n O(\epsilon_n^{-1})O_P(n^{-1/2})O(\epsilon_n^{\theta\wedge 1-1}) = O_P(n^{-1/2}\epsilon_n^{\theta\wedge 1-1}).$$

Also, by (S.15), we have

$$\|R_2\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} = \epsilon_n\|\mu_X\|_{\mathcal{H}_X}\|(\Sigma_{ZZ}+\epsilon_n I)^{-1}\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} = \epsilon_n O_P(\epsilon_n^{\theta\wedge 1-1}) = O_P(\epsilon_n^{\theta\wedge 1}).$$

Hence,

$$\|R\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} \le \|R_1\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} + \|R_2\Sigma_{ZZ}^{\theta}\|_{\mathrm{OP}} = O_P(n^{-1/2}\epsilon_n^{\theta\wedge 1-1} + \epsilon_n^{\theta\wedge 1}),$$

which gives the desired result.

To prove assertion 3, we use the same decomposition $R = R_1 + R_2$ as in (S.13). Similar to (S.14), we have

$$\begin{aligned}\|R_1\Sigma_{ZZ}\|_{\mathrm{HS}} &\le \epsilon_n\|\mu_X(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}\|\Sigma_{ZZ}-\hat{\Sigma}_{ZZ}\|_{\mathrm{HS}}\|(\Sigma_{ZZ}+\epsilon_n I)^{-1}\Sigma_{ZZ}\|_{\mathrm{OP}} \\ &= \epsilon_n O(\epsilon_n^{-1})O_P(n^{-1/2})O(1) = O_P(n^{-1/2})\end{aligned}$$

since $\|\hat{\Sigma}_{ZZ}-\Sigma_{ZZ}\|_{\mathrm{HS}} = O_P(n^{-1/2})$ by Hilbert space central limit theorem. Different from assertion 2, regarding $R_2\Sigma_{ZZ}$, we have

$$\|R_2\Sigma_{ZZ}\|_{\mathrm{HS}} = \epsilon_n\|\mu_X\|_{\mathcal{H}_X}\|(\Sigma_{ZZ}+\epsilon_n I)^{-1}\Sigma_{ZZ}\|_{\mathrm{HS}} = \epsilon_n O_P(\epsilon_n^{-1/(2\alpha)}) = O_P(\epsilon_n^{1-1/(2\alpha)}),$$

where the second equality holds because, under Assumption 7, by Lemma 2,

$$\|(\Sigma_{ZZ}+\epsilon_n I)^{-1}\Sigma_{ZZ}\|_{\mathrm{HS}}^2 = \sum_{j=1}^{\infty}\left(\frac{\gamma_j}{\gamma_j+\epsilon_n}\right)^2 = O(\epsilon_n^{-1/\alpha}).$$

Hence,

$$\|R\Sigma_{ZZ}\|_{\mathrm{HS}} \le \|R_1\Sigma_{ZZ}\|_{\mathrm{HS}} + \|R_2\Sigma_{ZZ}\|_{\mathrm{HS}} = O_P(n^{-1/2} + \epsilon_n^{1-1/(2\alpha)}),$$

which gives the desired result. □

B.7 Proof of Lemma 4

Proof. We first decompose $\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R$ as follows:

$$\begin{aligned}&\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R \\ &= \hat{\Sigma}_{\ddot{X}Z}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger} - \mu_X\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} + \mu_X \\ &= \hat{\Sigma}_{\ddot{X}Z}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} \\ &\quad + \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \mu_X\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger} + \mu_X\end{aligned}$$

$$
\begin{aligned}
&= \hat{\Sigma}_{\ddot{X}Z}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} \\
&\quad + \Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger} \\
&= \left[\Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\right] + \left[\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\right] \\
&\quad + \left[(\hat{\Sigma}_{\ddot{X}Z} - \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ} - \tilde{\Sigma}_{\ddot{X}Z|Z})(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}\right] \\
&\quad + \left[\tilde{\Sigma}_{\ddot{X}Z|Z}[(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - (\Sigma_{ZZ}+\epsilon_n I)^{-1}]\right] + \left[\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\right],
\end{aligned}
$$

where $\tilde{\Sigma}_{\ddot{X}Z|Z} = E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}|Z) \otimes \tilde{\kappa}_Z(\cdot, Z)]$ and $\tilde{\kappa}_{\ddot{X}|Z}$ is defined by (11). By the triangular inequality,

$$
\begin{aligned}
&\|\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R\|_{\mathrm{OP}} \\
&\leq \|\Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\|_{\mathrm{OP}} \\
&\quad + \|\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} \\
&\quad + \|(\hat{\Sigma}_{\ddot{X}Z} - \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ} - \tilde{\Sigma}_{\ddot{X}Z|Z})(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} \\
&\quad + \|\tilde{\Sigma}_{\ddot{X}Z|Z}[(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - (\Sigma_{ZZ}+\epsilon_n I)^{-1}]\|_{\mathrm{OP}} + \|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}.
\end{aligned} \tag{S.16}
$$

We next analyze the five terms in (S.16) one by one. For the second term of (S.16), since

$$
\begin{aligned}
&\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1} \\
=&\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\Sigma_{ZZ}(\Sigma_{ZZ}+\epsilon_n I)^{-1} \\
=&\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\left[(\Sigma_{ZZ}+\epsilon_n I)\hat{\Sigma}_{ZZ} - \Sigma_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)\right](\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} \\
=&\epsilon_n\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}(\Sigma_{ZZ}+\epsilon_n I)^{-1}(\hat{\Sigma}_{ZZ} - \Sigma_{ZZ})(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1},
\end{aligned}
$$

we have

$$
\begin{aligned}
&\|\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} \\
\leq&\epsilon_n\|\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}\|\hat{\Sigma}_{ZZ} - \Sigma_{ZZ}\|_{\mathrm{OP}}\|(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}.
\end{aligned}
$$

By Hilbert-space central limit theorem, $\|\hat{\Sigma}_{ZZ} - \Sigma_{ZZ}\|_{\mathrm{OP}} = O_P(n^{-1/2})$. Also,

$$
\|(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} \leq \epsilon_n^{-1}\|I\|_{\mathrm{OP}} = \epsilon_n^{-1}.
$$

Regarding $\|\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}$, by $\Lambda_{\ddot{X}Z} = S_{\ddot{X}Z}\Sigma_{ZZ}^{1+\beta}$ in Assumption 6, we have

$$
\begin{aligned}
\|\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} =&\|S_{\ddot{X}Z}\Sigma_{ZZ}^{1+\beta}\Sigma_{ZZ}^{\dagger}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} \\
\leq&\|S_{\ddot{X}Z}\|_{\mathrm{OP}}\|\Sigma_{ZZ}^{\beta}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} \\
\leq&\|S_{\ddot{X}Z}\|_{\mathrm{OP}}\|(\Sigma_{ZZ}+\epsilon_n I)^{-1+\beta}\|_{\mathrm{OP}} = O(\epsilon_n^{\beta\wedge 1-1}).
\end{aligned} \tag{S.17}
$$

Hence the second term in (S.16) has the order of magnitude

$$
\begin{aligned}
\|\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1} - \Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}} =&\epsilon_n O(\epsilon_n^{\beta\wedge 1-1})O_P(n^{-1/2})O_P(\epsilon_n^{-1}) \\
=&O_P(n^{-1/2}\epsilon_n^{\beta\wedge 1-1}).
\end{aligned} \tag{S.18}
$$

For the first term in (S.16), we have

$$\begin{aligned}\Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}-\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger} =&\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}[\Sigma_{ZZ}-(\Sigma_{ZZ}+\epsilon_n I)](\Sigma_{ZZ}+\epsilon_n I)^{-1}\\ =&-\epsilon_n\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}(\Sigma_{ZZ}+\epsilon_n I)^{-1}.\end{aligned}$$

Hence, by (S.17) and Assumption 6, we have

$$\|\Lambda_{\ddot{X}Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}-\Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\|_{\mathrm{OP}}=O_P(\epsilon_n^{\beta\wedge 1}). \tag{S.19}$$

For the third term of (S.16), since

$$\begin{aligned}&\hat{\Sigma}_{\ddot{X}Z}-\Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}-\tilde{\Sigma}_{\ddot{X}Z|Z}\\ =&E_n[\hat{\kappa}_{\ddot{X}}(\cdot,\ddot{X})\otimes\hat{\kappa}_Z(\cdot,Z)]-B_{\ddot{X}|Z}E_n[\hat{\kappa}_Z(\cdot,Z)\otimes\hat{\kappa}_Z(\cdot,Z)]-E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\tilde{\kappa}_Z(\cdot,Z)]\\ =&E_n[\tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X})\otimes\hat{\kappa}_Z(\cdot,Z)]-B_{\ddot{X}|Z}E_n[\tilde{\kappa}_Z(\cdot,Z)\otimes\hat{\kappa}_Z(\cdot,Z)]-E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\tilde{\kappa}_Z(\cdot,Z)]\\ =&E_n\{[\tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X})-B_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot,Z)]\otimes\hat{\kappa}_Z(\cdot,Z)\}-E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\tilde{\kappa}_Z(\cdot,Z)]\\ =&E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\hat{\kappa}_Z(\cdot,Z)]-E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\tilde{\kappa}_Z(\cdot,Z)]\\ =&E_n\{\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes[\hat{\kappa}_Z(\cdot,Z)-\tilde{\kappa}_Z(\cdot,Z)]\}\\ =&-E_n(\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\{E_n[\kappa_Z(\cdot,Z)]-E[\kappa_Z(\cdot,Z)]\})\\ =&-E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)]\otimes\{E_n[\kappa_Z(\cdot,Z)]-E[\kappa_Z(\cdot,Z)]\}.\end{aligned}$$

Note that $E[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)]=0$. By Hilbert space central limit theorem, we have

$$\|E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)]\|_{\mathcal{G}_{\ddot{X}}}=O_P(n^{-1/2}),\quad \|E_n[\kappa_Z(\cdot,Z)]-E[\kappa_Z(\cdot,Z)]\|_{\mathcal{G}_Z}=O_P(n^{-1/2}).$$

Thus,

$$\begin{aligned}&\|\hat{\Sigma}_{\ddot{X}Z}-\Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}-\tilde{\Sigma}_{\ddot{X}Z|Z}\|_{\mathrm{OP}}\le\|\hat{\Sigma}_{\ddot{X}Z}-\Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}-\tilde{\Sigma}_{\ddot{X}Z|Z}\|_{\mathrm{HS}}\\ &=\|E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)]\|_{\mathcal{G}_{\ddot{X}}}\|E_n[\kappa_Z(\cdot,Z)]-E[\kappa_Z(\cdot,Z)]\|_{\mathcal{G}_Z}=O_P(n^{-1/2})O_P(n^{-1/2})=O_P(n^{-1}),\end{aligned}$$

which implies that

$$\begin{aligned}&\|(\hat{\Sigma}_{\ddot{X}Z}-\Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}-\tilde{\Sigma}_{\ddot{X}Z|Z})(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}\\ &\le\|\hat{\Sigma}_{\ddot{X}Z}-\Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}-\tilde{\Sigma}_{\ddot{X}Z|Z}\|_{\mathrm{OP}}\|(\hat{\Sigma}_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}=O_P(n^{-1})O_P(\epsilon_n^{-1})=O_P(n^{-1}\epsilon_n^{-1}).\end{aligned} \tag{S.20}$$

We now consider the fifth term of (S.16), $\|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{OP}}$. Note that this term is bounded by $\|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{HS}}$, and we calculate the expectation of its square as

$$\begin{aligned}E\|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{HS}}^2 =&E\|E_n[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\tilde{\kappa}_Z(\cdot,Z)](\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{HS}}^2\\ =&n^{-1}E\|[\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\otimes\tilde{\kappa}_Z(\cdot,Z)](\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\mathrm{HS}}^2\\ =&n^{-1}E[\|\tilde{\kappa}_{\ddot{X}|Z}(\cdot,\ddot{X}|Z)\|_{\mathcal{G}_{\ddot{X}}}^2\|(\Sigma_{ZZ}+\epsilon_n I)^{-1}\tilde{\kappa}_Z(\cdot,Z)\|_{\mathcal{G}_Z}^2],\end{aligned} \tag{S.21}$$

where the second equality holds because (X_i, Z_i) are i.i.d. samples for $i = 1, \ldots, n$, as well as

$$E[\tilde{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}|Z) \otimes \tilde{\kappa}_Z(\cdot, Z)] = E\{E[\tilde{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}|Z)] \otimes \tilde{\kappa}_Z(\cdot, Z)\} = 0.$$

Note that Assumption 5 implies that, for some uniform $C > 0$, we have

$$\begin{aligned}\|\tilde{\kappa}_{\ddot{X}|Z}(\cdot, \ddot{X}|Z)\|_{\mathcal{G}_{\ddot{X}}} &= \|\kappa_{\ddot{X}}(\cdot, \ddot{X}) - E[\kappa_{\ddot{X}}(\cdot, \ddot{X})|Z]\|_{\mathcal{G}_{\ddot{X}}} \\ &\leq 2\|\kappa_{\ddot{X}}(\cdot, \ddot{X})\|_{\mathcal{G}_{\ddot{X}}} + 2\|E[\kappa_{\ddot{X}}(\cdot, \ddot{X})|Z]\|_{\mathcal{G}_{\ddot{X}}} \leq 4C.\end{aligned}$$

It remains to analyze $\|(\Sigma_{ZZ} + \epsilon_n I)^{-1}\tilde{\kappa}_Z(\cdot, Z)\|^2_{\mathcal{G}_Z}$. By the Karhunen-Loéve expansion (25), we have

$$(\Sigma_{ZZ} + \epsilon_n I)^{-1}\tilde{\kappa}_Z(\cdot, Z) = \sum_{k=1}^{\infty}(\Sigma_{ZZ} + \epsilon_n I)^{-1}\gamma_k^{1/2}\xi_k\phi_k = \sum_{k=1}^{\infty}(\gamma_k + \epsilon_n)^{-1}\gamma_k^{1/2}\xi_k\phi_k,$$

we have

$$\|(\Sigma_{ZZ} + \epsilon_n I)^{-1}\tilde{\kappa}_Z(\cdot, Z)\|^2_{\mathcal{G}_Z} = \sum_{k=1}^{\infty}(\gamma_k + \epsilon_n)^{-2}\gamma_k\xi_k^2.$$

Note that $E(\xi_k^2) = 1$. Plugging the results back into (S.21), by Fubini's theorem, we have

$$\begin{aligned}E\|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ} + \epsilon_n I)^{-1}\|^2_{\text{HS}} =& n^{-1}16C^2E\left[\sum_{k=1}^{\infty}(\gamma_k + \epsilon_n)^{-2}\gamma_k\xi_k^2\right] \\ =& n^{-1}16C^2\sum_{k=1}^{\infty}(\gamma_k + \epsilon_n)^{-2}\gamma_k = O(n^{-1}\epsilon_n^{-(\alpha+1)/\alpha}),\end{aligned}$$

where the last equality holds because $\sum_{k=1}^{\infty}(\gamma_k + \epsilon_n)^{-2}\gamma_k = O(\epsilon_n^{-(\alpha+1)/\alpha})$ under Assumption 7 according to Lemma 2. By Chebyshev's inequality, we have

$$\|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}} \leq \|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ} + \epsilon_n I)^{-1}\|_{\text{HS}} = O_P(n^{-1/2}\epsilon_n^{-(\alpha+1)/(2\alpha)}). \quad \text{(S.22)}$$

Regarding the fourth term of (S.16), we have

$$\begin{aligned}&\|\tilde{\Sigma}_{\ddot{X}Z|Z}[(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - (\Sigma_{ZZ} + \epsilon_n I)^{-1}]\|_{\text{OP}} \\ &= \|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ} + \epsilon_n I)^{-1}(\Sigma_{ZZ} - \hat{\Sigma}_{ZZ})(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}} \\ &\leq \|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}}\|\Sigma_{ZZ} - \hat{\Sigma}_{ZZ}\|_{\text{OP}}\|(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}}.\end{aligned}$$

Note that the first part $\|\tilde{\Sigma}_{\ddot{X}Z|Z}(\Sigma_{ZZ}+\epsilon_n I)^{-1}\|_{\text{OP}}$ is exactly the fifth term, and its rate is given by (S.22). Same as the arguments in the second term of (S.16), we have $\|\hat{\Sigma}_{ZZ} - \Sigma_{ZZ}\|_{\text{OP}} = O_P(n^{-1/2})$ and $\|(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\text{OP}} = O_P(\epsilon_n^{-1})$. Therefore, the fourth term has the rate

$$\begin{aligned}\|\tilde{\Sigma}_{\ddot{X}Z|Z}[(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - (\Sigma_{ZZ} + \epsilon_n I)^{-1}]\|_{\text{OP}} =& O_P(n^{-1/2}\epsilon_n^{-(\alpha+1)/(2\alpha)})O_P(n^{-1/2})O_P(\epsilon_n^{-1}) \\ =& O_P(n^{-1}\epsilon_n^{-(3\alpha+1)/(2\alpha)}).\end{aligned} \quad \text{(S.23)}$$

Substituting (S.19), (S.18), (S.20), (S.23), (S.22) back into (S.16), we have the desired result for $\|\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R\|_{\mathrm{OP}}$.

For the rate of $\|\hat{B}_{Y|Z} - B_{Y|Z}\|_{\mathrm{OP}}$, using a similar triangle inequality argument, (S.16) becomes

$$\begin{aligned}
&\|\hat{B}_{Y|Z} - B_{Y|Z}\|_{\mathrm{OP}} = \|\hat{\Sigma}_{YZ}(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - \Sigma_{YZ}\Sigma_{ZZ}^{\dagger}\|_{\mathrm{OP}} \\
&\leq \|\Sigma_{YZ}(\Sigma_{ZZ} + \epsilon_n I)^{-1} - \Sigma_{YZ}\Sigma_{ZZ}^{\dagger}\|_{\mathrm{OP}} \\
&\quad + \|[\Sigma_{YZ}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ}(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - \Sigma_{YZ}(\Sigma_{ZZ} + \epsilon_n I)^{-1}\|_{\mathrm{OP}} \\
&\quad + \|(\hat{\Sigma}_{YZ} - \Sigma_{YZ}\Sigma_{ZZ}^{\dagger}\hat{\Sigma}_{ZZ} - \tilde{\Sigma}_{YZ|Z})(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1}\|_{\mathrm{OP}} \\
&\quad + \|\tilde{\Sigma}_{YZ|Z}[(\hat{\Sigma}_{ZZ} + \epsilon_n I)^{-1} - (\Sigma_{ZZ} + \epsilon_n I)^{-1}]\|_{\mathrm{OP}} \\
&\quad + \|\tilde{\Sigma}_{YZ|Z}(\Sigma_{ZZ} + \epsilon_n I)^{-1}\|_{\mathrm{OP}},
\end{aligned}$$

where

$$\tilde{\Sigma}_{YZ|Z} = E_n[\tilde{\kappa}_{Y|Z}(\cdot, Y|Z) \otimes \tilde{\kappa}_Z(\cdot, Z)],$$

and $\tilde{\kappa}_{Y|Z}$ is defined by (11). The remaining arguments are same as in $\|\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R\|_{\mathrm{OP}}$, with $\ddot{X}$ replaced by Y and $\Lambda_{\ddot{X}Z}$ replaced by Σ_{YZ}. □

B.8 Proof of Theorem 2

Proof. We first define two intermediate linear operators:

$$\begin{aligned}
\tilde{\Sigma}^{(1)}_{\ddot{X}Y|Z} =& E_n\left\{[\tilde{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) - B_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot, Z)] \otimes [\tilde{\kappa}_Y(\cdot, Y) - B_{Y|Z}\tilde{\kappa}_Z(\cdot, Z)]\right\}, \\
\tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z} =& E_n\left\{[\tilde{\kappa}_{\ddot{X}}(\cdot, \ddot{X}) - \hat{B}_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot, Z)] \otimes [\tilde{\kappa}_Y(\cdot, Y) - \hat{B}_{Y|Z}\tilde{\kappa}_Z(\cdot, Z)]\right\}.
\end{aligned}$$

Using these operators we make the following decomposition:

$$\hat{\Sigma}_{\ddot{X}Y|Z} - \Sigma_{\ddot{X}Y|Z} = \hat{\Sigma}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z} + \tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(1)}_{\ddot{X}Y|Z} + \tilde{\Sigma}^{(1)}_{\ddot{X}Y|Z} - \Sigma_{\ddot{X}Y|Z}$$

By the central limit theorem for Hilbert-space-valued random elements,

$$\sqrt{n}(\tilde{\Sigma}^{(1)}_{\ddot{X}Y|Z} - \Sigma_{\ddot{X}Y|Z}) \xrightarrow{\mathcal{D}} N(0, \Gamma_{\ddot{X}Y|Z}). \tag{S.24}$$

So we need to show that

$$\hat{\Sigma}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z} = o_p(n^{-1/2}), \quad \tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(1)}_{\ddot{X}Y|Z} = o_p(n^{-1/2}). \tag{S.25}$$

To show the first equation in (S.25), we begin with the decomposition

$$
\begin{aligned}
&\sqrt{n}(\hat{\Sigma}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z}) \\
&= \sqrt{n}\left(E_n\left\{[\hat{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) - \hat{B}_{\ddot{X}|Z}\hat{\kappa}_Z(\cdot,Z)] \otimes [\hat{\kappa}_Y(\cdot,Y) - \hat{B}_{Y|Z}\hat{\kappa}_Z(\cdot,Z)]\right\}\right. \\
&\qquad \left. - E_n\left\{[\tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) - \hat{B}_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot,Z)] \otimes [\tilde{\kappa}_Y(\cdot,Y) - \hat{B}_{Y|Z}\tilde{\kappa}_Z(\cdot,Z)]\right\}\right) \\
&= \sqrt{n}\big\{E_n[\hat{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \hat{\kappa}_Y(\cdot,Y) - \tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \tilde{\kappa}_Y(\cdot,Y)] \\
&\qquad - \hat{B}_{\ddot{X}|Z}E_n[\hat{\kappa}_Z(\cdot,Z) \otimes \hat{\kappa}_Y(\cdot,Y) - \tilde{\kappa}_Z(\cdot,Z) \otimes \tilde{\kappa}_Y(\cdot,Y)] \\
&\qquad - E_n[\hat{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \hat{\kappa}_Z(\cdot,Z) - \tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \tilde{\kappa}_Z(\cdot,Z)]\hat{B}^*_{Y|Z} \\
&\qquad + \hat{B}_{\ddot{X}|Z}E_n[\hat{\kappa}_Z(\cdot,Z) \otimes \hat{\kappa}_Z(\cdot,Z) - \tilde{\kappa}_Z(\cdot,Z) \otimes \tilde{\kappa}_Z(\cdot,Z)]\hat{B}^*_{Y|Z}\big\}.
\end{aligned}
\tag{S.26}
$$

We now consider each term on the right-hand side of (S.26) separately. By construction, the first term is

$$
\begin{aligned}
&\sqrt{n}E_n[\hat{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \hat{\kappa}_Y(\cdot,Y) - \tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \tilde{\kappa}_Y(\cdot,Y)] \\
&= \frac{1}{\sqrt{n}}\sum_{i=1}^n \left\{[\kappa_{\ddot{X}}(\cdot,\ddot{X}_i) - \hat{\mu}_{\ddot{X}}] \otimes [\kappa_Y(\cdot,Y_i) - \hat{\mu}_Y] - [\kappa_{\ddot{X}}(\cdot,\ddot{X}_i) - \mu_{\ddot{X}}] \otimes [\kappa_Y(\cdot,Y_i) - \mu_Y]\right\}.
\end{aligned}
$$

The right-hand side can be decomposed into eight terms, among which the two terms involving $\kappa_{\ddot{X}}(\cdot,\ddot{X}_i) \otimes \kappa_Y(\cdot,Y_i)$ and $-\kappa_{\ddot{X}}(\cdot,\ddot{X}_i) \otimes \kappa_Y(\cdot,Y_i)$ cancel out, leaving the following six terms:

$$
\begin{aligned}
\frac{1}{\sqrt{n}}\sum_{i=1}^n \big[&- \kappa_{\ddot{X}}(\cdot,\ddot{X}_i) \otimes \hat{\mu}_Y - \hat{\mu}_{\ddot{X}} \otimes \kappa_Y(\cdot,Y_i) + \hat{\mu}_{\ddot{X}} \otimes \hat{\mu}_Y \\
&+ \kappa_{\ddot{X}}(\cdot,\ddot{X}_i) \otimes \mu_Y + \mu_{\ddot{X}} \otimes \kappa_Y(\cdot,Y_i) - \mu_{\ddot{X}} \otimes \mu_Y\big].
\end{aligned}
$$

Next, applying the relations $n^{-1}\sum_{i=1}^n \kappa_{\ddot{X}}(\cdot,\ddot{X}_i) = \hat{\mu}_{\ddot{X}}$ and $n^{-1}\sum_{i=1}^n \kappa_Y(\cdot,Y_i) = \hat{\mu}_Y$, we further reduce the right-hand side above to

$$
\begin{aligned}
&\sqrt{n}(-\hat{\mu}_{\ddot{X}} \otimes \hat{\mu}_Y - \hat{\mu}_{\ddot{X}} \otimes \hat{\mu}_Y + \hat{\mu}_{\ddot{X}} \otimes \hat{\mu}_Y + \hat{\mu}_{\ddot{X}} \otimes \mu_Y + \mu_{\ddot{X}} \otimes \hat{\mu}_Y - \mu_{\ddot{X}} \otimes \mu_Y) \\
=&\sqrt{n}(-\hat{\mu}_{\ddot{X}} \otimes \hat{\mu}_Y + \hat{\mu}_{\ddot{X}} \otimes \mu_Y + \mu_{\ddot{X}} \otimes \hat{\mu}_Y - \mu_{\ddot{X}} \otimes \mu_Y) \\
=&-\sqrt{n}(\hat{\mu}_{\ddot{X}} - \mu_{\ddot{X}}) \otimes (\hat{\mu}_Y - \mu_Y).
\end{aligned}
$$

By Chebychev's inequality, we have $\hat{\mu}_{\ddot{X}} - \mu_{\ddot{X}} = O_P(n^{-1/2})$ and $\hat{\mu}_Y - \mu_Y = O_P(n^{-1/2})$. Hence the right-hand side is of the order $O_P(n^{-1/2})$. Thus we have proved

$$
\sqrt{n}E_n[\hat{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \hat{\kappa}_Y(\cdot,Y) - \tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \tilde{\kappa}_Y(\cdot,Y)] = O_P(n^{-1/2}).
$$

By the similar argument, we can show that

$$
\begin{aligned}
&\sqrt{n}E_n[\hat{\kappa}_Z(\cdot,Z) \otimes \hat{\kappa}_Y(\cdot,Y) - \tilde{\kappa}_Z(\cdot,Z) \otimes \tilde{\kappa}_Y(\cdot,Y)] = O_P(n^{-1/2}) \\
&\sqrt{n}E_n[\hat{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \hat{\kappa}_Z(\cdot,Z) - \tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) \otimes \tilde{\kappa}_Z(\cdot,Z)] = O_P(n^{-1/2}) \\
&\sqrt{n}E_n[\hat{\kappa}_Z(\cdot,Z) \otimes \hat{\kappa}_Z(\cdot,Z) - \tilde{\kappa}_Z(\cdot,Z) \otimes \tilde{\kappa}_Z(\cdot,Z)] = O_P(n^{-1/2}).
\end{aligned}
$$

Note that (28) and Assumption 6 imply that

$$B_{\ddot{X}|Z} = \Sigma_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger} = \Lambda_{\ddot{X}Z}\Sigma_{ZZ}^{\dagger} + \mu_X \Sigma_{ZZ}\Sigma_{ZZ}^{\dagger} = S_{\ddot{X}Z}\Sigma_{ZZ}^{\beta} + \mu_X I$$

is bounded and $B_{Y|Z} = \Sigma_{YZ}\Sigma_{ZZ}^{\dagger} = S_{YZ}\Sigma_{ZZ}^{\beta}$ is also bounded. By Corollary 4 and Lemma 4, we have $\hat{B}_{\ddot{X}|Z} = O_P(1)$ and $\hat{B}_{Y|Z} = O_P(1)$. Hence

$$\begin{aligned}
&\sqrt{n}(\hat{\Sigma}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z}) \\
&= O_P(n^{-1/2}) + O_P(1)O_P(n^{-1/2}) + O_P(n^{-1/2})O_P(1) + O_P(1)O_P(n^{-1/2})O_P(1) \\
&= O_P(n^{-1/2}) = o_P(1).
\end{aligned}$$

This proves the first equation in (S.25).

To show the second equation in (S.25), we begin by rewriting $\sqrt{n}(\tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(1)}_{\ddot{X}Y|Z})$ as

$$\begin{aligned}
&\sqrt{n}\Big(E_n\left\{[\tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) - \hat{B}_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot,Z)] \otimes [\tilde{\kappa}_Y(\cdot,Y) - \hat{B}_{Y|Z}\tilde{\kappa}_Z(\cdot,Z)]\right\} \\
&\quad - E_n\left\{[\tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) - B_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot,Z)] \otimes [\tilde{\kappa}_Y(\cdot,Y) - B_{Y|Z}\tilde{\kappa}_Z(\cdot,Z)]\right\}\Big).
\end{aligned} \tag{S.27}$$

Define random elements in $\epsilon_{\ddot{X}} \in \mathcal{G}_{\ddot{X}}$ and $\epsilon_Y \in \mathcal{G}_Y$ as follows

$$\epsilon_{\ddot{X}} = \tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) - B_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot,Z), \quad \epsilon_Y = \tilde{\kappa}_Y(\cdot,Y) - B_{Y|Z}\tilde{\kappa}_Z(\cdot,Z).$$

Then,

$$\begin{aligned}
\tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X}) - \hat{B}_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot,Z) &= \epsilon_{\ddot{X}} - (\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z})\tilde{\kappa}_Z(\cdot,Z) \\
\tilde{\kappa}_Y(\cdot,Y) - \hat{B}_{Y|Z}\tilde{\kappa}_Z(\cdot,Z) &= \epsilon_Y - (\hat{B}_{Y|Z} - B_{Y|Z})\tilde{\kappa}_Z(\cdot,Z)
\end{aligned}$$

Substituting these into (S.27), we have

$$\begin{aligned}
&\sqrt{n}(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})E_n[\tilde{\kappa}_Z(\cdot,Z) \otimes \epsilon_Y] + \sqrt{n}E_n[\epsilon_{\ddot{X}} \otimes \tilde{\kappa}_Z(\cdot,Z)](B_{Y|Z} - \hat{B}_{Y|Z})^* \\
&\quad + \sqrt{n}(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})E_n[\tilde{\kappa}_Z(\cdot,Z) \otimes \tilde{\kappa}_Z(\cdot,Z)](B_{Y|Z} - \hat{B}_{Y|Z})^*.
\end{aligned} \tag{S.28}$$

We now consider the three terms in (S.28) one by one. For the first term of (S.28), take η to be a number such that

$$\max\left\{0, \frac{\alpha-1}{2\alpha} - (\beta \wedge 1)\right\} < \eta < \frac{\alpha-1}{2\alpha}.$$

Then,

$$\begin{aligned}
&\sqrt{n}(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})E_n[\tilde{\kappa}_Z(\cdot,Z) \otimes \epsilon_Y] = (B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})\Sigma_{ZZ}^{\eta}\sqrt{n}E_n\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z) \otimes \epsilon_Y\right] \\
&= -(\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R)\Sigma_{ZZ}^{\eta}\sqrt{n}E_n\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z) \otimes \epsilon_Y\right] - R\Sigma_{ZZ}^{\eta}\sqrt{n}E_n\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z) \otimes \epsilon_Y\right]
\end{aligned}$$

By Corollary 1, we have

$$E(\epsilon_Y|Z) = E[\tilde{\kappa}_Y(\cdot,Y)|Z] - B_{Y|Z}\tilde{\kappa}_Z(\cdot,Z) = 0.$$

Hence

$$E\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y\right]=E\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes E(\epsilon_Y|Z)\right]=0.$$

Since

$$\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)=\left(\sum_{j=1}^{\infty}\gamma_j^{-\eta}\phi_j\otimes\phi_j\right)\left(\sum_{k=1}^{\infty}\xi_k\phi_k\right)=\sum_{j=1}^{\infty}\gamma_j^{-\eta}\xi_j\phi_j,$$

where ξ_j are uncorrelated with mean zero and variance γ_j, we have, by Assumption 7,

$$E\|\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\|_{\mathcal{G}_Z}^2=\sum_{j=1}^{\infty}\gamma_j^{-2\eta}E(\xi_j^2)=\sum_{j=1}^{\infty}\gamma_j^{1-2\eta}\preceq\sum_{j=1}^{\infty}j^{-\alpha(1-2\eta)}<\infty,$$

where the sum is finite because $\alpha(1-2\eta)>1$. Moreover, Assumptions 5 and 6 guarantee that $E\|\epsilon_Y\|_{\mathcal{G}_Y}^2<\infty$. By the central limit theorem for i.i.d. random elements in Hilbert spaces, we have

$$\sqrt{n}E_n\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y(\cdot)\right]\xrightarrow{\mathcal{D}}N(0,\Xi_{ZY}),$$

where $\Xi_{ZY}=E\left\{[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y(\cdot)]\otimes[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y(\cdot)]\right\}$. Hence

$$E_n\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y(\cdot)\right]=O_P(n^{-1/2}).$$

By Corollary 4, the conditions given for α and β guarantee that $\hat{B}_{\ddot{X}|Z}-B_{\ddot{X}|Z}-R=o_P(1)$. Also, we know that Σ_{ZZ}^{η} is bounded. Thus,

$$(\hat{B}_{\ddot{X}|Z}-B_{\ddot{X}|Z}-R)\Sigma_{ZZ}^{\eta}\sqrt{n}E_n\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y\right]=o_P(1)\sqrt{n}O_P(n^{-1/2})=o_P(1).$$

Taking the θ in (29) to be η, we have

$$\|R\Sigma_{ZZ}^{\eta}\|_{\mathrm{OP}}=O_P(n^{-1/2}\epsilon_n^{\eta\wedge1-1})+O_P(\epsilon_n^{\eta\wedge1}).\tag{S.29}$$

Since $\epsilon_n=o(1)$ and $\eta>0$, the second term on the right is $o_P(1)$. Substituting $\epsilon_n\asymp n^{-\frac{\alpha}{2\alpha(\beta\wedge1)+\alpha+1}}$ into the first term on the right, it becomes n^c where

$$c=-\frac{1}{2}-\frac{\alpha}{2\alpha(\beta\wedge1)+\alpha+1}(\eta-1)<-\frac{1}{2}-\frac{\alpha}{2\alpha(\beta\wedge1)+\alpha+1}\left\{\frac{\alpha-1}{2\alpha}-(\beta\wedge1)-1\right\}=0.$$

Hence the first term on the right-hand side of (S.29) is also $o_P(1)$, implying

$$R\Sigma_{ZZ}^{\eta}\sqrt{n}E_n\left[\Sigma_{ZZ}^{-\eta}\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y\right]=o_P(1)O_P(1)=o_P(1).\tag{S.30}$$

Thus we have proved that the first term in (S.28) is of the order $o_P(1)$; that is,

$$\sqrt{n}(B_{\ddot{X}|Z}-\hat{B}_{\ddot{X}|Z})E_n[\tilde{\kappa}_Z(\cdot,Z)\otimes\epsilon_Y]=o_P(1).\tag{S.31}$$

For the second term of (S.28), similarly, by Corollary 1, we have

$$E(\epsilon_{\ddot{X}}|Z)=E[\tilde{\kappa}_{\ddot{X}}(\cdot,\ddot{X})|Z]-B_{\ddot{X}|Z}\tilde{\kappa}_Z(\cdot,Z)=0.$$

Hence

$$E\left[\epsilon_{\ddot{X}} \otimes \tilde{\kappa}_Z(\cdot, Z)\right] = E\left[E(\epsilon_{\ddot{X}}|Z) \otimes \tilde{\kappa}_Z(\cdot, Z)\right] = 0.$$

Moreover, Assumptions 5 and 6 guarantee that $E\|\epsilon_{\ddot{X}}\|^2_{\mathcal{H}_{\ddot{X}}} < \infty$ and $E\|\tilde{\kappa}_Z(\cdot, Z)\|^2_{\mathcal{H}_Z} < \infty$. By the central limit theorem for i.i.d. random elements in Hilbert spaces, we have

$$\sqrt{n} E_n\left[\epsilon_{\ddot{X}}(\cdot) \otimes \tilde{\kappa}_Z(\cdot, Z)\right] \xrightarrow{\mathcal{D}} N(0, \Xi_{\ddot{X}Z}),$$

where $\Xi_{\ddot{X}Z} = E\left\{[\epsilon_{\ddot{X}}(\cdot) \otimes \tilde{\kappa}_Z(\cdot, Z)] \otimes [\epsilon_{\ddot{X}}(\cdot) \otimes \tilde{\kappa}_Z(\cdot, Z)]\right\}$. Hence

$$E_n\left[\epsilon_{\ddot{X}}(\cdot) \otimes \tilde{\kappa}_Z(\cdot, Z)\right] = O_P(n^{-1/2}).$$

Furthermore, by Corollary 4, the conditions given for α and β guarantee that $\hat{B}_{Y|Z} - B_{Y|Z} = o_P(1)$. Hence the second term of (S.28) becomes

$$\sqrt{n} E_n\left[\epsilon_{\ddot{X}} \otimes \tilde{\kappa}_Z(\cdot, Z)\right](B_{Y|Z} - \hat{B}_{Y|Z})^* = \sqrt{n} O_P(n^{-1/2}) o_P(1) = o_P(1). \tag{S.32}$$

We now consider the last term of (S.28). Note that

$$\begin{aligned}
&\sqrt{n}(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z}) E_n[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)](B_{Y|Z} - \hat{B}_{Y|Z})^* \\
&= (B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})\sqrt{n}\left\{E_n[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)] - \Sigma_{ZZ}\right\}(B_{Y|Z} - \hat{B}_{Y|Z})^* \\
&\quad - \sqrt{n}(\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R)\Sigma_{ZZ}(B_{Y|Z} - \hat{B}_{Y|Z})^* - \sqrt{n} R \Sigma_{ZZ}(B_{Y|Z} - \hat{B}_{Y|Z})^*.
\end{aligned} \tag{S.33}$$

For the first term in (S.33), since $\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R = o_P(1)$ by Corollary 4 and R is bounded by Lemma 3, we have $\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} = O_p(1)$. Furthermore, by the central limit theorem in Hilbert spaces, we have, under Assumption 5,

$$\sqrt{n}\left\{E_n[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)] - \Sigma_{ZZ}\right\} \xrightarrow{\mathcal{D}} N(0, \Xi_{ZZ}),$$

where $\Xi_{ZZ} = E\left\{[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)] \otimes [\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)]\right\}$. Hence

$$E_n[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)] - \Sigma_{ZZ} = O_P(n^{-1/2}).$$

Also, by Corollary 4, the conditions given for α and β guarantee that $\hat{B}_{Y|Z} - B_{Y|Z} = o_P(1)$. Therefore, the first term in (S.33) is of the order

$$\begin{aligned}
&(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})\sqrt{n}\left\{E_n[\tilde{\kappa}_Z(\cdot, Z) \otimes \tilde{\kappa}_Z(\cdot, Z)] - \Sigma_{ZZ}\right\}(B_{Y|Z} - \hat{B}_{Y|Z})^* \\
&= O_P(1)\sqrt{n} O_P(n^{-1/2}) o_P(1) = o_P(1).
\end{aligned}$$

For the second term in (S.33), since the conditions given for α and β guarantee that

$$\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R = o_P(n^{-1/4}), \qquad \hat{B}_{Y|Z} - B_{Y|Z} = o_P(n^{-1/4}),$$

we have

$$\sqrt{n}(\hat{B}_{\ddot{X}|Z} - B_{\ddot{X}|Z} - R)\Sigma_{ZZ}(B_{Y|Z} - \hat{B}_{Y|Z})^* = \sqrt{n} o_P(n^{-1/4}) O(1) o_p(n^{-1/4}) = o_P(1). \tag{S.34}$$

We now consider the third term in (S.33). By (30), we know that

$$\|R\Sigma_{ZZ}\|_{\text{HS}} = O_P(n^{-1/2}) + O_P(\epsilon_n^{1-1/(2\alpha)}).$$

Also, by Corollary 4, the conditions given for α and β guarantee that

$$\hat{B}_{Y|Z} - B_{Y|Z} = O_P(n^{-\frac{\alpha(\beta\wedge 1)}{2\alpha(\beta\wedge 1)+\alpha+1}}) = o_P(1), \quad \text{and} \quad \epsilon_n \asymp n^{-\frac{\alpha}{2\alpha(\beta\wedge 1)+\alpha+1}}.$$

Thus,

$$\sqrt{n}R\Sigma_{ZZ}(\hat{B}_{Y|Z} - B_{Y|Z}) = \sqrt{n}O_P(n^{-1/2})o_P(1) + \sqrt{n}O_P(n^{-\frac{\alpha(\beta\wedge 1)}{2\alpha(\beta\wedge 1)+\alpha+1}})O(n^{-\frac{\alpha-1/2}{2\alpha(\beta\wedge 1)+\alpha+1}}). \tag{S.35}$$

The first term on the right is obviously $o_P(1)$. The second term is of the form n^c where

$$\begin{aligned} c &= \frac{1}{2} - \frac{\alpha(\beta\wedge 1)+\alpha-1/2}{2\alpha(\beta\wedge 1)+\alpha+1} = \frac{2\alpha(\beta\wedge 1)+\alpha+1-2\alpha(\beta\wedge 1)-2\alpha+1}{2[2\alpha(\beta\wedge 1)+\alpha+1]} \\ &= \frac{2-\alpha}{2[2\alpha(\beta\wedge 1)+\alpha+1]} < 0 \end{aligned}$$

since $\alpha > 2$. Thus all three parts of the last term in (S.28) are of the order $o_P(1)$; that is,

$$\sqrt{n}(B_{\ddot{X}|Z} - \hat{B}_{\ddot{X}|Z})E_n[\tilde{\kappa}_Z(\cdot,Z)\otimes\tilde{\kappa}_Z(\cdot,Z)](B_{Y|Z} - \hat{B}_{Y|Z})^* = o_P(1). \tag{S.36}$$

Substituting (S.31), (S.32), and (S.36) into (S.28), we have

$$\sqrt{n}(\tilde{\Sigma}^{(2)}_{\ddot{X}Y|Z} - \tilde{\Sigma}^{(1)}_{\ddot{X}Y|Z}) = o_P(1). \tag{S.37}$$

This proves the second equation in (S.25), completing the proof of the theorem. □

B.9 Proof of Theorem 3

Proof. Following the argument in the proof of Theorem 3 of Tang and Li (2024), we have the expansion

$$n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}} = n\|\hat{\Sigma}_{\ddot{X}Y|Z} - \Sigma_{\ddot{X}Y|Z}\|^2_{\text{HS}} + 2n\langle\hat{\Sigma}_{\ddot{X}Y|Z} - \Sigma_{\ddot{X}Y|Z}, \Sigma_{\ddot{X}Y|Z}\rangle_{\text{HS}} + n\|\Sigma_{\ddot{X}Y|Z}\|^2_{\text{HS}}.$$

By Theorem 2, $\sqrt{n}(\hat{\Sigma}_{\ddot{X}Y|Z} - \Sigma_{\ddot{X}Y|Z}) \xrightarrow{\mathcal{D}} G$ where $G \sim N(0, \Gamma_{\ddot{X}Y|Z})$. By the continuous mapping theorem,

$$n\|\hat{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}} \xrightarrow{\mathcal{D}} \|G\|^2_{\text{HS}} + 2\sqrt{n}\langle G, \Sigma_{\ddot{X}Y|Z}\rangle_{\text{HS}} + n\|\Sigma_{\ddot{X}Y|Z}\|^2_{\text{HS}}.$$

Thus, under $H_1^{(n)}$,

$$n\|\breve{\Sigma}_{\ddot{X}Y|Z}\|^2_{\text{HS}} \xrightarrow{\mathcal{D}} \|G\|^2_{\text{HS}} + 2\langle G, \Sigma_1\rangle_{\text{HS}} + c^2. \tag{S.38}$$

Since $G \sim N(0, \Gamma_{\ddot{X}Y|Z})$ ane $(\lambda_1, v_1), (\lambda_2, v_2), \ldots$ are eigenvalue-eigenfunction pairs of $\Gamma_{\ddot{X}Y|Z}$, we see that $\langle G, v_j \rangle_{\mathrm{HS}} \sim N(0, \lambda_j)$ for $j = 1, 2, \ldots$. Therefore, the Karhunen-Loéve expansion for G is

$$G = \sum_{j=1}^{\infty} \sqrt{\lambda_j} Z_j v_j,$$

where $Z_1, Z_2, \ldots$ are independent standard normal random variables. See Example 11.4.1 of Kokoszka and Reimherr (2017) or Section 7.3 of Hsing and Eubank (2015) for details. Also, since $\Sigma_1 = \sum_{j=1}^{\infty} \sigma_j v_j$, we have $\sum_{j=1}^{\infty} \sigma_j^2 = \|\Sigma_1\|_{\mathrm{HS}}^2 = c^2$. In addition, since

$$\|G\|_{\mathrm{HS}}^2 = \sum_{j=1}^{\infty} \lambda_j Z_j^2, \quad \langle G, \Sigma_1 \rangle_{\mathrm{HS}} = \sum_{j=1}^{\infty} \sqrt{\lambda_j} Z_j \sigma_j,$$

we can rewrite the right-hand side of (S.38) as

$$\begin{aligned} \|G\|_{\mathrm{HS}}^2 + 2\langle G, \Sigma_1 \rangle_{\mathrm{HS}} + c^2 &= \sum_{j=1}^{\infty} \lambda_j Z_j^2 + 2 \sum_{j=1}^{\infty} \sqrt{\lambda_j} Z_j \sigma_j + \sum_{j=1}^{\infty} \sigma_j^2 \\ &= \sum_{j=1}^{\infty} \lambda_j (Z_j + \sigma_j / \sqrt{\lambda_j})^2 =: \sum_{j=1}^{\infty} \lambda_j \tilde{Z}_j^2, \end{aligned}$$

where $\tilde{Z}_1, \tilde{Z}_2, \ldots$ are independent $N(\sigma_j / \sqrt{\lambda_j}, 1)$, as desired. The local power result follows from the definition of convergence in distribution. □

C Additional Plots for Section 8

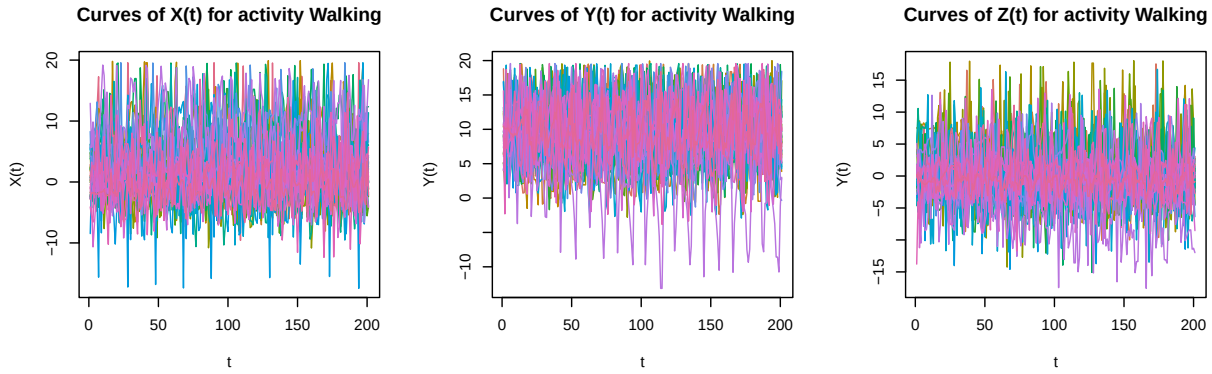


Figure S.1: Acceleration curves for Walking based on $n = 36$ users and $m = 200$ observations per user. The left, middle, and right panels show $X(t)$, $Y(t)$, and $Z(t)$, respectively, and each colored curve represents one user. The observations cover the first 10 seconds of data recorded at 20 Hz.

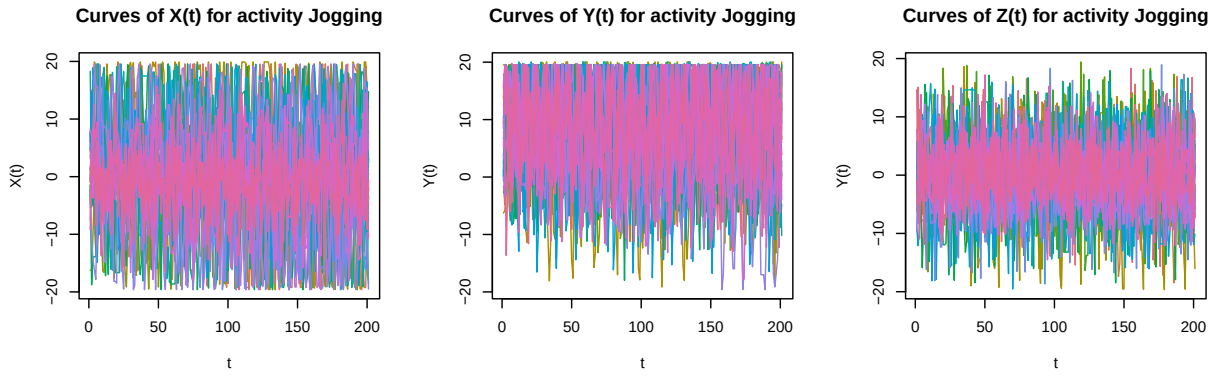


Figure S.2: Acceleration curves for Jogging based on $n = 32$ users. The panel arrangement and observation window are the same as in Figure S.1.

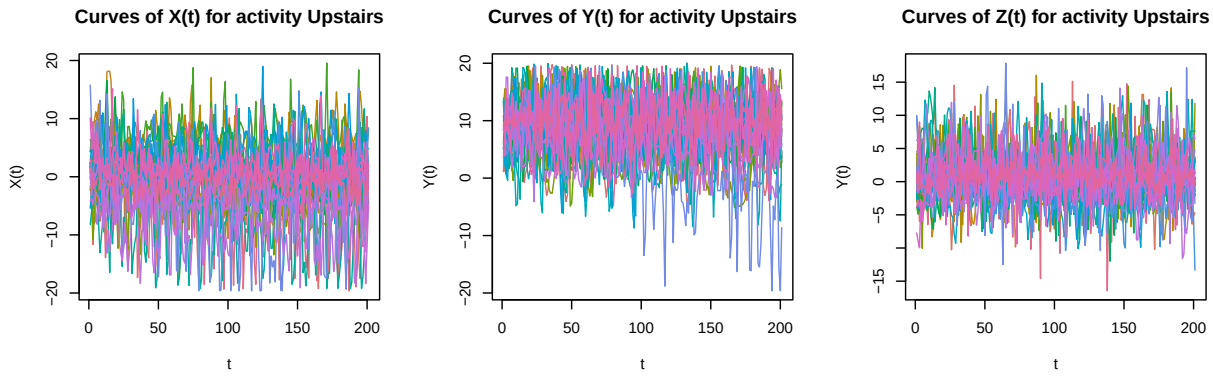


Figure S.3: Acceleration curves for Upstairs based on $n = 32$ users. The panel arrangement and observation window are the same as in Figure S.1.

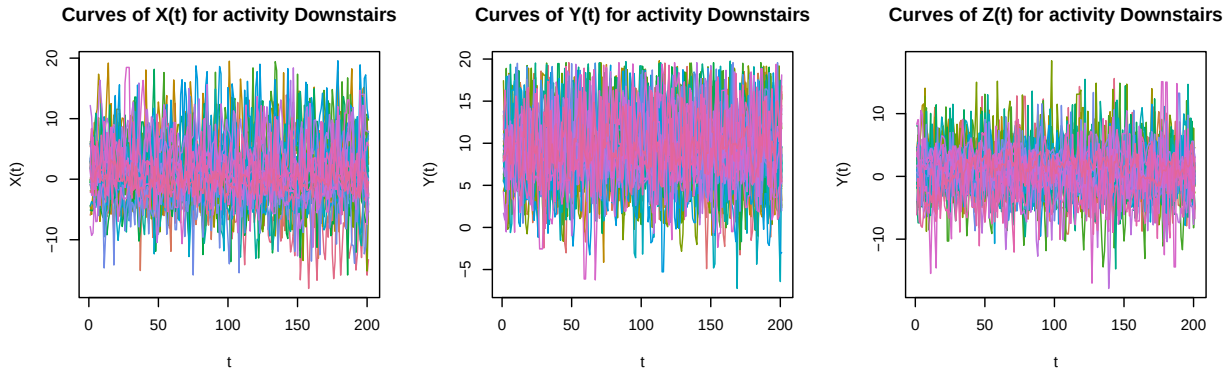


Figure S.4: Acceleration curves for Downstairs based on $n = 32$ users. The panel arrangement and observation window are the same as in Figure S.1.

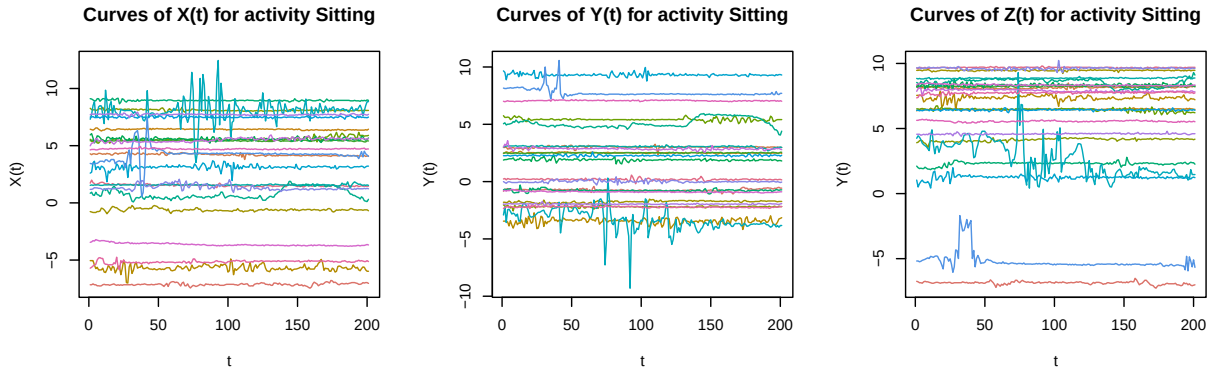


Figure S.5: Acceleration curves for Sitting based on $n = 23$ users. The panel arrangement and observation window are the same as in Figure S.1.

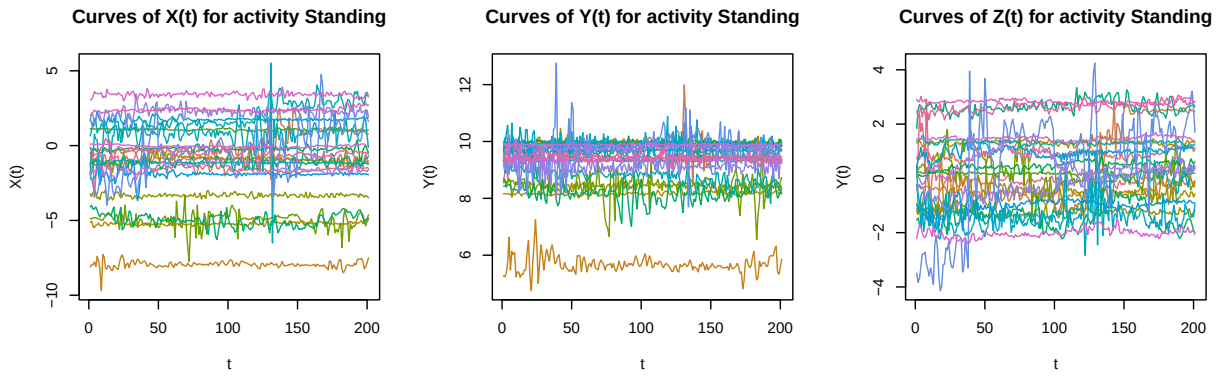


Figure S.6: Acceleration curves for Standing based on $n = 24$ users. The panel arrangement and observation window are the same as in Figure S.1.

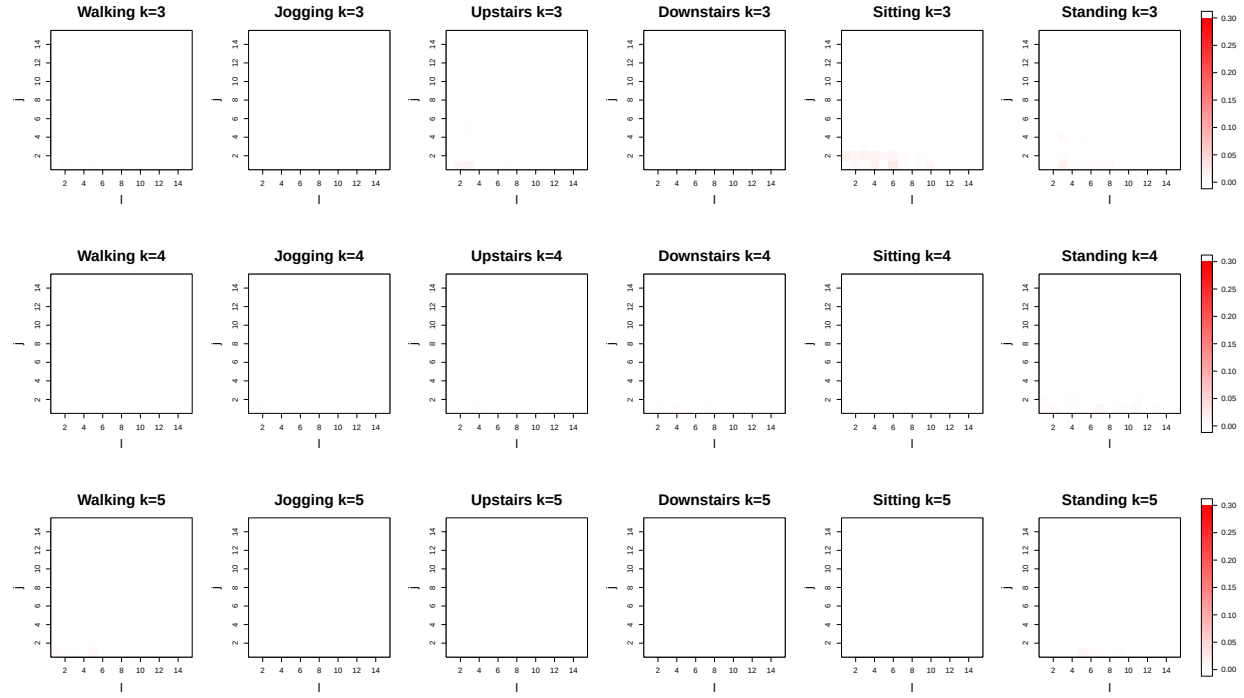


Figure S.7: Heatmaps of $s^{(3)}_{jk\ell}$ for $j, \ell \in \{1, \ldots, 15\}$, which measures a scaled third-order association involving the j-th component on the $\ddot{X}$ side and the k-th and ℓ-th components of Z. The three rows correspond to $k = 3$, 4, and 5. All panels use the same color scale from 0 to 0.3, with darker colors indicating larger values. Most values are close to zero.

D Application to WDI Dataset

We now apply our test to several indicators from the World Development Indicators (WDI) database (World Bank, 2025) to investigate whether pairs of development indicators are

k	Walking	Jogging	Upstairs	Downstairs	Sitting	Standing
1	0.0154	0.0042	0.0112	0.0096	0.2636	0.1413
2	0.0113	0.0038	0.0158	0.0080	0.1371	0.0790
3	0.0044	0.0027	0.0172	0.0029	0.0258	0.0174
4	0.0020	0.0030	0.0041	0.0075	0.0039	0.0096
5	0.0074	0.0025	0.0025	0.0027	0.0019	0.0108
6	0.0017	0.0014	0.0063	0.0026	0.0031	0.0038
7	0.0018	0.0016	0.0045	0.0027	0.0002	0.0010
8	0.0011	0.0013	0.0022	0.0012	0.0000	0.0015
9	0.0011	0.0016	0.0028	0.0009	0.0000	0.0013
10	0.0016	0.0012	0.0017	0.0013	0.0000	0.0005
11	0.0012	0.0020	0.0025	0.0017	0.0000	0.0007
12	0.0016	0.0014	0.0013	0.0010	0.0000	0.0002
13	0.0007	0.0012	0.0014	0.0013	0.0000	0.0002
14	0.0008	0.0006	0.0016	0.0015	0.0000	0.0002
15	0.0009	0.0012	0.0014	0.0015	0.0000	0.0000

Table S.1: Maximum values of $s^{(3)}_{jk\ell}$ over $j, \ell \in 1, \ldots, 15$ for each activity and $k \in 1, \ldots, 15$. For $k \geq 6$, all reported values are below 0.01.

conditionally independent given a country's GDP.[2] In other words, we examine whether the relationship between two socioeconomic factors can be explained solely by the level of economic development. The conditioning variable is `NY.GDP.MKTP.CD`, which represents GDP (current US$). The candidate factors considered in our analysis are listed in Table S.2.

	Variable	Explanation
1	`AG.LND.AGRI.ZS`	Agricultural land (% of land area)
2	`NV.AGR.TOTL.ZS`	Agriculture, forestry, and fishing, value added (% of GDP)
3	`SP.POP.GROW`	Population growth (annual %)
4	`FP.CPI.TOTL.ZG`	Inflation, consumer prices (annual %)
5	`FR.INR.RINR`	Real interest rate (%)
6	`SL.UEM.TOTL.NE.ZS`	Unemployment, total (% of total labor force) (national estimate)
7	`SP.DYN.LE00.IN`	Life expectancy at birth, total (years)

Table S.2: WDI factors included in the conditional independence analysis. The table gives the WDI variable code and description for each factor. GDP in current US dollars is used as the conditioning variable.

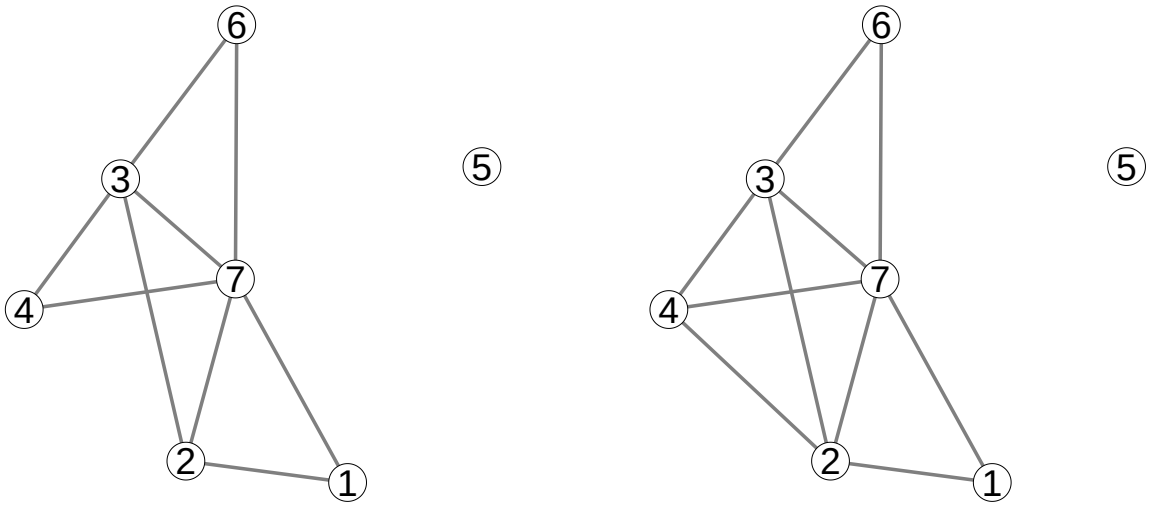
We use data from 2000 to 2023, so that $m = 23$, and treat the observations as a balanced design. For each pair of factors together with GDP, we remove all countries with missing values. The resulting sample sizes for all tests are reported in Table S.3. For each pair

[2]The dataset is available at `https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators`, version 123 (metadata last updated on October 9, 2025). We downloaded the CSV archive and used the file `WDICSV.csv`.

among the seven selected factors, we conduct our CCCO-based test for conditional independence given GDP at the significance level $\alpha = 0.05$. The resulting conditional dependency structure among the factors is shown in Figure S.8, where an edge between two factors indicates that they are not conditionally independent given GDP. All corresponding p-values are reported in Table S.5.

	1	2	3	4	5	6	7
1		163	188	58	70	57	189
2	163		166	53	65	56	167
3	188	166		59	70	58	194
4	58	53	59		19	41	59
5	70	65	70	19		19	71
6	57	56	58	41	19		58
7	189	167	194	59	71	58	

Table S.3: Sample sizes for the pairwise CCCO tests after countries with missing observations are removed. The row and column indices correspond to the factors in Table S.2. Sample sizes range from 19 to 194, reflecting differences in data availability across factor pairs.



(a) Results using weighted sum of $\chi^2_{(1)}$ distributions.

(b) Results with Welch-Satterthwaite approximation.

Figure S.8: Conditional dependency graphs for the seven WDI factors in Table S.2 given GDP. An edge indicates rejection of conditional independence at $\alpha = 0.05$. Panels (a) and (b) use the weighted sum of $\chi^2_{(1)}$ distributions and the Welch-Satterthwaite approximation, respectively. The graphs differ only in the edge between Factors 2 and 4, while Factor 5 is isolated in both panels.

	1	2	3	4	5	6	7
1		0.0013	0.1893	0.2925	0.1553	0.6596	0.0036
2	0.0013		0.0000	0.1029	0.2221	0.3105	0.0000
3	0.1893	0.0000		0.0469	0.1373	0.0132	0.0000
4	0.2925	0.1029	0.0469		0.2475	0.1898	0.0428
5	0.1553	0.2221	0.1373	0.2475		0.3454	0.2258
6	0.6596	0.3105	0.0132	0.1898	0.3454		0.0242
7	0.0036	0.0000	0.0000	0.0428	0.2258	0.0242	

Table S.4: CCCO p-values computed using the weighted sum of $\chi^2_{(1)}$ distributions. The row and column indices correspond to the factors in Table S.2. Values below 0.05 correspond to the edges in Figure S.8a.

	1	2	3	4	5	6	7
1		0.0000	0.1484	0.2795	0.0962	0.7570	0.0000
2	0.0000		0.0000	0.0479	0.1740	0.3002	0.0000
3	0.1484	0.0000		0.0097	0.0733	0.0006	0.0000
4	0.2795	0.0479	0.0097		0.2148	0.1392	0.0084
5	0.0962	0.1740	0.0733	0.2148		0.3512	0.1786
6	0.7570	0.3002	0.0006	0.1392	0.3512		0.0021
7	0.0000	0.0000	0.0000	0.0084	0.1786	0.0021	

Table S.5: CCCO p-values computed using the Welch–Satterthwaite approximation. The row and column indices correspond to the factors in Table S.2. Values below 0.05 correspond to the edges in Figure S.8b.

As shown in Figure S.8, life expectancy at birth (Factor 7) is conditionally dependent on each of the other factors except the real interest rate (Factor 5) given GDP. This suggests that the relationships between life expectancy and these factors cannot be explained solely by GDP. In contrast, the real interest rate (Factor 5) appears to be conditionally independent of all other factors given GDP, indicating that its associations with the remaining variables are largely mediated through GDP. Note that, comparing Figures S.8a and S.8b, the dependence results are only different between Factors 2 and 4, but the corresponding p-value with Welch-Satterthwaite approximation is 0.0479, which is very close to $\alpha = 0.05$. We should mention that the sample sizes associated with Factors 4, 5, and 6 are relatively small. Therefore, the conclusions should be interpreted with caution. In particular, the power of the asymptotic test may be limited when the sample size is small, and the presence of missing data may introduce additional bias. Further investigation may therefore be warranted to better understand the relationships among these variables.

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